

Two-coin counterexamples to the proposed generalized Shepp–Olkin thresholds

All finite orders above one, exact rational certificates,
and the fixed-mean entropy phase diagram

With Part II (added 29 September 2026): *Nine bits, ten bits*, sharp dimension
boundaries for product-Bernoulli entropy

With Part III (added 29 September 2026): *Beyond nine bits*, exact
alphabet-size criteria for product entropy

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Abstract

Hillion and Johnson proposed that the Rényi entropy of a sum of independent Bernoulli variables is jointly concave in the Bernoulli parameters up to order 2, and that the corresponding Tsallis threshold is approximately 3.65986. We give a counterexample to both proposed numerical thresholds. In fact, for every finite real $q > 1$, both entropies fail even quasiconcavity for the sum of *two* independent Bernoulli variables, with all parameters strictly between zero and one. The obstruction is a mean-preserving interpolation $(p + t, p - t)$ near the all-zero corner. Its power sum has a negative second variation at the balanced point when p is sufficiently small.

The counterexamples admit a fully explicit dyadic family. At $q = 2$, one particularly small certificate has midpoint $(1/10, 1/10)$ and endpoints $(3/20, 1/20)$ and $(1/20, 3/20)$; the Tsallis midpoint deficit is exactly $181/80000$. We also determine the sharp balanced-point curvature transition, classify every fixed-mean two-coin entropy maximizer for $q > 1$, prove concavity on all two-coin fixed-mean segments for $0 < q \leq 1$, and identify the minimum number of coins needed for failure. The transition near $q = 1$ occurs on an exponentially small parameter scale.

The all-order disproof is elementary and does not depend on numerical experiments. Reproducible exact and symbolic checks accompany the article. The general joint-concavity question for $0 < q < 1$ is not settled here, and bibliographic priority and the proofs have not been independently refereed.

[Added 29 September 2026, batch 43: Part II settles this question for the joint bit vector (collision-free weighted sums) instead of the sum: Rényi entropy is jointly concave in every dimension, Tsallis entropy for every $0 < q < 1$ exactly up to nine bits. For the ordinary sum it stays open here.]

Selection record. A single OS-backed uniform draw selected *Discrete probability*, area 58 of 96. The supplied manifest’s 71 excluded research packages were reviewed before the problem was chosen. The complete area list and original draw are included in the archive.

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1 The published problem and the scope of the result

1.1 Bernoulli sums and entropy

Let $X_1, \dots, X_n$ be independent, with $\mathbb{P}(X_i = 1) = p_i$ and $\mathbb{P}(X_i = 0) = 1 - p_i$. Write

$$S_{\mathbf{p}} = X_1 + \dots + X_n, \quad f_k(\mathbf{p}) = \mathbb{P}(S_{\mathbf{p}} = k), \quad \mathbf{p} \in [0, 1]^n. \quad (1.1)$$

The distribution is often called Poisson–binomial. Throughout this article, q is a *finite real order*, logarithms are natural, and $0^q = 0$ for $q > 0$. Define the power sum and the two entropies by

$$F_q(\mathbf{p}) = \sum_{k=0}^n f_k(\mathbf{p})^q, \quad (1.2)$$

$$H_{R,q}(\mathbf{p}) = \frac{\log F_q(\mathbf{p})}{1 - q}, \quad H_{T,q}(\mathbf{p}) = \frac{1 - F_q(\mathbf{p})}{q - 1}, \quad q \neq 1. \quad (1.3)$$

Both approach the Shannon entropy $H_1(\mathbf{p}) = -\sum_k f_k(\mathbf{p}) \log f_k(\mathbf{p})$ as $q \rightarrow 1$. For $q > 1$, both functions in (1.3) are strictly *decreasing* functions of F_q .

Joint concavity means

$$H((1 - \theta)\mathbf{x} + \theta\mathbf{y}) \geq (1 - \theta)H(\mathbf{x}) + \theta H(\mathbf{y}) \quad (0 \leq \theta \leq 1). \quad (1.4)$$

It tests every affine path in the parameter cube, including paths on which some coordinates increase and others decrease. It is not the same as concavity along the diagonal $p_1 = \dots = p_n$, and it is not the same as concavity in each coordinate separately.

1.2 The target

Hillion and Johnson proved joint concavity for Shannon entropy [1, Theorem 1.2]. The target here is the *proposed numerical threshold assertion adjoining Conjecture 4.2* of that paper’s arXiv version: the proposed upper orders are 2 for Rényi entropy and 3.65986... for Tsallis entropy. The latter is the root greater than one of $2 - 4q + 2^q = 0$. Johnson repeated these proposals in his survey’s open-problems section [2, Section 8, item 5]. The statement is about all Bernoulli parameter interpolations, not just monotone ones.

Theorem 1.1 (Main disproof). *For every finite real $q > 1$, each of the maps*

$$(p_1, p_2) \mapsto H_{R,q}(p_1, p_2), \quad (p_1, p_2) \mapsto H_{T,q}(p_1, p_2)$$

fails quasiconcavity on $(0, 1)^2$, and therefore fails joint concavity there. Counterexamples can be chosen on a fixed-mean affine segment contained in $(0, 1/2)^2$, with rational, indeed dyadic, coordinates.

The distinction between *threshold values* and *existence of a threshold interval* matters. Theorem 1.1 refutes the proposed values, but does not by itself establish concavity for every $0 < q < 1$. If the initial-interval formulation of Conjecture 4.2 is correct, its finite threshold must instead be 1. Equivalently, because the Shannon case is known, the largest finite order for which universal joint concavity can hold is exactly 1; the full set of admissible orders below 1 is not determined here.

[Added 29 September 2026, batch 43: for the ordinary sum studied here this is still undetermined in this report. Part II determines it for the joint vector $(X_1, \dots, X_n)$: Rényi entropy is jointly concave for all $0 < q \leq 2$ in every dimension, and Tsallis entropy for all $0 < q < 1$ exactly when $n \leq 9$ (Section 11.2). An unverified external preprint claims the ordinary-sum threshold one; see there.]

1.3 Source and novelty audit

The original arXiv PDF, including page 7 with the conjecture, was inspected. The survey independently confirms the proposed constants. The later monotonicity paper concerns a different property [3], while the Bernoulli-sum Rényi inequalities of Madiman, Melbourne and Roberto concern variance, convolution and entropy-power comparisons [4]. No earlier resolution of the specific numerical-threshold claim was located in the searches performed on 20 September 2026. This is a bounded search result, not a guarantee of priority. The exact theorem proved below is independent of whether the same counterexample has appeared elsewhere. A detailed source and search audit is included in `notes/sources.md`.

2 A complete rational counterexample at order two

Before proving an all-order theorem, it is useful to give a certificate that requires only arithmetic with three probabilities.

Take the two endpoint parameter vectors and their midpoint

$$\mathbf{x} = \left(\frac{3}{20}, \frac{1}{20}\right), \quad \mathbf{y} = \left(\frac{1}{20}, \frac{3}{20}\right), \quad \mathbf{m} = \frac{\mathbf{x} + \mathbf{y}}{2} = \left(\frac{1}{10}, \frac{1}{10}\right). \quad (2.1)$$

The endpoint distributions agree because exchanging the two independent coins does not change their sum. Direct multiplication gives

$$f(\mathbf{x}) = f(\mathbf{y}) = \frac{1}{400}(323, 74, 3), \quad f(\mathbf{m}) = \frac{1}{100}(81, 18, 1). \quad (2.2)$$

Every probability and every Bernoulli parameter is positive. Both sums have mean $1/5$. Their collision probabilities are

$$F_2(\mathbf{m}) = \frac{81^2 + 18^2 + 1}{10000} = \frac{55088}{80000}, \quad (2.3)$$

$$F_2(\mathbf{x}) = F_2(\mathbf{y}) = \frac{323^2 + 74^2 + 3^2}{400^2} = \frac{54907}{80000}. \quad (2.4)$$

Thus the midpoint has *larger* collision probability, by $181/80000$. Since $H_{T,2} = 1 - F_2$, its Jensen deficit is exactly

$$\frac{H_{T,2}(\mathbf{x}) + H_{T,2}(\mathbf{y})}{2} - H_{T,2}(\mathbf{m}) = \frac{181}{80000} > 0. \quad (2.5)$$

For Rényi entropy, $H_{R,2} = -\log F_2$, so the corresponding deficit is

$$\log \frac{55088}{54907} > 0, \quad \text{approximately } 0.003291061655. \quad (2.6)$$

The approximate decimal is illustrative; strict positivity follows from the integer inequality $55088 > 54907$.

This one example already refutes both proposed thresholds: the order 2 is in each proposed concavity range. It also shows why concavity in the probability vector is irrelevant. Although $\mathbf{m}$ is the midpoint of the *coin parameters*, $f(\mathbf{m})$ is not the midpoint of the endpoint probability vectors; that midpoint is simply $f(\mathbf{x})$.

3 Counterexamples for every finite order above one

3.1 The mean-preserving two-coin path

Fix $0 < p \leq 1/2$ and consider

$$p_1(t) = p + t, \quad p_2(t) = p - t, \quad |t| \leq p. \quad (3.1)$$

The three probabilities are

$$f_0(t) = (1 - p)^2 - t^2, \quad f_1(t) = 2p(1 - p) + 2t^2, \quad f_2(t) = p^2 - t^2. \quad (3.2)$$

Their sum is one and their mean is $2p$ for every t . Let

$$A = (1 - p)^2, \quad B = 2p(1 - p), \quad C = p^2, \quad a = q - 1. \quad (3.3)$$

The path is even in t . Writing $u = t^2$ isolates its essential dependence:

$$G_{q,p}(u) = (A - u)^q + (B + 2u)^q + (C - u)^q, \quad 0 \leq u \leq p^2, \quad (3.4)$$

so that $F_q(p + t, p - t) = G_{q,p}(t^2)$. For $q > 1$,

$$G'_{q,p}(u) = q[-(A - u)^a + 2(B + 2u)^a - (C - u)^a]. \quad (3.5)$$

Lemma 3.1 (A sufficient small-mean condition). *Suppose $q > 1$ and*

$$0 < p < \frac{1}{2(1 + 2^{1/(q-1)})}. \quad (3.6)$$

Then $G_{q,p}$ is strictly decreasing on $[0, p^2]$. Consequently, for every $0 < |t| \leq p$, both entropies in (1.3) are strictly larger at $(p + t, p - t)$ than at (p, p) .

Proof. On the stated interval,

$$A - u \geq 1 - 2p, \quad B + 2u \leq 2p, \quad C - u \geq 0.$$

Because $a > 0$, (3.5) gives

$$G'_{q,p}(u) \leq q[-(1 - 2p)^a + 2(2p)^a] < 0.$$

The strict inequality is exactly (3.6). The power function is continuously differentiable at zero for $q > 1$, so this argument also applies at the endpoint. The entropy conclusion follows because both entropies strictly decrease as functions of F_q . $\square$

3.2 An explicit dyadic construction

Theorem 3.2 (Uniform constructive witness). *For $q > 1$, set*

$$m = 3 + \left\lceil \frac{1}{q - 1} \right\rceil, \quad p = 2^{-m}. \quad (3.7)$$

The points

$$\mathbf{x}_q = (3p/2, p/2), \quad \mathbf{y}_q = (p/2, 3p/2), \quad \mathbf{m}_q = (p, p) \quad (3.8)$$

satisfy, for $H = H_{R,q}$ and for $H = H_{T,q}$,

$$H(\mathbf{m}_q) < H(\mathbf{x}_q) = H(\mathbf{y}_q). \quad (3.9)$$

Proof. Write $a = q - 1 > 0$. Since $m \geq 4$, we have $p \leq 1/16 < 1/4$ and therefore

$$2 \left(\frac{2p}{1-2p} \right)^a \leq 2(4p)^a = 2^{1+a(2-m)} \quad (3.10)$$

$$= 2^{1-a(1+\lceil 1/a \rceil)} \leq 2^{-a} < 1. \quad (3.11)$$

Hence $(1-2p)^a > 2(2p)^a$, the condition of Lemma 3.1. Apply that lemma with $t = p/2$. The equality between the two endpoints follows from symmetry. All their coordinates lie strictly between zero and $1/2$. $\square$

A reparametrization makes the source's unit-time convention explicit:

$$\mathbf{p}(\theta) = (p(1/2 + \theta), p(3/2 - \theta)), \quad 0 \leq \theta \leq 1.$$

This is an affine Bernoulli interpolation whose slopes are p and $-p$. The midpoint $\theta = 1/2$ violates (1.4).

Proof of Theorem 1.1. Quasiconcavity would require $H((\mathbf{x} + \mathbf{y})/2) \geq \min\{H(\mathbf{x}), H(\mathbf{y})\}$. Equation (3.9) contradicts this weaker inequality. $\square$

Corollary 3.3 (Monotone reparametrizations cannot repair the witness). *Any strictly increasing scalar transform of either entropy still fails quasiconcavity on the same segment. More generally, for fixed $q > 1$, every strictly decreasing function of F_q has the same strict midpoint defect.*

Proof. A strictly monotone transformation preserves the strict ordering and the equality of the two endpoint values in (3.9). $\square$

3.3 Why the construction is genuinely all-order

The proof does not sample q , rely on a graph, or assume that q is an integer. Its only order-dependent ingredient is the monotonicity of x^{q-1} for $q > 1$. For rational $a = A_0/B_0 > 0$ in lowest terms, the sufficient condition for the dyadic witness can also be verified by the integer test

$$(2^m - 2)^{A_0} > 2^{A_0+B_0}. \quad (3.12)$$

This follows by raising $(1-2p)^a > 2(2p)^a$ to the positive integer power B_0 and clearing denominators. The archive records such exact certificates at orders as close to one as $1001/1000$. These finite checks are a useful independent check on the formula, but the preceding inequality proof covers all real orders.

4 The exact balanced-point curvature transition

The dyadic construction deliberately uses a conservative small value of p . A sharp local analysis both enlarges the counterexample region and explains the behavior near the Shannon order.

Definition 4.1. For finite $q > 0$ and $0 < p \leq 1/2$, let

$$D_q(p) = (1-p)^{2(q-1)} - 2(2p(1-p))^{q-1} + p^{2(q-1)}. \quad (4.1)$$

At $t = 0$, all three first derivatives in (3.2) vanish. Their second derivatives are $-2, 4, -2$. Thus

$$\left. \frac{d^2}{dt^2} F_q(p+t, p-t) \right|_{t=0} = -2qD_q(p), \quad F'_q(0) = 0. \quad (4.2)$$

Applying the chain rule to (1.3) yields

$$\left. \frac{d^2}{dt^2} H_{T,q}(p+t, p-t) \right|_{t=0} = \frac{2q}{q-1} D_q(p), \quad (4.3)$$

$$\left. \frac{d^2}{dt^2} H_{R,q}(p+t, p-t) \right|_{t=0} = \frac{2q}{(q-1)F_q(p,p)} D_q(p). \quad (4.4)$$

The logarithmic chain-rule square term is zero here. The equality of signs in these two formulas is a special feature of this stationary path, not a general equivalence between Rényi and Tsallis concavity.

Theorem 4.2 (Sharp curvature boundary). *For $q > 1$, write $a = q - 1$ and define*

$$p_c(q) = \frac{1}{1 + \exp(\operatorname{arcosh}(2^a)/a)}. \quad (4.5)$$

Along (3.1), the balanced point $t = 0$ is a strict local entropy minimum when $0 < p < p_c(q)$, and a strict local entropy maximum when $p_c(q) < p \leq 1/2$. At $p = p_c(q)$ the second derivative vanishes, but the balanced point remains a strict local maximum with a negative quartic term. These statements hold for both entropies.

Proof. Let $x = p/(1-p) \in (0, 1]$ and $y = x^a$. Then

$$D_q(p) = (1-p)^{2a}(y^2 - 2^{a+1}y + 1). \quad (4.6)$$

The roots of the quadratic are

$$y_- = 2^a - \sqrt{2^{2a} - 1}, \quad y_+ = 2^a + \sqrt{2^{2a} - 1},$$

with $0 < y_- < 1 < y_+$ and $y_- y_+ = 1$. Hence, in the range $0 < y \leq 1$, the sign is positive exactly when $y < y_-$. Since $y_- = \exp(-\operatorname{arcosh}(2^a))$, this is equivalent to $p < p_c(q)$. Equations (4.3)–(4.4) prove the nonzero curvature statements.

For the zero-curvature case, all A, B, C are strictly positive. Taylor expansion gives

$$G_{q,p}(t^2) = G_{q,p}(0) - qD_q(p)t^2 + \frac{q(q-1)}{2}E_q(p)t^4 + O(t^6), \quad (4.7)$$

where

$$E_q(p) = A^{q-2} + 4B^{q-2} + C^{q-2} > 0.$$

When $D_q(p) = 0$, the entropy differences are respectively

$$-\frac{q}{2}E_q(p)t^4 + O(t^6), \quad -\frac{qE_q(p)}{2F_q(p,p)}t^4 + O(t^6).$$

Both are negative for sufficiently small nonzero t . □

Example 4.3 (Order two). Here $D_2(p) = 1 - 6p + 6p^2$ and

$$p_c(2) = \frac{3 - \sqrt{3}}{6} = 0.2113248654 \dots \quad (4.8)$$

At $p = 1/10$, the Tsallis directional second derivative is $46/25 > 0$. The full Hessian there is

$$\nabla^2 H_{T,2}(1/10, 1/10) = \frac{1}{25} \begin{pmatrix} -73 & -96 \\ -96 & -73 \end{pmatrix}. \quad (4.9)$$

Its symmetric and antisymmetric eigenvalues are $-169/25$ and $23/25$. Thus both coordinate directions and the diagonal direction have negative curvature, while the mean-preserving direction $(1, -1)$ has positive curvature. Testing only the former directions misses the obstruction.

5 All fixed-mean maximizers for two coins

We now classify the maximizers, not just the curvature at the balanced point. The classification applies simultaneously to the two entropies because their maxima correspond to the same minima of F_q .

5.1 Reduction and the two boundaries

Complementing both Bernoulli variables replaces the sum S by $2 - S$ and reverses its probability vector, without changing either entropy. Therefore it suffices to treat total mean $s \in [0, 1]$. Put $p = s/2$. Every pair of parameters with this mean has the form $(p + t, p - t)$ for $|t| \leq p$.

Define, in addition to (4.5),

$$p_b(q) = \frac{1}{2(1 + 2^{1/(q-1)})}. \quad (5.1)$$

The subscript b refers to the boundary of the parameter segment, and the subscript c to its center.

Lemma 5.1 (Strict convexity in squared imbalance). *For $q > 1$ and $0 < p \leq 1/2$, $G_{q,p}$ is strictly convex on $[0, p^2]$. Its endpoint derivatives are*

$$G'_{q,p}(0) = -qD_q(p), \quad (5.2)$$

$$G'_{q,p}(p^2) = q[2(2p)^{q-1} - (1 - 2p)^{q-1}]. \quad (5.3)$$

Moreover $0 < p_b(q) < p_c(q) < 1/2$.

Proof. In the interior,

$$G''_{q,p}(u) = q(q-1)[(A-u)^{q-2} + 4(B+2u)^{q-2} + (C-u)^{q-2}] > 0. \quad (5.4)$$

Continuity extends strict convexity to the closed interval. The endpoint formulas follow from (3.5); its derivative exists at the endpoints because $q > 1$. Equation (5.3) is zero at $p = p_b(q)$ and changes from negative to positive as p increases. At this p , strict increase of G' implies $G'(0) < G'(p^2) = 0$. Consequently $D_q(p_b(q)) > 0$, and Theorem 4.2 gives $p_b(q) < p_c(q)$. Positivity and the remaining bounds follow directly from the formulas. $\square$

Theorem 5.2 (Fixed-mean phase classification). *Fix a finite $q > 1$, and let $0 < s \leq 1$, $p = s/2$. Up to exchanging the two coins, the maximizer of either entropy at mean s is unique, and is as follows.*

- (i) If $0 < p \leq p_b(q)$, the maximizing pair is $(2p, 0)$.
- (ii) If $p_b(q) < p < p_c(q)$, the maximizing pair is $(p + \sqrt{u_*}, p - \sqrt{u_*})$, where $u_* \in (0, p^2)$ is the unique solution of

$$(A - u_*)^{q-1} + (C - u_*)^{q-1} = 2(B + 2u_*)^{q-1}. \quad (5.5)$$

- (iii) If $p_c(q) \leq p \leq 1/2$, the maximizing pair is (p, p) .

For $1 < s < 2$, complement the maximizing pairs at mean $2 - s$. At $s = 0$ and $s = 2$ the unique possible distributions are deterministic.

Proof. Minimize the strictly convex function $G_{q,p}$ on its compact interval. When $p \leq p_b(q)$, its right-end derivative is nonpositive, and its derivative is strictly negative before that endpoint. Its unique minimum is therefore at $u = p^2$. When $p \geq p_c(q)$, its left-end derivative is nonnegative, and its derivative is strictly positive thereafter, so its unique minimum is at $u = 0$. Between the boundaries, $G'(0) < 0 < G'(p^2)$; continuity and strict increase of G' give a unique interior zero, exactly (5.5). The only duplication in the map from u to parameter pairs comes from the two signs of t , which exchange the coins. Complement symmetry proves the remaining mean range. $\square$

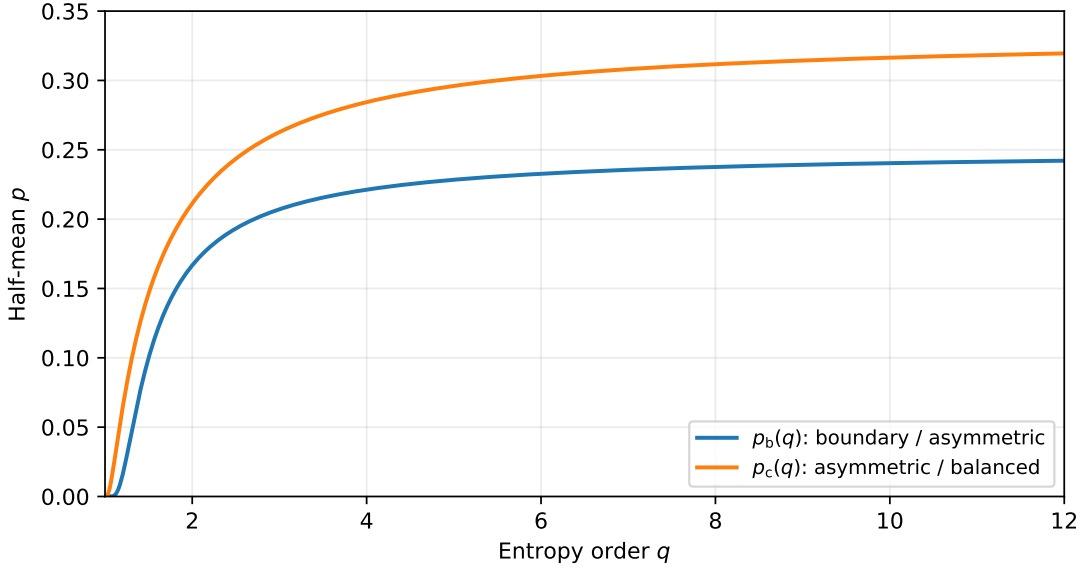


Figure 1: The exact boundaries (5.1) and (4.5), plotted numerically. Below the lower curve the optimizer is a boundary pair; between the curves it is an unequal interior pair; above the upper curve it is the balanced pair. The half-mean is $p = (p_1 + p_2)/2 \leq 1/2$. The plot illustrates Theorem 5.2; it is not used in its proof.

The intermediate regime is important. Failure of concavity does not mean that the best distribution must always have a deterministic summand. There is an entire interval of means for which the unique unordered maximizer consists of two distinct, strictly interior Bernoulli parameters.

5.2 Closed formulas at order two

Let $s = p_1 + p_2$ and $r = p_1 p_2$. The probability vector is $(1 - s + r, s - 2r, r)$, so direct expansion gives

$$F_2(s, r) = 1 - 2s + 2s^2 + (2 - 6s)r + 6r^2 \quad (5.6)$$

$$= 6 \left(r - \frac{3s - 1}{6} \right)^2 + \frac{5 - 6s + 3s^2}{6}. \quad (5.7)$$

For $0 \leq s \leq 1$, the feasible product interval is $0 \leq r \leq s^2/4$. Thus the exact optimizer is

$$r_* = \max \left\{ 0, \min \left\{ \frac{s^2}{4}, \frac{3s - 1}{6} \right\} \right\}. \quad (5.8)$$

This gives the boundary regime $s \leq 1/3$, the unequal interior regime

$$\frac{1}{3} < s < 1 - \frac{1}{\sqrt{3}},$$

and the balanced regime above it. In the middle regime,

$$(p_1, p_2) = \frac{1}{2} \left(s \pm \sqrt{s^2 - 2s + \frac{2}{3}}, \quad s \mp \sqrt{s^2 - 2s + \frac{2}{3}} \right), \quad (5.9)$$

and the minimum collision probability equals $(5 - 6s + 3s^2)/6$. The notation in (5.9) means the two opposite sign choices, which exchange the coordinates.

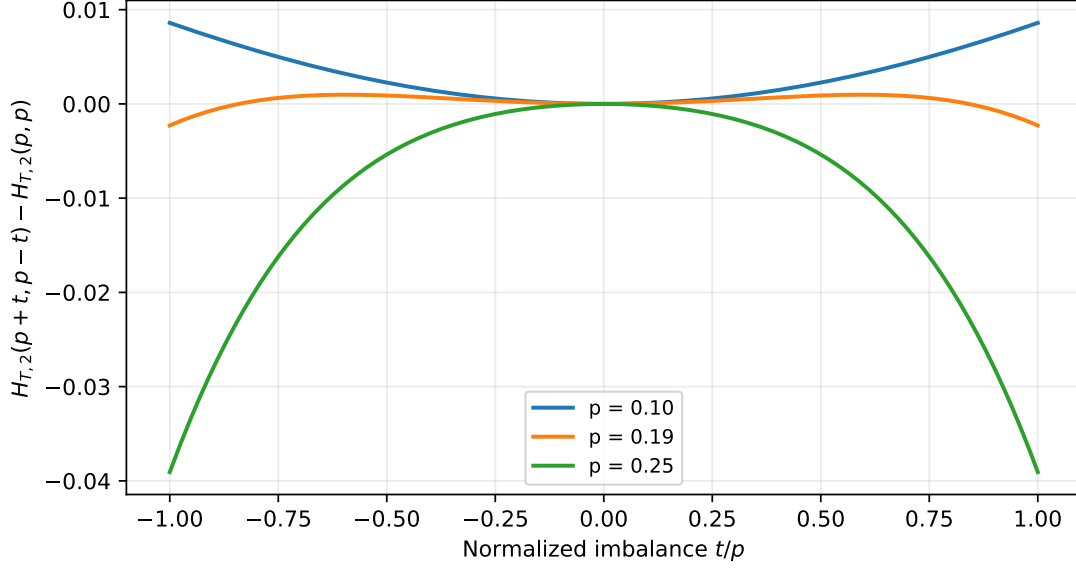


Figure 2: Exact order-two Tsallis entropy differences from the balanced value, evaluated from (5.10). The three half-means lie in the boundary, unequal-interior and balanced regimes, respectively. A positive value away from zero is a strict midpoint obstruction when the two symmetric parameter points are used as endpoints.

On the path (3.1), the complete collision identity is

$$F_2(p+t, p-t) - F_2(p, p) = -2(1 - 6p + 6p^2)t^2 + 6t^4. \quad (5.10)$$

It supplies a second way to check the rational certificate: $p = 1/10$ and $t = 1/20$ give $F_2(p, p) - F_2(p+t, p-t) = 181/80000$.

6 The Shannon transition and subcritical fixed-mean concavity

6.1 Monotonicity and asymptotics of the boundaries

Proposition 6.1. *Both $p_b(q)$ and $p_c(q)$ are strictly increasing for $q > 1$. With $a = q - 1$ and $L = \log 2$, as $a \downarrow 0$,*

$$p_b(1+a) = \frac{1}{2}e^{-L/a}(1 + O(e^{-L/a})), \quad (6.1)$$

$$p_c(1+a) = \exp\left(-\sqrt{\frac{2L}{a}} + O(\sqrt{a})\right). \quad (6.2)$$

As $a \rightarrow \infty$,

$$p_b(1+a) = \frac{1}{4} - \frac{L}{8a} + O(a^{-2}), \quad p_c(1+a) = \frac{1}{3} - \frac{2L}{9a} + O(a^{-2}). \quad (6.3)$$

Proof. Monotonicity of p_b follows directly from (5.1). Set $g(a) = \operatorname{arcosh}(e^{La})$, with $g(0) = 0$. For $a > 0$,

$$g'(a) = \frac{L}{\sqrt{1 - e^{-2La}}}, \quad g''(a) = -\frac{L^2 e^{-2La}}{(1 - e^{-2La})^{3/2}} < 0.$$

Strict concavity and $g(0) = 0$ imply that $g(a)/a$ strictly decreases. Equation (4.5) then makes p_c strictly increasing. For small a , the expansion $\cosh z = 1 + z^2/2 + O(z^4)$ gives $g(a) = \sqrt{2La}(1 + O(a))$.

q	$p_b(q)$	$p_c(q)$
1.01	3.944305×10^{-31}	7.597324×10^{-6}
1.05	4.768367×10^{-7}	0.004986586
1.1	0.000487805	0.022610828
1.25	0.029411765	0.081406962
1.5	0.100000000	0.146446609
2	0.166666667	0.211324865
3	0.207106781	0.262751066
4	0.221246667	0.284370123
10	0.240377712	0.316443755
100	0.249124818	0.331779267

Table 1: Numerical boundary values from the exact formulas; the CSV in the archive retains more digits. For $q = 1.01$ the balanced-point instability requires a half-mean below about 7.6×10^{-6} , whereas the simple boundary-optimizer condition is far more restrictive.

Substitution into (4.5) gives (6.2), while (6.1) is immediate. For large a ,

$$g(a) = (a + 1)L + O(2^{-2a}).$$

Expanding the two explicit boundary formulas in $1/a$ gives (6.3). $\square$

The distinction between the two exponentially small scales is substantial. The explicit dyadic construction guarantees a finite witness without solving an equation, but is not close to the sharp local scale when q is near one. A coarse scan of moderate probabilities can therefore miss the instability near Shannon entropy. This is an explanation of the sampling issue, not a claim about how the original conjecture was tested.

6.2 A nonuniform limit of curvature

For any fixed $p \in (0, 1/2]$, expansion of (4.1) around $a = 0$ gives

$$D_{1+a}(p) = -2a \log 2 + O_p(a^2). \quad (6.4)$$

Indeed the coefficient of a is $2 \log(1 - p) - 2 \log(2p(1 - p)) + 2 \log p = -2 \log 2$. Consequently each of (4.3) and (4.4) approaches $-4 \log 2$ as $q \downarrow 1$.

In contrast, if $q > 1$ is fixed and $p \downarrow 0$, then $D_q(p) \rightarrow 1$ and $F_q(p, p) \rightarrow 1$, so both curvatures approach

$$\frac{2q}{q - 1} > 0. \quad (6.5)$$

Thus convergence to the Shannon curvature is not uniform near the parameter boundary. There is no contradiction between the Shannon theorem and counterexamples at every order immediately above one.

6.3 A positive result below and at one

Although general joint concavity for $0 < q < 1$ is not established here, the fixed-mean two-coin problem admits a complete result.

Theorem 6.2 (Concavity on every two-coin fixed-mean segment). *For each $0 < q \leq 1$ and each mean $s \in (0, 2)$, Rényi and Tsallis entropy are strictly concave along the nondegenerate interior of the two-coin parameter segment $p_1 + p_2 = s$, using their common Shannon value at $q = 1$. They extend as concave functions to the closed segment. Their unique maximizer is the balanced pair $(s/2, s/2)$.*

Proof. By complement symmetry assume $s \leq 1$ and use $p = s/2$, $u = t^2$. First suppose $0 < q < 1$, so $a = q - 1 \in (-1, 0)$. The factorization (4.6) is still valid at $u = 0$. Since $2^a < 1$,

$$y^2 - 2^{a+1}y + 1 = (y - 2^a)^2 + 1 - 2^{2a} > 0.$$

Thus $D_q(p) > 0$ and $G'(0) = -qD_q(p) < 0$. Equation (5.4) now has a negative prefactor $q(q-1)$, so $G''(u) < 0$ throughout the interior. Hence $G'(u) < 0$ there. With $K(t) = G(t^2)$, the chain rule gives

$$K''(t) = 2G'(t^2) + 4t^2G''(t^2) < 0. \quad (6.6)$$

Tsallis entropy is $(K-1)/(1-q)$, so it is strictly concave. Rényi entropy is $(\log K)/(1-q)$, and $(\log K)'' = K''/K - (K'/K)^2 < 0$, giving the same conclusion. Evenness and strict decrease of G in u make $t = 0$ the unique maximum. Continuity of x^q at zero extends concavity to the closed segment.

For $q = 1$, write $J(u) = -\sum_{k=0}^2 f_k(u) \log f_k(u)$, with f_k as in (3.4). Direct differentiation gives

$$J'(u) = \log \frac{f_0(u)f_2(u)}{f_1(u)^2}, \quad (6.7)$$

$$J''(u) = -\frac{1}{f_0(u)} - \frac{4}{f_1(u)} - \frac{1}{f_2(u)} < 0. \quad (6.8)$$

For any two Bernoulli parameters v, w , the identity $f_1 = v(1-w) + w(1-v)$ and the arithmetic–geometric mean inequality imply $f_1^2 \geq 4f_0f_2$. Thus $J'(u) \leq -\log 4 < 0$. Applying the chain rule as in (6.6) proves strict concavity in t , and continuity of $x \log x$ at zero handles the endpoints. At $t = 0$, $f_1^2 = 4f_0f_2$, so the second derivative is exactly $-4 \log 2$. $\square$

This theorem concerns a one-dimensional family of directions in a *two*-dimensional parameter cube. It is not a proof of joint concavity on that cube for $q < 1$, much less for arbitrary numbers of coins.

[Added 29 September 2026, batch 43: for the joint vector of two bits, instead of their sum, Part II proves joint concavity on the whole square for every $0 < q < 1$ (Theorem 12.2, any $n \leq 9$). That is a statement about a different function and implies nothing about the sum; see Section 11.2.]

7 Minimum dimension and separate coordinate concavity

7.1 Tsallis entropy remains concave in each coordinate

Proposition 7.1. *For every n and every finite $q > 0$, Tsallis entropy of a Bernoulli sum is concave in each parameter separately, with all other parameters fixed. Nevertheless it fails joint concavity when $n = 2$ and $q > 1$.*

Proof. Remove coin i , and let g_k be the probability mass function of the remaining sum, extended by zero outside its support. With $z = p_i$,

$$f_k(z) = (1-z)g_k + zg_{k-1}, \quad d_k = g_{k-1} - g_k.$$

For $q \neq 1$ and $0 < z < 1$, direct differentiation gives

$$\frac{d^2}{dz^2} H_{T,q} = -q \sum_k d_k^2 f_k(z)^{q-2} \leq 0. \quad (7.1)$$

Indices at which both g_k and g_{k-1} vanish are omitted; their contribution is identically zero. The formula for $q = 1$ is $-\sum_k d_k^2 / f_k(z) \leq 0$. Continuity extends the conclusion to $z = 0, 1$. The failure of joint concavity is Theorem 1.1. $\square$

In particular, one coin cannot be a Tsallis counterexample at any finite positive order. The minimum number of coins needed for failure is exactly two for every $q > 1$. Formula (4.9) gives a tangible matrix explanation: a symmetric matrix may have negative diagonal entries while possessing a positive eigenvalue.

7.2 The one-coin Rényi calculation

Proposition 7.2 (Minimum number of coins for Rényi failure). *For $1 < q \leq 2$, Rényi entropy of one Bernoulli variable is concave, so the minimum number of coins needed for a counterexample is two. For every finite $q > 2$, one coin already gives failure.*

Proof. Put $a = q - 1$, $b = 1 - z$, and $C(z) = z^q + b^q$. Then

$$H''_{R,q}(z) = -\frac{C''(z)C(z) - C'(z)^2}{aC(z)^2}.$$

For $0 < a \leq 1$, set $x = z/b > 0$ and $w = \log x$. Expanding the numerator gives

$$C''C - (C')^2 = qb^{2a}N_a(x), \quad (7.2)$$

$$N_a(x) = ax^{a-1}(1+x^2) + 2(a+1)x^a - x^{2a} - 1 \quad (7.3)$$

$$= 2x^a[a \cosh w + (a+1) - \cosh(aw)]. \quad (7.4)$$

Convexity of $\cosh$ gives $\cosh(aw) \leq a \cosh w + (1-a) \cosh 0$. Hence $N_a(x) \geq 4ax^a > 0$, proving strict concavity in the interior. Continuity gives concavity on $[0, 1]$.

If $q > 2$, as $z \downarrow 0$ we have $C \rightarrow 1$, $C' \rightarrow -q$, and $C'' \rightarrow q(q-1)$. Therefore

$$H''_{R,q}(z) \rightarrow \frac{q}{q-1} > 0.$$

This supplies a one-coin convex interval. Combining these facts with Theorem 1.1 proves the minimum-dimension statements. $\square$

For example, at $q = 2$ the one-coin identity simplifies to

$$H''_{R,2}(z) = -\frac{8z(1-z)}{(z^2 + (1-z)^2)^2} \leq 0.$$

The elementary one-coin obstruction above order two is consistent with the source's original test family; the two-coin antisymmetric direction is what forces the bound down to one.

[Added 29 September 2026, batch 43: Part II's Theorem 19.2 re-derives the one-coin statement by a second route and extends it additively to the joint vector of n bits: there Rényi entropy is jointly concave in every dimension exactly for $0 < q \leq 2$.]

8 Higher dimensions and structural consequences

8.1 Strictly interior counterexamples with any number of coins

Appending deterministic zero coins immediately transfers a two-coin counterexample to every larger dimension. A stronger statement avoids boundary parameters altogether.

Theorem 8.1 (Instability at small binomial parameter vectors). *Fix $n \geq 2$ and a finite $q > 1$. For all sufficiently small $p > 0$, each entropy has positive second variation at $(p, \dots, p)$ in the direction $(1, -1, 0, \dots, 0)$. Consequently every dimension $n \geq 2$ admits a strict interior midpoint counterexample of this form.*

Proof. Let $W \sim \text{Bin}(n-2, p)$ be the background sum, independent of the first two coins, and write $g_k = \mathbb{P}(W = k)$ with zero extension. Convolution (3.2) with g shows that

$$f_k(t) = f_k(0) - t^2(g_k - 2g_{k-1} + g_{k-2}), \quad (8.1)$$

where $f(0)$ is the $\text{Bin}(n, p)$ mass function. All first derivatives at zero vanish. Thus the Tsallis second variation is

$$\frac{2q}{q-1} \sum_{k=0}^n f_k(0)^{q-1} (g_k - 2g_{k-1} + g_{k-2}), \quad (8.2)$$

and the Rényi second variation is this quantity divided by $F_q(0) > 0$. As $p \downarrow 0$, the $k=0$ summand in (8.2) tends to one. For every $k \geq 1$, $f_k(0)^{q-1} \rightarrow 0$, while the bracket is bounded. The finite sum therefore tends to one. Both second variations are positive for sufficiently small p . Choose $|t|$ sufficiently small compared with p . Smoothness gives a strict local entropy minimum at $t=0$, and exchanging the first two coins makes the two nonzero endpoints have equal entropy. All coordinates are interior. $\square$

The proof is dimensionwise: the allowed size of p may depend on n and q . No uniform-in- n neighborhood is asserted.

8.2 Parameter majorization and the geometry of the counterexample

At a fixed small mean, averaging the two unequal parameter vectors in Theorem 3.2 *decreases* entropy. Thus neither entropy is Schur-concave as a function of the Bernoulli *parameters* for $q > 1$. Nor is it Schur-convex on the whole parameter square: at mean one, Theorem 4.2 shows that the balanced point $(1/2, 1/2)$ is a strict local maximum, reversing the required Schur-convex ordering. These are statements about parameter vectors, not about majorization of probability mass vectors.

There is an elementary distributional interpretation of the direction. Increasing $u = t^2$ transfers mass u from outcome zero and mass u from outcome two to outcome one. The mean stays fixed, while

$$\text{Var}(S) = 2p(1-p) - 2u \quad (8.3)$$

decreases. When p is small, the mass at zero is overwhelmingly the largest. At orders above one, reducing that dominant mass can reduce the power sum more than the loss of a rare outcome increases it. The resulting entropy increase is therefore compatible with a decrease in variance. This interpretation is encoded exactly, rather than assumed heuristically, in (3.5).

8.3 What the result does not contradict

The Shannon joint-concavity theorem is untouched. So are positive results about monotone changes of one parameter, including the separate-coordinate Tsallis inequality (7.1). The later Hillion–Johnson question about increasing parameters towards $1/2$ is a monotonicity question [3, Section 4]; the present path has slopes of opposite signs and is not a counterexample to that question. Likewise, convolution entropy-power inequalities such as those studied in [4] are not Jensen inequalities in the Bernoulli parameters.

9 Verification, computation and reproducibility

9.1 Three independent layers

The mathematical argument is deliberately independent of a search program. The main disproof consists of the three-probability identity (3.2), the derivative bound in Lemma 3.1, and the explicit

arithmetic estimate in Theorem 3.2. The rational order-two certificate can be checked without differentiation or logarithm evaluation.

The archive adds the following checks.

- (1) `code/verify_exact.py` uses only Python’s standard library and exact rational arithmetic. The recorded run passed **12,512 checks**: 24 probability calculations were compared with independent exhaustive outcome enumeration; the collision polynomial was tested on 441 rational parameter pairs; 1,991 integer-order Jensen examples, 16 rational-order sufficient-condition certificates, 11 higher-dimensional witnesses, and 9,999 fixed-mean optimizer comparisons were checked. The remaining checks cover normalization and the main explicit certificate.
- (2) `code/verify_symbolic.py` independently differentiates symbolic expressions using SymPy. It checks the general center-curvature and quartic identities, the collision polynomial, the one-coin order-two formula and the exact Hessian (4.9).
- (3) `code/make_figures.py` evaluates the explicit boundaries at 80-decimal working precision and produces the numerical CSV and vector figures. These numerical illustrations play no role in the proof.

A floating-point Hessian search was used during exploration to find promising directions. Its script and log are retained for provenance, but no theorem relies on an eigenvalue returned by floating-point linear algebra. In particular, the exploratory log is not an exact certificate.

9.2 Rebuilding

From the archive’s top-level directory, run

```
python code/verify_exact.py
python code/verify_symbolic.py
python code/make_figures.py
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

Only the last two optional verification/illustration programs require third-party Python libraries; the exact certificate checker has none. A prebuilt PDF, the numerical table and the two figure PDFs are included, so rebuilding the article itself does not require running Python. The dependency versions used are recorded in `results/environment.txt`.

[Added 29 September 2026, batch 43: this section describes Part I’s files. Part II’s checks and their shipped names are in Section 25. The article build now also reads `data/02-product-entropy-profile_table.tex` and `data/02-product-entropy-profile_plot.csv` and needs the `pgfplots` package; it still needs no Python.]

10 Conclusions and remaining questions

The proposed numerical generalized Shepp–Olkin thresholds are false in the unrestricted affine-interpolation setting. The failure is already visible with two independent coins, at a fixed mean, away from every face of the parameter square. It persists for every finite order above one and defeats even quasiconcavity. A single strict rational inequality at order two is sufficient to disprove both proposed values.

The counterexample is not merely isolated. Its sharp curvature boundary is (4.5); its fixed-mean maximizers undergo the three rigorously classified regimes of Theorem 5.2; and its approach to

the Shannon order has the singular scale (6.2). Below one, Theorem 6.2 gives a positive result on every two-coin fixed-mean segment, but not on every affine direction.

The most immediate remaining mathematical question is whether universal joint concavity holds for all $0 < q < 1$. An affirmative result would complete a corrected finite-order threshold theorem with threshold one. A second, distinct question is what happens after restricting all interpolation slopes to the same sign. The present construction deliberately uses a mixed-sign direction and does not settle that restriction. A third is to classify fixed-mean entropy maximizers for three or more Bernoulli parameters; Theorem 8.1 rules out a universal balanced-maximizer principle above one, but is not such a classification.

[Added 29 September 2026, batch 43: the first question is open in this report for the ordinary sum studied here. For the joint bit vector, equivalently a weighted sum with distinct subset sums, Part II answers it: Rényi entropy yes in every dimension; Tsallis entropy yes exactly when $n \leq 9$, with the sharp fixed-order criterion $n\Psi(q) \leq 1$ and failure at $q = 20/41$ from ten bits on (Section 11.2). A preprint by Wang [5] claims the ordinary-sum threshold is exactly one; it is recorded there as an unverified external claim, and this question is not marked answered on its strength. The second and third questions are not addressed by Part II.]

Scope of the order parameter. All threshold and minimum-dimension assertions in this article concern finite positive orders. In particular, the direct limit of the Tsallis functional as $q \rightarrow \infty$ is the constant zero functional, since $0 \leq H_{T,q} \leq 1/(q-1)$. That limiting functional should not be silently included in a finite-order threshold statement. No classification at order zero is asserted here.

Research status. The article presents self-contained proofs, not just computational suggestions, but it is an AI-assisted research draft. No independent referee report, author confirmation, or proof-assistant verification is claimed. The novelty claim is limited to what the documented literature search established. These limitations concern review and priority; they do not replace or weaken the explicit mathematical statements and proofs.

Part II. Nine bits, ten bits: sharp dimension boundaries for product-Bernoulli entropy

Sections 1–10 and Appendices A–B form the original report (Part I) of 20 September 2026, unchanged except for six dated pointers to this addition (one in the abstract), a two-line notice on the title page, contents entries for the two parts, labels on seven headings, a page-anchor fix around the title page, and two bibliography entries used only by Part II. Part II was added on 29 September 2026 in batch 43 of ProveIt’s incoming-report intake, from one manuscript written against the pinned state of Part I. It takes up Part I’s “most immediate remaining mathematical question”, whether universal joint concavity holds for all $0 < q < 1$, for a different observable: the joint bit vector $(X_1, \dots, X_n)$, equivalently a weighted sum whose subset sums are all distinct, and not Part I’s ordinary sum $X_1 + \dots + X_n$. For that observable it settles the question: Rényi entropy is jointly concave in every dimension, and Tsallis entropy is jointly concave for every $0 < q < 1$ exactly when $n \leq 9$. Part I’s question for the ordinary sum stays open in this report. Section 11 records the provenance, what is and is not settled, and the notation of the addition.

11 Provenance and conventions of Part II

11.1 One manuscript, one addition

Part II is batch-43 manuscript 01, *Nine Bits, Ten Bits: Sharp dimension boundaries for product-Bernoulli entropy*. A scalar curvature criterion, exact counterexamples, and mean-preserving weighted-sum obstructions, an AI-assisted research draft prepared for Vladimir Reshetnikov and dated 29 September 2026 (21 A4 pages as delivered). It arrived as the archive `proveit_product_entropy.zip` (inner directory `proveit_product_entropy/`, main file `article.tex`) in ProveIt commit 06bcc37a8 and was placed in faef2ed2a. It was written against ProveIt commit ac9107d74083fcfe8064b4aaeda7989612e6a5b6, reading Part I’s `article.tex` and `README.md` there (its source notes name the setup, the subunit fixed-mean theorem, the one-coin Rényi threshold, the separate-coordinate concavity and the higher-dimensional counterexample), through [the pinned source](#). Part I’s files are unchanged between that commit and the placement commit, so every statement the manuscript makes about “the repository report” refers to Part I as printed above.

The manuscript’s code and data are shipped beside Part I’s files with the prefix `02-product-entropy-` (Part I’s unprefix files count as 01): its three scripts and its Makefile in `code/`, its six recorded outputs (delivered in `results/`) in `data/`, and its source notes, delivered as `SOURCES_AND_STATUS.md`, as `02-product-entropy-SOURCES_AND_STATUS.md` at the report root. Its manuscript, delivery README, PDF, two figure PDFs and checksum file are not shipped; they survive in the arrival commit. Section 25 gives the shipped names and how to rerun the checks.

In Part II, “this article”, “here” and “we” refer to the manuscript, and “the repository report” to Part I. The manuscript contributes, in the order printed here: its question and scope, the notation and three summary theorems (Section 12); the rank-one Hessian (Section 13); the hyperbolic profile and its unique optimizer (Section 14); the inequality behind nine dimensions (Section 15); the exact orders $1/2$ and $1/3$, the quartic law and the derivative at $1/2$ (Section 16); the rational counterexamples at $q = 20/41$ (Section 17); the weighted-mean obstruction near equal weights (Section 18); strong concavity, the Rényi and above-one orders and quasiconcavity (Section 19); endpoint asymptotics (Section 20); the rational-order decision procedure (Section 21); the block-product criterion and a formalization plan (Section 22); ten research questions (Section 23); its verification, provenance and limitations (Section 24); and its reproduction appendix (Section 25).

Every result, proof, example, remark, limitation and question of the manuscript is printed. Its abstract reads:

Let $X_1, \dots, X_n$ be independent Bernoulli variables, and consider Tsallis entropy of the full bit vector, equivalently of a weighted sum whose subset sums are all distinct. We prove a sharp dimension theorem: joint concavity in the Bernoulli parameters holds for *every* order $0 < q < 1$ precisely when $n \leq 9$. Ten bits are the first obstruction. For each fixed order, concavity is equivalent to $n\Psi(q) \leq 1$, where an explicit scalar profile Ψ is evaluated at the unique solution of $\tanh((1-q)t) \tanh t = 1 - q$.

An elementary hyperbolic inequality gives $\Psi(q) < 1/9$. At $q = 1/2$, $\Psi = 1/10$ exactly, but its derivative is negative; orders just below $1/2$ already violate ten-dimensional concavity. A rational witness at $q = 20/41$ reduces local failure to a positive 96-digit integer. We also certify a finite rational midpoint inequality without floating-point assumptions. A second certificate preserves the mean of a genuinely weighted sum with positive weights arbitrarily close to one.

We determine the exceptional thresholds $N(1/2) = 10$ and $N(1/3) = 11$, prove a fourth-order bending law at the half-order degeneracies, obtain strong-concavity bounds and endpoint asymptotics, and give an exact algebraic decision procedure for rational orders. For comparison, product Rényi entropy is concave in every dimension exactly for $0 < q \leq 2$. Our subunit Tsallis counterexamples remain quasiconcave.

These results concern the collision-free weighted extension of the repository’s Bernoulli-entropy project, not a new solution of the ordinary unweighted Shepp–Olkin problem. All principal statements have self-contained proofs. The accompanying finite certificates are not a Lean formalization, and bibliographic priority and the proofs have not been independently reviewed.

11.2 What Part II settles in Part I, and what it does not

Part I closes (Section 10) with “the most immediate remaining mathematical question”: whether universal joint concavity holds for all $0 < q < 1$. There, as throughout Part I, the entropies are those of the *ordinary sum* $S_p = X_1 + \dots + X_n$, see (1.3), joint concavity is (1.4) on $[0, 1]^n$, and both Rényi and Tsallis entropy are meant. Part II answers the same question for the *joint vector* $(X_1, \dots, X_n)$, equivalently for every weighted sum $\sum_i w_i X_i$ whose 2^n subset sums are distinct:

- (i) **Rényi: yes, in every dimension.** $R_{q,n}$ is jointly concave on $[0, 1]^n$ for every n exactly when $0 < q \leq 2$ (Theorem 19.2). The subunit case is the additive consequence of the one-bit statement.
- (ii) **Tsallis: yes exactly up to nine bits.** $T_{q,n}$ is jointly concave for every $0 < q < 1$ if and only if $n \leq 9$ (Theorem 12.2); at a fixed order, if and only if $n\Psi(q) \leq 1$ (Theorem 12.3). For every $n \geq 10$ it fails at $q = 20/41$ with strictly interior rational parameters and a certified finite Jensen gap (Theorem 17.1, Proposition 17.3), and, with positive weights arbitrarily close to one, even at constant weighted mean (Theorem 18.2). These failures are of concavity only: product Tsallis entropy stays quasiconcave for $0 < q \leq 2$ (Proposition 19.4).
- (iii) **Above one, for comparison.** For the joint vector, Tsallis entropy with $n \geq 2$ bits is concave for $1 \leq q \leq 2$ and fails quasiconcavity for every $q > 2$ (Theorem 19.3), whereas the ordinary sum fails quasiconcavity for every $q > 1$ already with two coins (Theorem 1.1). The two observables behave differently on both sides of order one.

What is *not* settled:

- **Part I’s question for the ordinary sum.** Part II does not answer it, and neither does this report. The ordinary sum is the coarsening of the vector by Hamming weight, and entropy does not pass continuously from one to the other as weights coalesce (Section 18.3). Tempting false readings: “ten bits fail, so ordinary-sum Tsallis entropy fails below order one for $n \geq 10$ ” (not implied; nothing is proved here about failure of the ordinary sum below order one), and “nine bits are concave, so the ordinary sum is concave for $n \leq 9$ ” (not implied

either). Likewise Part I’s Theorem 6.2 (two coins, fixed-mean segments, ordinary sum) and the case $n = 2$ of Theorem 12.2 (two bits, all directions, joint vector) are statements about different functions, and neither implies the other.

- **An external claim, recorded and not verified.** The manuscript records the preprint of H. Wang [5] (arXiv:2609.27433v1, submitted 23 September 2026). By the manuscript’s account (its source notes, shipped as `02-product-entropy-SOURCES_AND_STATUS.md`, which consulted the abstract and HTML), that preprint states that the ordinary-sum threshold is exactly one and establishes subunit power-sum concavity; if correct, it would answer Part I’s question affirmatively for the ordinary sum. The manuscript does not rely on it or certify it. This report has not read or verified the preprint and does not mark Part I’s question answered on its strength: in this report the question remains open, with that claim recorded.
- **Partially colliding weights** (question Q4 of Section 23), and every other question of Part I’s Section 10: same-sign interpolation slopes and fixed-mean maximizers for three or more coins. Part II does not address them.

Part I’s delivered status file `STATUS.md` (20 September 2026) is shipped unchanged; its items “General joint concavity for every $0 < q < 1$ ” and “An all-order interval theorem for arbitrary n ” remain accurate for the ordinary sum.

11.3 Notation of Part II beside Part I

Part II keeps the manuscript’s symbols, with one exception (below). Its entropies use the same unnormalized conventions as Part I’s (1.3): $T_{q,n} = (Z_{q,n} - 1)/(1 - q)$ and $R_{q,n} = \log Z_{q,n}/(1 - q)$ are $H_{T,q}$ and $H_{R,q}$ applied to the joint vector instead of the sum, and for $n = 1$ they coincide with Part I’s. Several letters carry different meanings in the two parts (Table 2); within Part II the Part II meaning always applies.

Table 2: Symbols of Part II that differ from, or could be confused with, Part I’s.

Part II symbol	Meaning in Part II	Not to be read as (Part I)
observable	joint vector $(X_1, \dots, X_n)$, or a collision-free $S_w = \sum_i w_i X_i$	the ordinary sum S_p
$Z_{q,n}$	$\sum_x P_p(x)^q = \prod_i h_q(p_i)$, power sum of the vector	F_q , power sum of the sum, (1.2)
$T_{q,n}, R_{q,n}$ $h_q(p)$	Tsallis, Rényi entropy of the vector $p^q + (1 - p)^q$, one-bit power sum	$H_{T,q}, H_{R,q}$ of the sum equals Part I’s $C(z)$ at $z = p$ (proof of Proposition 7.2)
a	$1 - q$ (Sections 12–16, 19 for $q < 1$, 20); but $q - 1$ in the above-one part of the proof of Theorem 19.2	$a = q - 1$ throughout Part I; the sign flips
$b_q, d_q, \rho_q; b, D$	gradient and curvature of $\log h_q$; D is the diagonal matrix $\text{diag}(d_q(p_i))$ (block-diagonal in Theorem 22.1)	$D_q(p)$, the balanced-point quantity (4.1)
D_{cert}	the 96-digit integer of (17.3) (the manuscript’s D)	—
$\Psi(q), N(q)$	$\max_p \rho_q(p)$; $\lfloor 1/\Psi(q) \rfloor$, the <i>largest concave</i> dimension of $T_{q,n}$	Part I’s minimum number of coins for failure (Section 7); for the vector that is $N(q) + 1$
t, t_q, t_*	half log-odds $\frac{1}{2} \log(p/(1 - p)) $; its optimal value; the root of $t \tanh t = 1$	t , the imbalance on the path (3.1)
$\beta_a(t), B_q, p_q, p_*$	profile quotient, its root, optimizing parameters	p_b, p_c , the phase boundaries

Part II symbol	Meaning in Part II	Not to be read as (Part I)
$F_a(t); F(s); F$	a series in the proof of Lemma 15.1; a one-variable restriction (Section 17); the block product (Theorem 22.1)	F_q , the power sum
L	rational root bounds L, U (Section 17); $\log(4/q)$ (Theorem 20.1)	$\log 2$ (Proposition 6.1)
C	$t_*^2 - 1$ (Theorem 20.2)	p^2 (from Section 3 on); $C(z)$ (Proposition 7.2)
w, w_i	weights (Section 18); in the proof of Theorem 20.1, $w = z^{1-q}$	—
$m, k; P_{m,k}, K_{m,k,n}, y$	$q = m/k$; the polynomials of Theorem 21.1; $y = e^{-2t_q/k}$	m , the dyadic exponent (3.7); $K(t) = G(t^2)$; $y = x^a$ in Theorem 4.2
g	the one-variable function in the proof of Theorem 16.3	$g(a), g_k$
$\sigma, v, c, \delta, \varepsilon, \Delta, \Delta_w$	sign vector, directions, weight perturbation, step, Jensen gaps	—
$\mathbf{1}, \mathbf{p}$	all-ones vector; parameter vector $\mathbf{p}$	—

Renamed symbol. The 96-digit certificate integer is D_{cert} here; the manuscript, its verifier output (`data/02-product-entropy-checks.txt`) and its certificate file call it D . The renaming separates it from the matrix D of the same proof and from Part I’s $D_q(p)$; nothing else is renamed and no normalization changes.

11.4 Where the write had to choose

- **Placement and numbering.** Part II follows Part I’s last section and precedes Part I’s appendices, so no number of Part I moves (the `.aux` numbers of all of Part I’s labels were compared with a build of the committed text). The manuscript’s Section n is Section $n + 11$ here ($n = 1, \dots, 13$), and each numbered statement and equation keeps its position: its Theorem $n.m$ is Theorem $(n + 11).m$ and its equation $(n.m)$ is $(n + 11.m)$. Its Figures 1–2 are Figures 3–4 and its Table 1 is Table 3 (Table 2 is new); its questions keep the names Q1–Q10. Its Appendix A is Section 25.
- **Front matter.** The manuscript’s title page, abstract, contents and repository-baseline line are replaced by this section (the abstract is quoted above, the baseline in Section 11.1).
- **Bibliography.** The manuscript’s entry for Hillion–Johnson is the same paper as Part I’s [1] and is cited as such; its entries for Wang and Tsallis are added as [5] and [6]. Its two citations of Part I (“the repository report”) are now references to Part I’s sections and propositions, and the pinned URL of that entry is in Section 11.1.
- **Figures and files.** The two figures are redrawn with `pgfplots` from the shipped profile data (their captions say so), and the delivered file names in Section 17 are replaced by the shipped ones, with the delivered names beside them.
- **Overlaps with Part I.** The manuscript itself disclaims novelty for the ordinary-sum counterexample above one, its fixed-mean construction and the one-bit Rényi threshold (Section 24). Concretely: the one-bit case of Theorem 19.2 for $q > 1$ is Part I’s Proposition 7.2, proved there through the convexity of $\cosh$ and here through $\sinh(at) \leq a \sinh t$, a second route; the one-bit case of Theorem 19.3 is the remark after Part I’s Proposition 7.1 that one coin is never a Tsallis counterexample; and the separate-coordinate clause of Proposition 19.4 is the product-model counterpart of Proposition 7.1. The rest of Part II concerns the joint vector and has no counterpart in Part I.

- **Dated notes.** Text added at the write is marked “[Write, 29 September 2026, batch 43: ...]” in Part II and “[Added 29 September 2026, batch 43: ...]” in Part I; in the two figure captions and Section 25 the added words are in brackets. Unmarked substitutions in the manuscript’s text are only these: its two citations of Part I became references (“(Part I above)”, “(Part I, Section 10)” in Section 12, and “(Part I, Proposition 7.2, for $1 < q \leq 2$ and $q > 2$)” after Theorem 19.2); D became D_{cert} and “ D in the Hessian lemma” became “the matrix D in the Hessian lemma” in the proof of Theorem 17.1; and the file names in the proof of Proposition 17.3. No mathematical statement of the manuscript or of Part I was changed, and nothing of the manuscript was omitted.
- **Status.** The write read the manuscript’s proofs and spot-checked its principal constants numerically; that reading is not an independent proof review. Part II is AI-assisted and unrefereed, and nothing in it is formalized: the proof-assistant decomposition in Section 22 and question Q10 are proposals, and ProveIt has no Lean or Rocq development of Rényi or Tsallis entropy.

12 Research question, context, and precise scope

12.1 Why this extension is different

The ProveIt report *Two-coin counterexamples to the proposed generalized Shepp–Olkin thresholds* (Part I above) studies the entropy of the ordinary sum $X_1 + \cdots + X_n$. It gives counterexamples above order one and leaves the full subunit joint-concavity question unresolved in that report (Part I, Section 10). The present investigation changes the observable: instead of identifying bit strings with the same number of ones, it retains the full vector, or encodes that vector in a scalar weighted sum. This removes collisions between outcomes. The resulting probability model factorizes completely, but its Tsallis entropy need not be jointly concave. The first failing dimension turns out to be ten.

For historical context, Hillion and Johnson proved the ordinary-sum Shannon concavity theorem and proposed Rényi and Tsallis extensions [1]. A preprint submitted on 23 September 2026 by Wang states the sharp ordinary-sum range $0 < q \leq 1$ [5]. We record that current claim rather than treating the unweighted question as untouched. None of our proofs invokes its argument. Tsallis entropy is used in its standard unnormalized convention [6].

[Write, 29 September 2026, batch 43: the preprint’s claim is recorded here as the manuscript reports it. Neither the manuscript nor this report has verified its proof, so Part I’s ordinary-sum question is not marked answered in this report; see Section 11.2.]

The question resolved here is therefore the following independent extension:

For which dimensions and entropy orders is the entropy of an independent Bernoulli vector jointly concave in its success probabilities? Does that property survive a collision-free weighted encoding, including weights arbitrarily close to equal weights?

The second formulation is exactly equivalent to the first whenever the encoding is injective. It is not equivalent when subset sums collide.

12.2 Notation and definitions

Let $n \geq 1$ and let $X_i \sim \text{Bern}(p_i)$ be independent. Their joint mass function is

$$P_{\mathbf{p}}(x) = \prod_{i=1}^n p_i^{x_i} (1 - p_i)^{1-x_i}, \quad x \in \{0, 1\}^n, \quad \mathbf{p} \in [0, 1]^n. \quad (12.1)$$

For finite $q > 0$, define

$$h_q(p) = p^q + (1 - p)^q, \quad (12.2)$$

$$Z_{q,n}(\mathbf{p}) = \sum_x P_{\mathbf{p}}(x)^q = \prod_{i=1}^n h_q(p_i), \quad (12.3)$$

$$T_{q,n}(\mathbf{p}) = \frac{Z_{q,n}(\mathbf{p}) - 1}{1 - q}, \quad R_{q,n}(\mathbf{p}) = \frac{\log Z_{q,n}(\mathbf{p})}{1 - q} \quad (q \neq 1). \quad (12.4)$$

We set $0^q = 0$. At $q = 1$ both entropies mean the Shannon limit

$$T_{1,n}(\mathbf{p}) = R_{1,n}(\mathbf{p}) = \sum_i [-p_i \log p_i - (1 - p_i) \log(1 - p_i)], \quad (12.5)$$

with $0 \log 0 = 0$. All logarithms are natural.

Joint concavity means concavity on the *parameter cube*:

$$H((1 - \theta)x + \theta y) \geq (1 - \theta)H(x) + \theta H(y), \quad x, y \in [0, 1]^n, \quad 0 \leq \theta \leq 1. \quad (12.6)$$

It is stronger than concavity in each coordinate. Concavity of entropy as a function of an unrestricted mass vector does not settle this question: $\mathbf{p} \mapsto P_{\mathbf{p}}$ is not affine.

Definition 12.1. Weights $w_1, \dots, w_n \in \mathbb{R}$ are *collision-free* if $x \mapsto \sum_i w_i x_i$ is injective on $\{0, 1\}^n$. Equivalently, no nonzero vector in $\{-1, 0, 1\}^n$ has weighted sum zero.

If the weights are collision-free, $S_w = \sum_i w_i X_i$ has exactly the same nonzero probabilities as the vector, merely relabeled. Thus its entropies are (12.4). The weights $w_i = 2^{i-1}$ are an example. All subsequent theorems about collision-free sums concern ordinary *discrete* entropy, not differential entropy.

12.3 Main conclusions at a glance

Theorem 12.2 (Sharp universal dimension). *For $n \geq 1$, the following are equivalent:*

- (i) $T_{q,n}$ is jointly concave on $[0, 1]^n$ for every $0 < q < 1$;
- (ii) $n \leq 9$.

For $n \leq 9$ it is in fact strongly concave for each fixed such q . For $n = 10$ failure occurs at $q = 20/41$, with strictly interior rational parameters and a finite rational midpoint certificate. Every $n > 10$ also fails at that same order.

Theorem 12.3 (Complete fixed-order criterion). *For $0 < q < 1$, let $a = 1 - q$ and let $t_q > 0$ be the unique solution of*

$$\tanh(at_q) \tanh t_q = a. \quad (12.7)$$

Put

$$B_q = \frac{\sinh(at_q)}{\cosh t_q}, \quad \Psi(q) = \frac{qB_q^2}{a + B_q^2}. \quad (12.8)$$

Then $T_{q,n}$ is jointly concave if and only if $n\Psi(q) \leq 1$. Consequently the largest concave dimension is

$$N(q) = \left\lfloor \frac{1}{\Psi(q)} \right\rfloor. \quad (12.9)$$

In particular, $N(1/2) = 10$ and $N(1/3) = 11$.

Theorem 12.4 (Weighted and Rényi consequences). *The preceding conclusions hold unchanged for every collision-free weighted sum. At $q = 20/41$ there are ten positive weights arbitrarily close to one for which Tsallis concavity fails even on an affine hyperplane of constant $\mathbb{E}S_w$.*

For every n , $R_{q,n}$ is jointly concave exactly when $0 < q \leq 2$. For $n \geq 2$, $T_{q,n}$ is concave throughout $1 \leq q \leq 2$ and fails concavity for every $q > 2$. For $n = 1$, Tsallis entropy is concave for all finite $q > 0$.

Theorems 12.2–12.4 are proved below; their statement here introduces no additional hypotheses.

13 A rank-one description of the entire Hessian

13.1 Local curvature in one scalar statistic

Fix $0 < q < 1$. Define, for $0 < p < 1$,

$$b_q(p) = \frac{h'_q(p)}{h_q(p)}, \quad d_q(p) = -(\log h_q(p))'', \quad \rho_q(p) = \frac{b_q(p)^2}{d_q(p)}. \quad (13.1)$$

Since

$$h''_q(p) = -q(1-q)[p^{q-2} + (1-p)^{q-2}] < 0, \quad (13.2)$$

we have $d_q = b_q^2 - h''_q/h_q > 0$ and

$$\rho_q(p) = \frac{h'_q(p)^2}{h'_q(p)^2 - h_q(p)h''_q(p)} \in [0, 1]. \quad (13.3)$$

Let $b = (b_q(p_i))_i$ and $D = \text{diag}(d_q(p_i))$.

Lemma 13.1 (Rank-one Hessian). *Throughout the open cube,*

$$\frac{1-q}{Z_{q,n}} \nabla^2 T_{q,n} = bb^\top - D. \quad (13.4)$$

The Hessian is negative semidefinite exactly when $\sum_i \rho_q(p_i) \leq 1$. If the sum is less than one, it is negative definite. If the sum equals one, it has a one-dimensional kernel spanned by $D^{-1}b$. If the sum exceeds one, it has exactly one positive and $n-1$ negative eigenvalues.

Proof. Differentiate $\log Z_{q,n} = \sum_i \log h_q(p_i)$. Its gradient is b and its Hessian is $-D$, so differentiating the exponential gives $\nabla^2 Z_{q,n} = Z_{q,n}(bb^\top - D)$. Congruence by $D^{-1/2}$ transforms the last matrix to $uu^\top - I$, where $u = D^{-1/2}b$ and $\|u\|^2 = \sum_i \rho_q(p_i)$. Its eigenvalues are -1 on the orthogonal complement of u and $\|u\|^2 - 1$ in direction u . Congruence preserves inertia. At equality $u \neq 0$, and the stated kernel follows by direct multiplication. $\square$

This lemma identifies the obstruction, not just its sign: there can be only one unstable direction. A useful explicit positive-curvature direction when $\sum_i \rho_q(p_i) > 1$ is $v = D^{-1}b$, since

$$v^\top (bb^\top - D)v = \left(\sum_i \rho_q(p_i) \right) \left(\sum_i \rho_q(p_i) - 1 \right) > 0. \quad (13.5)$$

13.2 From local curvature to global concavity

Proposition 13.2. *Suppose $\Psi(q) = \max_{0 < p < 1} \rho_q(p)$ exists. Then $T_{q,n}$ is concave on $[0, 1]^n$ if and only if $n\Psi(q) \leq 1$.*

Proof. If the inequality holds, Lemma 13.1 makes the Hessian negative semidefinite everywhere in the open cube. Restricting to any line segment there and integrating its nonpositive second derivative proves concavity. Continuity of (12.4) extends it to the closed cube by approaching the endpoints from the interior.

Conversely, let p_* attain the maximum. If $n\Psi(q) > 1$, the Hessian at $(p_*, \dots, p_*)$ has a positive direction. For a sufficiently small nonzero displacement in that direction, Taylor expansion gives

$$\frac{T(\mathbf{p} + \varepsilon v) + T(\mathbf{p} - \varepsilon v)}{2} - T(\mathbf{p}) > 0.$$

This contradicts concavity. Existence and uniqueness up to reflection of the maximizing parameter are proved next. $\square$

14 The hyperbolic profile and its unique optimizer

14.1 Exact reduction

Lemma 14.1. *Set $a = 1 - q$ and $t = \frac{1}{2}|\log(p/(1-p))|$. With*

$$\beta_a(t) = \frac{\sinh^2(at)}{\cosh^2 t}, \quad (14.1)$$

we have

$$\rho_q(p) = \frac{q\beta_a(t)}{a + \beta_a(t)}. \quad (14.2)$$

Proof. By reflection symmetry, write $p = e^{-t}/(2 \cosh t)$ and $1 - p = e^t/(2 \cosh t)$. The three relevant expressions are

$$\begin{aligned} h_q(p) &= \frac{2 \cosh(qt)}{(2 \cosh t)^q}, \\ h'_q(p) &= 2q(2 \cosh t)^a \sinh(at), \\ -h''_q(p) &= 2qa(2 \cosh t)^{1+a} \cosh((1+a)t). \end{aligned}$$

Consequently

$$\frac{h'_q(p)^2}{-h_q(p)h''_q(p)} = \frac{q}{a} \frac{\sinh^2(at)}{\cosh((1-a)t) \cosh((1+a)t)}.$$

Use $\cosh((1-a)t) \cosh((1+a)t) = \cosh^2 t + \sinh^2(at)$ and then $\rho = r/(1+r)$ for the displayed ratio r . Simplification gives (14.2). $\square$

Proposition 14.2 (Unique critical point). *For each $0 < q < 1$, $t \mapsto \sinh(at)/\cosh t$ has a unique maximum on $[0, \infty)$, attained at t_q in (12.7). Thus ρ_q has exactly two maximizers,*

$$p_q = \frac{1 - \tanh t_q}{2}, \quad 1 - p_q. \quad (14.3)$$

The function Ψ is positive and real analytic on $(0, 1)$.

Proof. The ratio vanishes at zero and tends to zero at infinity because $a < 1$. For $t > 0$ its logarithmic derivative is $a \coth(at) - \tanh t$, whose sign is the sign of $a - \tanh(at) \tanh t$. The product of the two hyperbolic tangents is strictly increasing from zero to one. It crosses a exactly once, proving all the maximizing statements. Analytic dependence follows from the implicit function theorem: the derivative with respect to t of $\tanh(at) \tanh t - a$ is strictly positive at its root. $\square$

Combining this proposition and Proposition 13.2 proves the fixed-order criterion in Theorem 12.3.

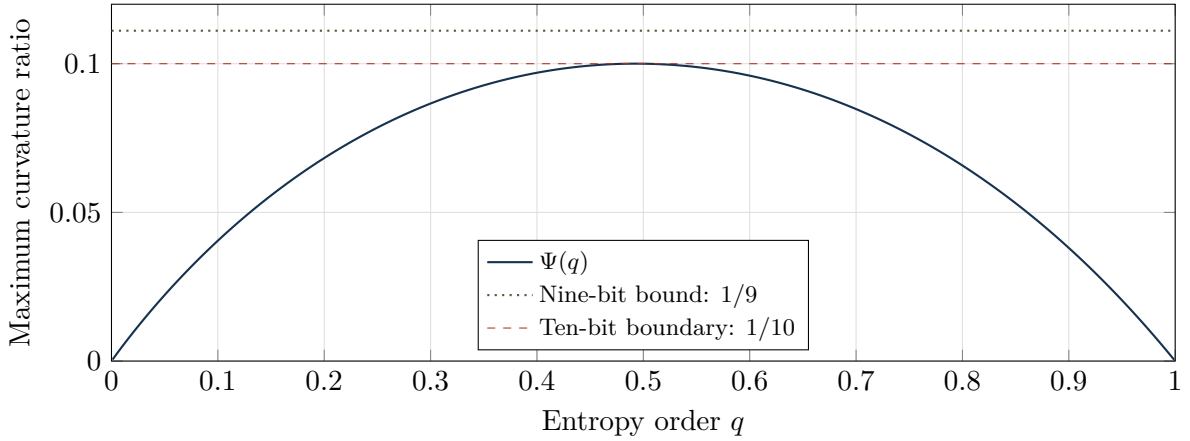


Figure 3: The scalar profile, numerically plotted. The analytic bound $\Psi(q) < 1/9$ proves the nine-bit result. The very small excursion above $1/10$ is resolved in Figure 4. The plot is illustrative, not a numerical proof. [Write: redrawn with `pgfplots` from the shipped `data/02-product-entropy-profile_plot.csv`, the manuscript's own data for this figure; the delivered Matplotlib PDF is not shipped.]

15 The elementary inequality behind nine dimensions

The key estimate is sufficiently explicit that the universal dimension bound does not require numerical maximization.

Lemma 15.1 (Uniform hyperbolic inequality). *For $0 < a < 1$ and every real t ,*

$$(2 - a^2) \sinh^2(at) < a^2 \cosh^2 t. \quad (15.1)$$

Proof. Twice the difference between the right and left sides is

$$F_a(t) = a^2(\cosh 2t + 1) - (2 - a^2)(\cosh 2at - 1).$$

Its power series is absolutely convergent. The constant term is $2a^2$. The terms of degrees two and four are respectively $-2a^2(1 - a^2)t^2$ and $\frac{2}{3}a^2(1 - a^2)^2t^4$. For every $k \geq 2$, the coefficient of t^{2k} is

$$\frac{2^{2k}a^2}{(2k)!} [1 - (2 - a^2)a^{2k-2}] \geq 0,$$

because $(2 - a^2)a^{2k-2} \leq (2 - a^2)a^2 \leq 1$. Discarding terms of degree at least six and putting $u = (1 - a^2)t^2$ gives

$$F_a(t) \geq 2a^2 \left(1 - u + \frac{u^2}{3} \right) = 2a^2 \left(\frac{(u - 3/2)^2}{3} + \frac{1}{4} \right) \geq \frac{a^2}{2} > 0.$$

This proves the strict inequality. □

Theorem 15.2 (A uniform scalar bound). *For $0 < q < 1$,*

$$0 < \Psi(q) < \frac{q(1-q)}{2 + q(1-q)} \leq \frac{1}{9}. \quad (15.2)$$

Thus $T_{q,n}$ is concave whenever

$$(n-1)q(1-q) \leq 2. \quad (15.3)$$

In particular it is concave for every subunit order when $n \leq 9$.

Proof. Lemma 15.1 gives $\beta_a(t) < a^2/(2 - a^2)$. The right side of (14.2) is strictly increasing in β_a , so

$$\rho_q(p) < \frac{a(1-a)}{2+a-a^2} = \frac{q(1-q)}{2+q(1-q)}.$$

The maximum is attained, hence the strict inequality survives maximization. Now $q(1-q) \leq 1/4$ gives the bound $1/9$. Condition (15.3) implies that n times the middle fraction is at most one; strictness then yields $n\Psi(q) < 1$. $\square$

This proves the positive half of Theorem 12.2. Notice that the bound is designed to separate the integers nine and ten, not to equal the true maximum of Ψ . No assertion that $1/9$ is the sharp continuous upper bound is being made.

16 Two exact orders and an exceptional ten-bit boundary

16.1 The half-order profile

Theorem 16.1. *At $q = 1/2$, $\Psi(q) = 1/10$. The maximizing parameters are*

$$p_* = \frac{2 - \sqrt{3}}{4}, \quad 1 - p_*. \quad (16.1)$$

Consequently $T_{1/2,n}$ is concave if and only if $n \leq 10$.

Proof. Set $a = 1/2$ and $y = \cosh t \geq 1$. Then

$$\beta_{1/2}(t) = \frac{y-1}{2y^2} \leq \frac{1}{8},$$

with equality exactly at $y = 2$, as follows either by differentiation or from $y^2 - 4y + 4 \geq 0$. Substitution in (14.2) gives $\Psi = (\frac{1}{2} \cdot \frac{1}{8}) / (\frac{1}{2} + \frac{1}{8}) = 1/10$. Since $\tanh(\operatorname{arcosh} 2) = \sqrt{3}/2$, the maximizing parameters follow. $\square$

Theorem 16.2 (The third-order counterpart). *At $q = 1/3$, $\Psi(q) = 1/11$, and the maximizing parameters are*

$$p_{1/3} = \frac{5 - 2\sqrt{5}}{10}, \quad 1 - p_{1/3}. \quad (16.2)$$

Thus $T_{1/3,n}$ is concave exactly when $n \leq 11$.

Proof. Write $t = 3u$ and $s = \sinh u \geq 0$. For $a = 2/3$,

$$\frac{\sinh(at)}{\cosh t} = \frac{\sinh 2u}{\cosh 3u} = \frac{2s}{1 + 4s^2} \leq \frac{1}{2},$$

with equality exactly when $s = 1/2$. Therefore $\beta = 1/4$ and $\Psi = (\frac{1}{3} \cdot \frac{1}{4}) / (\frac{2}{3} + \frac{1}{4}) = 1/11$. At equality $u = \log((1 + \sqrt{5})/2)$, so $e^{2t} = 9 + 4\sqrt{5}$ and $p = 1/(1 + e^{2t}) = (5 - 2\sqrt{5})/10$. $\square$

16.2 Degenerate Hessians still allow strict concavity

Theorem 16.3 (Half-order degeneracy and fourth-order bending). *The function $T_{1/2,10}$ is strictly concave on its closed cube, but not strongly concave. Its interior Hessian is negative definite except at the 2^{10} points whose coordinates all belong to $\{p_*, 1 - p_*\}$. At such a point, let $\sigma_i = 1$ when $p_i = p_*$ and $\sigma_i = -1$ otherwise. Then*

$$\nabla^2 T_{1/2,10}(\mathbf{p}) = \frac{81}{4} (\sigma\sigma^\top - 10I). \quad (16.3)$$

Its null direction is σ . Along that direction,

$$T_{1/2,10}(\mathbf{p} + s\sigma) = \frac{211}{16} + \frac{405\sqrt{3}}{4}s - 6480s^4 + O(s^5). \quad (16.4)$$

Proof. Equality in $\sum_i \rho \leq 10\Psi = 1$ requires equality in each of the ten scalar inequalities. Theorem 16.1 identifies the points. At them,

$$h^2 = \frac{3}{2}, \quad Z = \frac{243}{32}, \quad b_i = \frac{2}{\sqrt{3}}\sigma_i, \quad d_i = \frac{40}{3}.$$

Substitution in (13.4) gives (16.3).

Every nontrivial line segment in the open cube has strictly negative second derivative except possibly at finitely many points. Therefore its restriction is strictly concave. A line segment on the boundary lies in a minimal face of dimension $m < 10$; its deterministic coordinates contribute factors $h(0) = h(1) = 1$, reducing the restriction to $T_{1/2,m}$, whose Hessian is negative definite in that face's relative interior. This proves strict concavity on the closed cube. The zero Hessian directions rule out any positive strong-concavity constant.

By reflection of the high coordinates, the left side of (16.4) equals $g(p_* + s)$, where

$$g(p) = 2 \left[(1 + 2\sqrt{p(1-p)})^5 - 1 \right].$$

Direct differentiation at p_* gives $g = 211/16$, $g' = 405\sqrt{3}/4$, $g'' = g''' = 0$, and $g^{(4)} = -155520$. Taylor's theorem proves the expansion. $\square$

The same finite-degeneracy argument proves strict, but not strong, concavity of $T_{1/3,11}$. We do not need its fourth derivative.

16.3 Why ten bits fail just below one half

Theorem 16.4 (A nonzero profile derivative). *The profile satisfies*

$$\Psi'(1/2) = \frac{9 - 4\sqrt{3}\log(2 + \sqrt{3})}{25} < 0. \quad (16.5)$$

Consequently $\Psi(q) > 1/10$ throughout a nonempty interval immediately below $1/2$.

Proof. At the maximizing t_q , the t derivative of the profile vanishes, so its q derivative is obtained by differentiating $q\beta/(1 - q + \beta)$ while holding t fixed. At $q = 1/2$, $t_0 = \operatorname{arcosh} 2$, $\beta = 1/8$, and

$$\left. \frac{\partial \beta}{\partial q} \right|_{t=t_0} = -2t_0 \coth(t_0/2)\beta = -\frac{\sqrt{3}t_0}{4}.$$

The quotient rule gives (16.5). To verify the strict sign without decimals, use $t_0 = 2 \operatorname{artanh}(1/\sqrt{3})$ and the positive power series:

$$4\sqrt{3}t_0 = 8 \sum_{k=0}^{\infty} \frac{1}{(2k+1)3^k} > 8 \left(1 + \frac{1}{9} + \frac{1}{45} \right) = \frac{136}{15} > 9.$$

Analyticity and the nonzero derivative give the final assertion. $\square$

This proves that the integer nine in Theorem 12.2 is optimal. The next section supplies an explicit order and finite certificate, rather than leaving the counterexample existential.

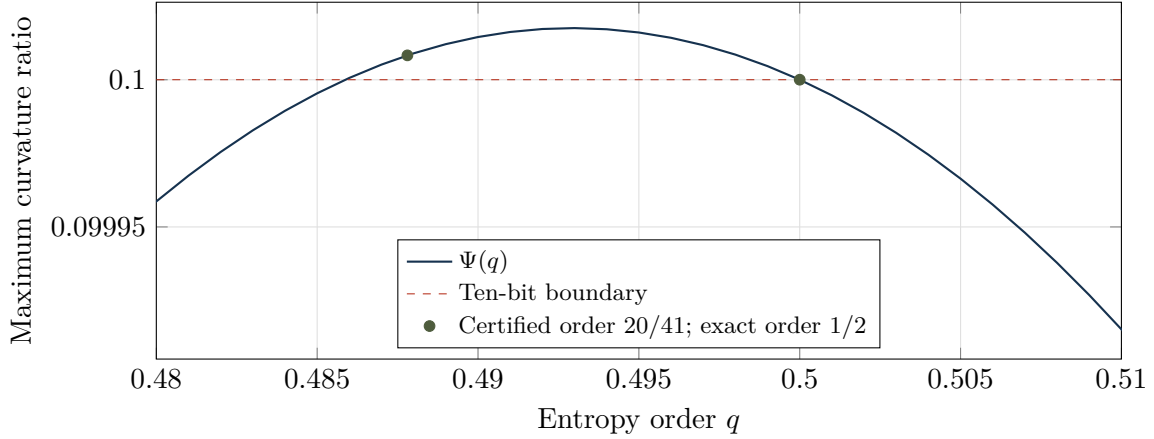


Figure 4: The narrow failure region near $q = 1/2$. The value at $1/2$ and the sign of the derivative there are proved exactly. The rational order $20/41$ is independently certified. The shape elsewhere in this plot is numerical and does not prove global unimodality. [Write: redrawn with `pgfplots` from the shipped `data/02-product-entropy-profile_plot.csv`, whose step in q is 0.001 ; the delivered plot evaluated Ψ at step 10^{-5} on this window and is not shipped. The marked value at $20/41$ is $\Psi(20/41) \approx 0.100008270243$ from Table 3.]

17 Exact rational counterexamples

17.1 A 96-digit integer certifies local failure

Theorem 17.1 (Rational curvature certificate). *Set*

$$q_0 = \frac{20}{41}, \quad p_0 = \frac{15^{41}}{15^{41} + 16^{41}}. \quad (17.1)$$

At $\mathbf{p}_0 = (p_0, \dots, p_0) \in (0, 1)^{10}$, the second derivative of $T_{q_0, 10}(\mathbf{p}_0 + s\mathbf{1})$ at $s = 0$ is strictly positive. The same construction works with all $n \geq 10$ coordinates equal to p_0 .

Proof. Put $y = 15/16$. Since $p_0/(1 - p_0) = y^{41}$, (14.2) becomes

$$\beta = \frac{y^{20}(1 - y^{21})^2}{(1 + y^{41})^2}, \quad \rho = \frac{20\beta}{21 + 41\beta}. \quad (17.2)$$

Thus $10\rho > 1$ is equivalent to $53\beta > 7$. Clearing denominators reduces this to $D_{\text{cert}} > 0$, where

$$D_{\text{cert}} = 53 \cdot 15^{20} 16^{20} (16^{21} - 15^{21})^2 - 7(16^{41} + 15^{41})^2. \quad (17.3)$$

Its exact decimal expansion, written in consecutive blocks, is

$$\begin{aligned} &45454170259471782223845613664512 \\ &86206963512698735119030706120924 \\ &25935633848945080535442406921433. \end{aligned} \quad (17.4)$$

Equivalently, one may verify positivity directly from (17.3); the code checks that expression and its full 96-digit decimal string. At an equal-parameter point $\mathbf{b}$ is a constant multiple of $\mathbf{1}$ and the matrix D in the Hessian lemma is scalar diagonal, so $\mathbf{1}$ is the positive direction exactly when $n\rho > 1$. The inequality for ten implies it for every larger n . $\square$

Remark 17.2. The approximate quantities

$$p_0 \approx 0.0662310671864749, \quad \rho_{q_0}(p_0) \approx 0.100008231403062$$

help locate the witness but play no role in its proof. In particular, $\rho_{q_0}(p_0)$ is a value at our rational parameter, not the exact maximum $\Psi(q_0)$, which is slightly larger.

17.2 A finite midpoint inequality, not only a Hessian test

Define the one-variable restriction

$$F(s) = \frac{[(p_0 + s)^{20/41} + (1 - p_0 - s)^{20/41}]^{10} - 1}{21/41}. \quad (17.5)$$

Let $\varepsilon = 1/1000$. Both vectors $\mathbf{p}_0 \pm \varepsilon \mathbf{1}$ have strictly interior rational coordinates.

Proposition 17.3 (Certified finite Jensen deficit). *The midpoint deficit*

$$\Delta = \frac{F(\varepsilon) + F(-\varepsilon)}{2} - F(0) \quad (17.6)$$

satisfies the exact rational bounds

$$\frac{8379718}{10^{14}} < \Delta < \frac{8379719}{10^{14}}. \quad (17.7)$$

In particular these two endpoints explicitly violate joint concavity.

Exact certificate method. For every rational $r \in [0, 1]$, the program constructs dyadic rational bounds L, U of denominator 2^{96} such that

$$L^{41} \leq r^{20} \leq U^{41}, \quad U - L = 2^{-96}. \quad (17.8)$$

Integer binary search establishes these inequalities without numerical root extraction. Since $x \mapsto x^{41}$ is increasing on the nonnegative reals, they prove $L \leq r^{20/41} \leq U$.

Apply (17.8) to $r = p_0 + s$ and $1 - p_0 - s$ for each of $s = -\varepsilon, 0, \varepsilon$. Add the bounds, raise them to the tenth power, and form the outward-rounded difference

$$\frac{41}{21} \left(\frac{Z_+ + Z_-}{2} - Z_0 \right).$$

Here a lower bound uses lower endpoint values and an upper midpoint value; an upper bound uses the opposite choices. The resulting rational bounds, recorded in `data/02-product-entropy-certificates.json` (delivered as `results/certificates.json`), are strictly inside the rational interval (17.7). Every comparison is an integer comparison after clearing positive denominators. The complete dependency-free verifier is included as `code/02-product-entropy-verify_exact.py` (delivered as `code/verify_exact.py`). $\square$

This proposition is a finite computer-assisted certificate. The all-dimension theorem and the exact local witness have separate analytic proofs; neither depends on trusting a floating-point calculation.

17.3 A constant Hamming-mean version

Replace the midpoint vector by

$$\bar{\mathbf{p}} = (\underbrace{p_0, \dots, p_0}_5, \underbrace{1 - p_0, \dots, 1 - p_0}_5), \quad \sigma = (1, 1, 1, 1, 1, -1, -1, -1, -1, -1). \quad (17.9)$$

The path $\bar{\mathbf{p}} + s\sigma$ has $\sum_i p_i = 5$ for all s . Reflection invariance $h_q(1 - p) = h_q(p)$ makes its entropy exactly $F(s)$, so Proposition 17.3 supplies a constant-mean *Hamming-count* counterexample. This is not yet a claim about constant mean for a nonconstant weighted sum; that stronger statement requires the adjustment in the next section.

18 Weighted means and arbitrarily small perturbations of equal weights

18.1 Injective weights close to one

Lemma 18.1. *For $n \geq 1$ and $0 < \delta < (2^n - 1)^{-1}$, the weights*

$$w_i = 1 + \delta 2^{i-1}, \quad 1 \leq i \leq n, \quad (18.1)$$

are positive and collision-free.

Proof. If two bit strings have different Hamming weights, the difference of their weighted sums has a nonzero integer part and a perturbation of absolute value at most $\delta(2^n - 1) < 1$, so it cannot be zero. If they have the same Hamming weight, equality would require equality of their binary encodings $\sum_i 2^{i-1} x_i$, which forces the strings to agree. $\square$

18.2 An exactly mean-preserving direction

Theorem 18.2 (Weighted-mean obstruction). *Let $q_0, p_0, \bar{p}, \sigma$ be as in (17.1) and (17.9). For every sufficiently small $\delta > 0$, there is a rational direction when δ is rational along which the mean of the weighted sum with weights (18.1) is constant but its Tsallis entropy has positive second derivative at $\bar{p}$. Thus arbitrarily small positive perturbations of equal weights admit mean-preserving failures of subunit-order concavity.*

Proof. Set $n = 10$. Since $\sum_{i=1}^5 2^{i-1} = 31$ and $\sum_{i=6}^{10} 2^{i-1} = 992$, define

$$c = \frac{w \cdot \sigma}{w \cdot \mathbf{1}} = \frac{-961\delta}{10 + 1023\delta}, \quad v = \sigma - c\mathbf{1}. \quad (18.2)$$

Then $w \cdot v = 0$, so $\mathbb{E}S_w$ is constant along $\bar{p} + sv$. At $\bar{p}$ we have $b = b_0\sigma$ and $D = d_0I$, with $b_0^2/d_0 = \rho_{q_0}(p_0) = \rho$. Also $\sigma \cdot \mathbf{1} = 0$, whence

$$b \cdot v = 10b_0, \quad \|v\|^2 = 10(1 + c^2).$$

The Hessian quadratic form has the sign of

$$10\rho - 1 - c^2. \quad (18.3)$$

By Theorem 17.1, $10\rho - 1 > 0$, whereas $c \rightarrow 0$ as $\delta \rightarrow 0$. Hence (18.3) is positive for all sufficiently small δ . For such δ , Lemma 18.1 identifies the weighted entropy with the product entropy. Interior parameters permit a nondegenerate interval of the path. $\square$

Proposition 18.3 (Explicit weighted-mean certificate). *Take $\delta = 10^{-6}$, so $c = -961/10001023$, and use $v = \sigma - c\mathbf{1}$ and $\varepsilon = 1/1000$. The endpoints $\bar{p} \pm \varepsilon v$ are strictly interior rational vectors, have identical weighted mean, and have Tsallis midpoint deficit Δ_w satisfying*

$$\frac{8378699}{10^{14}} < \Delta_w < \frac{8378700}{10^{14}}. \quad (18.4)$$

Exact certificate. The verifier checks $w \cdot v = 0$ exactly and verifies $10\rho > 1 + c^2$ with rational arithmetic. It also checks the 1024 subset sums are distinct. For the finite inequality, reflection of the last five coordinates gives the power sum

$$Z(s) = [h_{q_0}(p_0 + (1 - c)s)h_{q_0}(p_0 + (1 + c)s)]^5.$$

The outward dyadic root bounds (17.8), followed by rational multiplication and subtraction, yield the asserted interval. The exact enclosing rationals are stored with the first certificate. $\square$

18.3 What happens as the weights coalesce

For every allowed $\delta > 0$, the weights encode all 2^{10} bit strings separately, so their entropy equals the same product entropy, independently of the magnitude of δ . At $\delta = 0$, many probabilities are added together: the observable becomes the ordinary Hamming sum. This is a discontinuity of discrete entropy as outcomes merge, not a discontinuity of the finite probability vector before merging. Consequently there is no valid continuity argument in the weights that transfers unweighted concavity to all nearby weighted models. Theorem 18.2 exhibits the obstruction even after imposing the weighted-mean constraint itself.

No conclusion is drawn here about the minimum failing dimension among *all* possible weight systems with collisions. The sharp number ten applies to the collision-free class and to independent bit vectors.

19 Strong concavity, quasiconcavity, and all orders above one

19.1 A quantitative Jensen inequality

Theorem 19.1 (Strong concavity). *For $0 < q < 1$, if $n\Psi(q) < 1$, then*

$$-\nabla^2 T_{q,n}(\mathbf{p}) \succeq 4q(1 - n\Psi(q))I \quad (\mathbf{p} \in (0, 1)^n). \quad (19.1)$$

Consequently for all $x, y \in [0, 1]^n$ and $0 \leq \theta \leq 1$,

$$\begin{aligned} T_{q,n}((1 - \theta)x + \theta y) &\geq (1 - \theta)T_{q,n}(x) + \theta T_{q,n}(y) \\ &\quad + 2q(1 - n\Psi(q))\theta(1 - \theta)\|x - y\|^2. \end{aligned} \quad (19.2)$$

Proof. Weighted Cauchy–Schwarz gives $(b \cdot v)^2 \leq (\sum_i \rho_i) v^\top D v \leq n\Psi(q) v^\top D v$. Therefore

$$-\nabla^2 T \succeq \frac{Z}{1 - q} (1 - n\Psi(q))D.$$

For $0 < q < 1$, $Z \geq 1$, $h_q(p) \leq 2^{1-q}$, and convexity of x^{q-2} gives $p^{q-2} + (1 - p)^{q-2} \geq 2^{3-q}$. Hence

$$d_q(p) \geq -\frac{h_q''(p)}{h_q(p)} \geq 4q(1 - q).$$

This proves (19.1). Apply ordinary concavity to $T_{q,n}(x) + 2q(1 - n\Psi(q))\|x\|^2$ and rearrange. Interior approximation extends the result to boundary endpoints. $\square$

Theorem 15.2 makes this apply to every $n \leq 9$ and every fixed subunit order. The constant is explicit but is not claimed optimal.

19.2 Rényi additivity and its exact concavity range

Theorem 19.2. *For each $n \geq 1$, $R_{q,n}$ is jointly concave on $[0, 1]^n$ exactly for $0 < q \leq 2$. It is strictly concave in the interior throughout that range.*

Proof. By (12.3),

$$R_{q,n}(\mathbf{p}) = \sum_i \frac{\log h_q(p_i)}{1 - q}.$$

When $0 < q < 1$, $h_q'' < 0$ gives $(\log h_q)'' < 0$. The Shannon case is immediate from the binary entropy second derivative $-1/[p(1 - p)]$.

For $1 < q \leq 2$, put $a = q - 1 \in (0, 1]$ and use the same half-log-odds variable t . Direct substitution in $hh'' - h'^2$ shows its sign is the sign of

$$a \cosh^2 t - \sinh^2(at). \quad (19.3)$$

For completeness the positive common factor comes from

$$\frac{hh'' - h'^2}{4q(2 \cosh t)^{-2a}} = a \cosh((1+a)t) \cosh((1-a)t) - (1+a) \sinh^2(at) = a \cosh^2 t - \sinh^2(at).$$

Convexity of $\sinh$ on $[0, \infty)$ gives $\sinh(at) \leq a \sinh t$, so (19.3) is positive. Thus $\log h_q$ is strictly convex and division by $1 - q < 0$ gives strict concavity. Reflection treats negative log odds.

If $q > 2$, as $p \downarrow 0$ we have $h \rightarrow 1$, $h' \rightarrow -q$ and $h'' \rightarrow q(q-1)$, so $(\log h)'' \rightarrow -q < 0$. Consequently $(\log h/(1-q))'' \rightarrow q/(q-1) > 0$, disproving one-coordinate concavity. Holding all other coordinates fixed proves failure for every n . $\square$

The binary portion of this theorem is also present in the repository report (Part I, Proposition 7.2, for $1 < q \leq 2$ and $q > 2$); the product conclusion is the additive consequence, not the main novelty claim of this article.

Theorem 19.3 (Remaining Tsallis orders). *For $n = 1$, $T_{q,1}$ is concave for every $q > 0$. For $n \geq 2$, it is concave for $1 \leq q \leq 2$ and fails even quasiconcavity for every $q > 2$.*

Proof. For one bit and $q \neq 1$,

$$T_{q,1}''(p) = -q[p^{q-2} + (1-p)^{q-2}] < 0.$$

The Shannon limit has the same sign. For $1 < q \leq 2$, Theorem 19.2's proof gives convexity of $\log Z = \sum_i \log h(p_i)$; its exponential Z is convex. Since $T = (1 - Z)/(q - 1)$, it is concave.

For $q > 2$, choose $p > 0$ sufficiently small that $(\log h_q)''(p) < 0$. Keep all but two parameters fixed and use the path $(p + s, p - s)$. The logarithmic first derivative of Z vanishes at zero, and its second derivative is $2(\log h_q)''(p) < 0$. Thus $Z''(0) < 0$ and $T''(0) > 0$. Exchanging the two parameters makes the endpoint entropies equal. The midpoint is a strict local minimum, which violates quasiconcavity as well as concavity. $\square$

19.3 Why the subunit counterexample is not a quasiconcavity failure

Proposition 19.4. *For every n and $0 < q \leq 2$, product Tsallis entropy is quasiconcave, even when it is not concave. It is also concave in every coordinate separately for every $q > 0$.*

Proof. For $q \neq 1$,

$$T_{q,n} = \frac{\exp((1-q)R_{q,n}) - 1}{1-q}, \quad (19.4)$$

which is a strictly increasing function of $R_{q,n}$. A monotone transform of a concave function has convex superlevel sets, giving quasiconcavity through order two. The Shannon case is immediate. With all but coordinate i fixed, the second derivative of T is a positive product of the other h_q factors times the negative binary second derivative displayed above. $\square$

Thus the positive Jensen gaps below one compare the midpoint to the *average* endpoint entropy, not to their minimum. Conflating the two would falsely strengthen the ten-bit theorem. All product entropies here still have their unique maximum at $p_i = 1/2$ for all i , since this separately maximizes each binary contribution in the relevant monotone direction.

20 Endpoint asymptotics and recovery of concavity

The optimal dimension is finite for each fixed $q \in (0, 1)$, but tends to infinity at both endpoints. This explains why the ten-bit obstruction occupies an interior range of orders rather than extending to either endpoint.

20.1 The small-order regime

Theorem 20.1. *As $q \downarrow 0$, with $L = \log(4/q)$,*

$$t_q = \frac{1}{2}L + O(qL), \quad (20.1)$$

$$\Psi(q) = \frac{q}{2} - \frac{q^2}{4} \log \frac{4}{q} + O(q^3 L^2), \quad (20.2)$$

$$\frac{1}{\Psi(q)} = \frac{2}{q} + \log \frac{4}{q} + O(qL^2), \quad (20.3)$$

$$N(q) = \frac{2}{q} + \log \frac{4}{q} + O(1). \quad (20.4)$$

Proof. Let $z = e^{-2t_q}$ and $w = z^{1-q}$. The critical-point equation is exactly

$$q(1 + zw) = (2 - q)(z + w). \quad (20.5)$$

First $t_q \rightarrow \infty$: otherwise a bounded subsequence and $q \rightarrow 0$ would give $\tanh^2 t = 1$ for finite t . Thus $z, w \rightarrow 0$. Because $0 < z \leq w < 1$, (20.5) bounds w above and below by positive constant multiples of q . Hence $\log z = \log q + O(1)$ and $qt_q \rightarrow 0$. It follows that $w/z = \exp(-q \log z) = 1 + O(qL)$ and then, using (20.5) again,

$$z = \frac{q}{4}(1 + O(qL)), \quad t_q = \frac{1}{2}L + O(qL).$$

The hyperbolic quotient has the exact representation

$$\beta = e^{-2qt_q} \frac{(1 - w)^2}{(1 + z)^2}.$$

The first factor is $1 - qL + O(q^2 L^2)$. For the second, $z + w = q/2 + O(q^2)$ follows directly from (20.5), and a logarithmic expansion gives $1 - q + O(q^2 L)$. Therefore

$$\beta = 1 - q(L + 1) + O(q^2 L^2). \quad (20.6)$$

Substitution in $\Psi = q\beta/(1 - q + \beta)$ yields (20.2); the order- q^2 constant cancels. Taking its reciprocal gives (20.3). Finally, taking a floor changes the expression by less than one, and $qL^2 \rightarrow 0$. $\square$

The optimizer itself approaches the parameter boundary, with $p_q = z/(1 + z) \sim q/4$. The maximum of the local curvature ratio therefore escapes to rare-event probabilities as the entropy order tends to zero.

20.2 The Shannon endpoint

Theorem 20.2. *Let $t_* > 0$ be the unique solution of $t_* \tanh t_* = 1$, and set*

$$C = t_*^2 \operatorname{sech}^2 t_* = t_*^2 - 1. \quad (20.7)$$

Then, with $a = 1 - q \downarrow 0$,

$$t_q = t_* + \frac{t_*}{3}a^2 + O(a^4), \quad (20.8)$$

$$\Psi(q) = Ca - C(1 + C)a^2 + O(a^3), \quad (20.9)$$

$$\frac{1}{\Psi(q)} = \frac{1}{Ca} + \frac{1 + C}{C} + O(a), \quad (20.10)$$

$$N(q) = \frac{1}{C(1 - q)} + O(1). \quad (20.11)$$

Numerically, $t_* \approx 1.19967864025773$ and $C \approx 0.439228839890645$.

Proof. Divide the critical-point equation by a and extend analytically to $a = 0$:

$$\frac{\tanh(at)}{a} \tanh t = 1. \quad (20.12)$$

At zero the left side is $t \tanh t$, strictly increasing from zero to infinity. Its derivative at t_* is $\tanh t_* + t_* \operatorname{sech}^2 t_* = t_* > 0$. The implicit function theorem supplies an even analytic solution in a near zero. For positive a this is the unique optimizer already established. Expanding $\tanh(at)/a = t - a^2 t^3/3 + O(a^4)$ gives (20.8).

Now $\beta = \sinh^2(at_q)/\cosh^2 t_q = Ca^2 + O(a^4)$, and

$$\Psi = \frac{(1 - a)\beta}{a + \beta} = Ca - C(1 + C)a^2 + O(a^3).$$

Taking the reciprocal and then the floor gives the remaining formulas. The identity $C = t_*^2 - 1$ follows from $\tanh t_* = 1/t_*$. $\square$

Corollary 20.3 (Recovery near both endpoints). *For each fixed dimension n , product Tsallis entropy is concave for all sufficiently small positive orders and for all orders sufficiently close to one from below. For every $n \geq 10$, it nonetheless fails at $q = 20/41$.*

Proof. Both endpoint expansions give $\Psi(q) \rightarrow 0$. The fixed-order criterion and Theorem 17.1 finish the proof. Alternatively, (15.3) supplies explicit sufficient endpoint regions. $\square$

We have proved the existence of these concave and nonconcave regions, not that the nonconcave orders form one connected interval. That stronger assertion requires further control of Ψ .

21 Exact algebraic decisions at rational orders

21.1 A unique polynomial root

Theorem 21.1. *Let $q = m/k \in (0, 1)$, with positive integers $m < k$. There is exactly one root $y_q \in (0, 1)$ of*

$$P_{m,k}(y) = m(1 + y^{2k-m}) - (2k - m)(y^k + y^{k-m}). \quad (21.1)$$

At that root put

$$\beta(y) = \frac{y^m(1 - y^{k-m})^2}{(1 + y^k)^2}. \quad (21.2)$$

Then

$$\Psi(m/k) = \frac{m\beta(y_q)}{k - m + k\beta(y_q)}. \quad (21.3)$$

For a given n , concavity is equivalent to

$$K_{m,k,n}(y_q) \leq 0, \quad (21.4)$$

where

$$K_{m,k,n}(y) = (nm - k)y^m(1 - y^{k-m})^2 - (k - m)(1 + y^k)^2. \quad (21.5)$$

In particular this is an exact, decidable algebraic question.

Proof. Set $y = e^{-2t_q/k}$. The hyperbolic critical equation becomes

$$\frac{(1 - y^{k-m})(1 - y^k)}{(1 + y^{k-m})(1 + y^k)} = \frac{k - m}{k}.$$

Clearing the positive denominator gives (21.1). The change of variable is a bijection from $t \in (0, \infty)$ to $y \in (0, 1)$, so uniqueness follows from Proposition 14.2. The same substitution in the hyperbolic quotient gives $\beta(y)$. Equation (21.3) is (14.2) with $q = m/k$. Multiplying $n\Psi \leq 1$ by positive denominators gives (21.4).

For decidability, compute the rational polynomial greatest common divisor of P and K . Exact real-root counting determines whether this gcd has a root in $(0, 1)$; if so, it must be the unique y_q , and equality holds. If not, $K(y_q) \neq 0$. Isolate y_q by successively smaller rational intervals and evaluate K by interval arithmetic. Continuity and its nonzero value guarantee eventual strict sign separation. Rational polynomial real-root counting itself is available by the Euclidean remainder sign-variation procedure, so every step uses finite exact algebra. $\square$

21.2 Certified bounds without general algebraic-number software

For rational $0 < \ell \leq y_q \leq u < 1$, monotonicity of each positive factor yields

$$\frac{\ell^m(1 - u^{k-m})^2}{(1 + u^k)^2} \leq \beta(y_q) \leq \frac{u^m(1 - \ell^{k-m})^2}{(1 + \ell^k)^2}. \quad (21.6)$$

The fractional map in (21.3) is increasing, giving rational bounds $L \leq \Psi \leq U$. Whenever

$$\left\lfloor \frac{1}{U} \right\rfloor = \left\lfloor \frac{1}{L} \right\rfloor,$$

the common integer is a certified value of $N(q)$. Exact equality cases cannot generally be settled by progressively tighter strict intervals alone. The orders $1/2$ and $1/3$ illustrate this point: their exact boundary values require the analytic arguments in Theorems 16.1 and 16.2, or the gcd branch of the complete decision procedure.

The included standard-library verifier implements the interval branch for the nonexceptional entries in Table 3. It does *not* purport to implement the general gcd/root-counting branch; that branch is an algorithmic theorem, not a claimed completed software feature. In particular, no arbitrary timeout is being interpreted as an exact equality decision.

22 A block-product principle and formalization architecture

22.1 The matrix argument survives beyond binary factors

Theorem 22.1 (Block-product curvature criterion). *For $1 \leq j \leq r$, let $U_j \subseteq \mathbb{R}^{d_j}$ be open and convex, and let $f_j : U_j \rightarrow (0, \infty)$ be twice continuously differentiable. Assume $D_j = -\nabla^2 \log f_j$ is positive definite everywhere. Define*

$$F(x_1, \dots, x_r) = \prod_j f_j(x_j), \quad b_j = \nabla \log f_j(x_j), \quad \lambda_j = b_j^\top D_j^{-1} b_j.$$

Order q	$\Psi(q)$ (approximate)	Exact $N(q)$
1/100	0.004849618892	206
1/10	0.040484994049	24
1/4	0.078512965127	12
1/3	0.090909090909	11
2/5	0.096959419208	10
20/41	0.100008270243	9
1/2	0.100000000000	10
3/5	0.095978801240	10
3/4	0.076273741422	13
9/10	0.038042258451	26
99/100	0.004329556562	230

Table 3: Fixed-order dimension thresholds. Decimal profile values are approximations. The integers are certified using rational root isolation and outward bounds, except the two equality cases $1/2$ and $1/3$, which follow from exact analytic formulas.

Then $\nabla^2 F$ is negative semidefinite at $(x_j)_j$ exactly when

$$\sum_j \lambda_j \leq 1. \quad (22.1)$$

If this sum exceeds one, there is exactly one positive eigenvalue. If the separate suprema $M_j = \sup_{x_j \in U_j} \lambda_j$ are finite, F is concave on the product domain exactly when $\sum_j M_j \leq 1$.

Proof. Concatenating the b_j and forming the block-diagonal D yields $\nabla^2 F = F(bb^\top - D)$. Congruence by $D^{-1/2}$ gives the same rank-one matrix as in Lemma 13.1. For the global statement, the supremum of the sum over a Cartesian product is the sum of the separate suprema: approximate each within any prescribed positive error. Thus a sum of suprema exceeding one produces a point of positive curvature. The reverse implication follows from the local criterion and restriction to line segments. $\square$

No blanket assertion about all categorical distributions follows without calculating their block quantities M_j . The theorem isolates exactly which estimate would be needed to generalize the nine-bit phenomenon. [Added 29 September 2026, batch 57: Part III computes these block quantities for categorical simplices: $M_j = \Theta_{k_j}(q)$ for $0 < q < 1$ (Definition 30.2), attained only at one large and $k_j - 1$ equal probabilities (Theorem 29.1), so T_q is concave on $\prod_j \Delta_{k_j}$ if and only if $\sum_j \Theta_{k_j}(q) \leq 1$ (Theorem 30.3).]

22.2 Suggested proof-assistant decomposition

The repository supplies motivation and a location for integration, but this article does not depend on trusting its unformalized entropy claims. A formalization can be organized around the following boundaries.

The finite probability layer defines the independent bit mass function and proves (12.3) by induction on the number of bits. An injective relabeling lemma transfers it to collision-free weighted sums. This layer is finite algebra and can be separated from calculus.

The linear-algebra layer proves that $bb^\top - D \preceq 0$ if and only if $b^\top D^{-1}b \leq 1$ for a positive diagonal matrix, using weighted Cauchy–Schwarz. Full spectral machinery is unnecessary for the main concavity theorem; inertia is only needed for the stronger one-positive-eigenvalue statement.

The analytic layer differentiates h_q , identifies the Hessian, proves the hyperbolic identity, and establishes the positive-coefficient inequality in Lemma 15.1. A route through absolutely convergent real power series keeps the proof faithful to the article. The optimizer's uniqueness follows from monotonicity; the implicit function and asymptotic arguments can be formalized later without blocking the dimension theorem.

Finally, an exact arithmetic checker should validate (17.3) and the dyadic inequalities (17.8), and a proved bridge should turn those certificates into the real inequalities claimed here. Executing Python successfully is not that bridge. The main nine-bit proof and rational ten-bit curvature witness form a smaller initial target than the entire paper.

23 Further research questions

The following are concrete continuations rather than claims proved in this article.

- Q1. Global shape of the scalar profile.** Is Ψ strictly unimodal on $(0, 1)$? An affirmative answer would prove that the nonconcave orders in each fixed dimension form at most one interval. The graphs suggest this, but our derivative calculation is only local.
- Q2. An exact description of the ten-bit failure interval.** Determine the other solution of $\Psi(q) = 1/10$ besides $q = 1/2$, prove there are no others, and bound the interior maximum with certified transcendental interval methods. The algebraic certificate at 20/41 provides a rigorous interior point, not the complete interval.
- Q3. Exceptional rational orders.** Classify rational q for which $1/\Psi(q)$ is an integer, extending $1/2$ and $1/3$. The polynomial pair (P, K) in Theorem 21.1 turns each instance into a gcd problem, but an infinite classification would require additional structure.
- Q4. Weights with collisions.** Find the minimum failing dimension when arbitrary positive weights, including partially colliding subset sums, are allowed. Is nine-bit concavity valid for every such weight system below order one? Our nine-bit theorem proves this only for collision-free weights.
- Q5. A combinatorial classification of coarsenings.** Which partitions of the bit cube preserve parameter concavity of subunit power sums? The ordinary Hamming layers and singleton partitions are fundamentally different examples. A classification could explain when aggregation removes the rank-one obstruction.
- Q6. Sharp constrained concavity.** For fixed weights w , determine the threshold after restricting the Hessian to $w^\perp$. The explicit adjustment $v = \sigma - c\mathbf{1}$ proves failure near equal weights but does not give a complete constrained analogue of $N(q)$.
- Q7. Higher alphabets and nonidentical blocks.** Compute the block supremum M_j for categorical probability simplices or structured finite distributions. Determine whether sharp integer dimension transitions analogous to nine versus ten persist, and how they depend on alphabet size. [Added 29 September 2026, batch 57: answered for categorical simplices in Part III. The block supremum is $\Theta_k(q)$ (Theorem 30.3); at $q = 1/2$ three factors with a common alphabet are concave exactly for alphabets of at most seventeen symbols (Theorem 31.1); concavity for every choice of alphabets holds exactly when $nq \leq 1$ (Theorem 33.2); above order one a compression-matrix criterion replaces the additive budget (Theorem 35.2). Structured finite families other than full simplices are not treated (Part III, question AQ8).]
- Q8. Optimal quantitative stability.** Determine the best strong-concavity constant, classify null directions at every algebraic equality order, and compute the first nonzero bending term. The half-order quartic law provides an exactly solvable test case.

Q9. Certified decision software. Implement the complete rational-order gcd/root-counting branch and produce compact proof certificates for $N(m/k)$. Seek degree reduction from reciprocal symmetries of $P_{m,k}$ before using general-purpose real-root isolation.

Q10. Machine-checked core theorem. Formalize the rank-one criterion, the hyperbolic coefficient inequality, and the 96-digit certificate in Lean. Then add a verified dyadic-root checker for the finite midpoint examples. This would establish a clear kernel-checked boundary for the result inside ProveIt.

24 Verification, provenance, and limitations

24.1 What is established, and how

The all-order nine-bit theorem, the unique scalar optimizer, exact half- and third-order thresholds, derivative sign, weighted-mean obstruction, strong-concavity bound, order classification, endpoint asymptotics, and block criterion have mathematical proofs in the text. The finite decimal-enclosure inequalities are separately certified by exact rational computation. The profile figures are numerical illustrations only.

The package’s dependency-free verifier completed **17,927 exact arithmetic checks**. These include independently enumerated joint mass functions versus the product formula, collision-free binary encodings, finite coefficient checks, the cleared-denominator identity and positive integer, both finite midpoint certificates, exact mean preservation, and rational enclosures for the nonexceptional table entries. The finite coefficient checks support the implementation but do not replace the infinite-series proof.

An additional optional symbolic script checks the special-order identities, the rank-one determinant identity in small dimensions, and the half-order Taylor coefficients. Its symbolic simplifications are independent implementation checks, not kernel certification of the real-analytic arguments.

24.2 Relation to the existing repository

The mathematical baseline was read at commit `ac9107d74083fcfe8064b4aaeda7989612e6a5b6`. The relevant report is located at

```
SetTheory/Cardinals/docs/reports/
log-concavity-and-unimodality/
bernoulli-entropy-quasiconcavity/
```

The ordinary-sum $q > 1$ counterexample, its fixed-mean construction, and the binary Rényi threshold are not claimed as new contributions here. The new target is the product/collision-free weighted model, especially the sharp universal dimension, the nearby-weight mean constraint, and the exact scalar threshold theory. We did not modify the GitHub repository.

[Write, 29 September 2026, batch 43: the report read at that commit is Part I as printed above. Between the pinned commit and the placement commit `faef2ed2a` its files are unchanged; the placement only added this manuscript’s code, data and source notes beside them. Part I’s paragraph in its Section 10 that poses the subunit question now carries a dated pointer to this part.]

24.3 Priority and reliability statement

The inspected sources do not state the nine-versus-ten product-bit result or the formulas developed here. Targeted searches on 29 September 2026 did not establish an earlier occurrence, but several queries returned weakly relevant material; they are not a comprehensive priority search. We

therefore present these as candidate original results with explicit proofs, not as a bibliographically certified breakthrough. The recent ordinary-sum preprint is acknowledged to avoid mistaking a change of model for a fresh solution of that question.

The manuscript has not been independently refereed and has not been formalized in Lean. Its general weighted conclusions require collision-freeness, except where the specifically constructed weights are proved to have that property. No result here resolves the general partially colliding weighted problem. These scope restrictions are part of the mathematical statements, not merely implementation caveats.

25 Reproducing Part II

[Write, 29 September 2026, batch 43: this section is the manuscript’s Appendix A, “Reproducing the package”, rewritten for the names under which its files are shipped in this report. Every statement the manuscript makes about its package is kept; the changes are in brackets.]

The principal files [of the delivered package] are `article.tex` and `article.pdf` [here: Part II is printed in the report’s `article.tex` and `article.pdf`]. The bibliography is inline [here: merged into the report’s bibliography], and the two vector figures and generated LaTeX table are included [here: the table is shipped as `data/02-product-entropy-profile_table.tex` and input by the article; the two figure PDFs are not shipped, and Figures 3 and 4 are drawn at build time with `pgfplots` from `data/02-product-entropy-profile_plot.csv`]. A standard pdfLaTeX installation [with `pgfplots`] suffices to rebuild the article; no network access is required. [In this report, from a *copy* of the report directory:]

```
python code/02-product-entropy-verify_exact.py
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The exact verifier uses only the Python standard library and can be run from any working directory. It writes `results/certificates.json` and `results/profile_table.tex` [under the report root, beside Part I’s own files in `results/`; the recorded copies are shipped as `data/02-product-entropy-certificates.json` and `data/02-product-entropy-profile_table.tex`, and its standard output as `data/02-product-entropy-checks.txt`]. Its decimal strings are for readability; all accept/reject decisions use integer or rational arithmetic.

Optional checks and figure generation are available separately:

```
python code/02-product-entropy-verify_symbolic.py
python code/02-product-entropy-make_figures.py
```

Their dependencies and the versions actually used are recorded in `README.md` [of the delivered package, which is not shipped: Python 3.13.5, SymPy 1.14.0, Matplotlib 3.10.8 and pdfTeX 1.40.26]. Prebuilt figures make these commands unnecessary for compiling the supplied manuscript [here the figures need no Python at all]. No third-party papers, fonts, or software distributions are bundled.

[The symbolic script prints its result, recorded as `data/02-product-entropy-symbolic_checks.txt`; do not redirect it to `results/symbolic_checks.txt`, which is Part I’s record. The figure script writes `results/profile_plot.csv` and `figures/profile.pdf`, `figures/ten_bit_detail.pdf` under the report root. The shipped `code/02-product-entropy-Makefile` names the delivered paths, so its `check`, `symbolic` and `figures` targets would run Part I’s scripts `code/verify_exact.py`, `code/verify_symbolic.py` and `code/make_figures.py`; it is shipped as delivered and should not be used in this directory. On 29 September 2026 the exact and symbolic scripts were rerun on a copy of the report directory (Python 3.14.4; SymPy 1.14.0): both passed, with 17,927 and 14 checks, and every output agreed with the recorded file apart from line endings.]

Part III. Beyond nine bits: exact alphabet-size criteria and the Shannon boundary for product entropy

Part III was added on 29 September 2026 in batch 57 of ProveIt’s incoming-report intake, from one manuscript written against the pinned state of Parts I and II. It answers Part II’s question Q7 (Section 23) for full categorical simplices: it evaluates the block supremum of Part II’s block-product criterion (Theorem 22.1) for every finite alphabet, and so replaces the independent bits of Part II by independent categorical variables with arbitrary, possibly unequal, finite alphabets. Its binary case is Part II’s. Like Part II it concerns the joint vector, not Part I’s ordinary sum. Parts I and II are unchanged except for two dated notes in Part II (after Theorem 22.1 and at question Q7), a line on the title page, a contents entry, five macros and the `cleveref` package in the preamble, and a remark on the bibliography entry [5]. Section 26 records the provenance, what is and is not settled, and the notation of the addition.

26 Provenance and conventions of Part III

26.1 One manuscript, one addition

Part III is batch-57 manuscript 04, *Beyond Nine Bits: Exact Alphabet-Size Criteria and the Shannon Boundary for Product Entropy*, subtitled “A solution of the categorical-simplex part of the repository’s higher-alphabet research question”, a research manuscript prepared with ChatGPT for Vladimir Reshetnikov and dated 29 September 2026 (24 A4 pages as delivered). It arrived as the archive `ProveIt_Alphabet_Entropy.zip` (inner directory `ProveIt_Alphabet_Entropy/`, main file `article.tex`) in ProveIt commit 6ad01e9c8 and was placed in 70b39943a. It was written against ProveIt commit 1085b506d65e207a05b7e9c861bb1fe88432fe38, reading this report’s `article.tex` and `README.md` there, through [the pinned report](#). Between that commit and the placement commit the report’s files are unchanged (the placement only added the eighteen 03-alphabet-entropy- files), so every statement the manuscript makes about “the repository”, “the repository report” or *Nine Bits*, *Ten Bits* refers to Parts I and II as printed above, and its section numbers 22 and 23 are those of this report.

The manuscript’s code and data are shipped with the prefix 03-alphabet-entropy- (Part I’s unprefix files count as 01, Part II’s are 02): its four scripts and its Makefile in `code/`; its figure data, its requirements file and eight of the nine records of its `verification/` directory (certificates, numerical diagnostics, symbolic results, three console transcripts, build log, environment) in `data/`; and its claim ledger, source notes and the ninth record, a layout check, at the report root. Its two figures are shipped as delivered, as `figures/03-alphabet-entropy-alphabet_profiles.pdf` and `figures/03-alphabet-entropy-shannon_window.pdf`. Its manuscript, delivery README and PDF are not shipped, and its checksum file (23 of 23 entries verified at placement) was retired; they survive in the arrival commit.

In Part III, “this article”, “here” and “we” refer to the manuscript; “the repository”, “the ProveIt report” and its bibliography key `repo` refer to Parts I–II. The manuscript contributes, in the order printed here: its target, contribution and provenance (Section 27); the setting (Section 28); the extremal power-sum product (Section 29); the subunit criterion (Section 30); seventeen versus eighteen symbols (Section 31); rational-order decisions (Section 32); large alphabets (Section 33); the Shannon window (Section 34); orders above one (Section 35); quantum product states (Section 36); verification and a Lean plan (Section 37); nine research questions (Section 38); its conclusion (Section 39); and its appendix of coefficients (Section 40). Every result, proof, example, remark, limitation and question of the manuscript is printed. Its abstract reads:

We determine exactly when the Tsallis entropy of an independent joint vector is concave in its marginal probability vectors, allowing arbitrary and unequal finite alphabets. Below order one, the answer is an additive curvature budget, with one explicitly computable scalar per alphabet. Its optimizer is unique up to permutation and has one large probability and equal remaining probabilities. This evaluates the block suprema left open by a recent ProveIt report. At order $1/2$, three factors are concave for every common alphabet of at most seventeen symbols, whereas eighteen symbols already admit a rational interior counterexample; both sides have exact certificates.

We also give a mixed-alphabet matrix criterion above order one, transfer the classification to quantum product-state parameters, and derive a convergent large-alphabet expansion. The Shannon limit is nonuniform: when $q = 1 - \lambda/\log^2(k-1)$, the maximal block curvature tends to $\lambda/(4+\lambda)$. A uniform two-term expansion yields the local concavity boundary through order $\log^{-4}(k-1)$. All principal statements have proofs. Exact arithmetic certificates, symbolic and numerical checks, a formalization plan, and further research questions accompany the manuscript.

Its title page adds:

Scope and status. Concavity refers to affine interpolation of the *marginals*, not arbitrary mixtures of joint laws and not the entropy of their ordinary sum. These are conventional mathematical proofs in an unrefereed manuscript; no Lean verification is claimed. The extension is assessed against the inspected repository and primary-source records, not asserted to have globally certified bibliographic priority.

26.2 What Part III settles in Part II, and what it does not

- **Question Q7 of Part II, for full categorical simplices: answered.** For a block Δ_k the supremum M_j of Theorem 22.1 is $\Theta_k(q)$ (Definition 30.2), attained exactly at the k permutations of one large and $k-1$ equal probabilities (Theorem 29.1); so for $0 < q < 1$ the product Tsallis entropy is concave on $\prod_i \Delta_{k_i}$ if and only if $\sum_i \Theta_{k_i}(q) \leq 1$ (Theorem 30.3). Sharp integer transitions persist and depend on the alphabet: at $q = 1/2$, three factors are concave exactly for common alphabets of at most seventeen symbols (Theorem 31.1; Table 5 lists the largest concave factor count for twelve alphabet sizes, its caption two more), and concavity for *every* finite choice of alphabets holds exactly when $nq \leq 1$ (Theorem 33.2). The sentence after Theorem 22.1, “No blanket assertion about all categorical distributions follows without calculating their block quantities M_j ”, stays true as written; Part III does that calculation for full simplices.
- **The binary case is Part II.** For $k = 2$ the block quantity τ_q of Lemma 30.1 is Part II’s $\rho_q = b_q^2/d_q$ (13.1): both are $g(D^{-1}g)$ for the one-dimensional binary block, and this ratio does not depend on how the segment is parametrized. Hence $\Theta_2(q) = \Psi(q)$, and Theorem 30.3 for binary alphabets is Part II’s Theorem 12.3. The placement audit of this intake checked it numerically, independently of both manuscripts’ code: $\Theta_2 = 1/11, 1/10$ and $0.0657638785\dots$ at $q = 1/3, 1/2$ and 0.8 , matching $N(1/3) = 11$ and $N(1/2) = 10$ (Theorems 16.1 and 16.2); the binary entry of Table 5 is Part II’s ten bits. The Shannon-end constant $V_2 = t_0^2/4 - 1 = 0.4392288\dots$ of Proposition 33.4 is Part II’s $C = t_*^2 - 1$ (Theorem 20.2): at $k = 2$, $c = 0$ and $t_0 = 2t_*$. Its limit $\Theta_k(q)/q \rightarrow (k-1)/k$ at $q \rightarrow 0$ gives $1/2$, the leading term of Part II’s $N(q) \sim 2/q$ (Theorem 20.1). The binary statements of Proposition 35.3 are Part II’s Theorems 19.2 and 19.3; the manuscript itself calls these recovered, not new.
- **Not settled.** Structured finite families other than full simplices (exponential families, grouped categories, constrained density matrices: question AQ8), and the other half of Q7’s “structured finite distributions”. Global unimodality of Θ_k or of Part II’s Ψ (Part II’s Q1, Part III’s AQ1): the local uniqueness in the Shannon window (Corollary 34.2) is not global unimodality. Partially colliding weights and coarsenings (Part II’s Q4–Q5, Part III’s AQ5).

Part I's question for the *ordinary sum*, which Part III does not address: its concavity is in the marginals of the independent joint law, not under mixtures of joint laws, and not for the ordinary sum. The preprint of H. Wang [5] is recorded by this manuscript too from its abstract only; this report still has not read or verified it.

26.3 Notation of Part III beside Parts I and II

Part III keeps the manuscript's symbols except for two renamings (below). Its entropies use the unnormalized conventions of Parts I and II, applied to the independent joint law of categorical variables: for binary alphabets F_q , T_q and H_q are Part II's $Z_{q,n}$, $T_{q,n}$ and $R_{q,n}$. Several letters carry other meanings in Parts I and II (Table 4); within Part III the Part III meaning always applies.

Table 4: Symbols of Part III that differ from, or could be confused with, those of Parts I and II.

Part III symbol	Meaning in Part III	Not to be read as (Parts I, II)
observable	independent joint law of n categorical variables with marginals $p^{(i)} \in \Delta_{k_i}$; concavity under affine interpolation of the marginals	Part I's ordinary sum S_p ; a mixture of joint laws
$k, k_i; n$	alphabet sizes; number of factors	Part II's k in $q = m/k$ (Theorem 21.1); there n counts bits
$\Delta_k, E_k; \Delta$	probability simplex and its tangent space; the midpoint deficit of Proposition 31.2	Δ, Δ_w , Part II's Jensen gaps
$S_q(p)$	$\sum_j p_j^q$, power sum of one marginal	Part I's $S_p = X_1 + \dots + X_n$, the ordinary sum
F_q	$\prod_i S_q(p^{(i)})$, power sum of the joint law; Part II's $Z_{q,n}$ for binary alphabets	Part I's F_q , power sum of the ordinary sum (1.2)
T_q, H_q	Tsallis and Rényi entropy of the joint law (28.3)	Part I's $H_{T,q}, H_{R,q}$ of the sum; Part II's $T_{q,n}, R_{q,n}$ (the binary case)
$\Pi_\alpha(p), \Pi_k^*(\alpha), \Pi$	$S_{1-\alpha}(p)S_{1+\alpha}(p)$, its maximum over Δ_k , and its value at the current p (the manuscript's $R_\alpha, \mathcal{R}_k, R$)	Part II's Rényi entropy $R_{q,n}$
α	$1 - q$ below order one; $q - 1$ in Section 35	Part I's $a = q - 1$; Part II's $a = 1 - q$
$c; t; x$	$\log(k - 1)$; the logarithm of the large-to-small probability ratio (Theorem 29.1); the large probability. In Section 28 and (30.11), $t \in [0, 1]$ is the interpolation parameter	Part II's half log-odds t (at $k = 2$, $c = 0$ and t here is twice Part II's); Part I's imbalance t
$A, g, D; D_i, D_0$	tangent bilinear form, log-gradient, $-D^2 \log f = A + g \otimes g$; blocks; a diagonal matrix in the proof of Theorem 35.2. A also names two sums in the proof of Theorem 29.1 and a function in the proof of Proposition 31.2	Part II's diagonal D has the same role for bits; Part I's $D_q(p)$ (4.1)
ω	$g(A^{-1}g)$ in Lemma 30.1 (the manuscript's K)	—
K	$k - 1$ (Sections 31–33 and 40)	Part II's $K_{m,k,n}, (P, K)$; Part I's $K(t)$
$\tau_q(p), \sigma$	block curvature $g(D^{-1}g)$; its sum over blocks	Part II's $\rho_q(p)$ and λ_j are the binary case; Part II's σ is a sign vector
$\Theta_k(q)$	$\max \tau_q$ on Δ_k , the alphabet profile	$\Theta_2 = \Psi$ (Part II); M_j of Theorem 22.1 for a full simplex
$\kappa_q, \Gamma_k(q), M(\cdot), \gamma_i$	above-one block quantity, its maximum, compression matrix (Section 35)	Part II's block suprema M_j

Part III symbol	Meaning in Part III	Not to be read as (Parts I, II)
C_q ; C ; $C(\lambda)$	large-alphabet constant (33.1); a diagonal matrix (proof of Theorem 35.2); a coefficient (Corollary 34.2)	Part II's $C = t_*^2 - 1$, which is V_2 here; Part I's $C = p^2$
V_k	$t_0^2/4 - 1$, the largest variance of $\log p_j$ on k symbols	$V_2 = C$ of Part II
λ , λ_n	Shannon-window scale $q = 1 - \lambda/c^2$; its critical value	Part II's block quantity λ_j
h	$h(z)$ in the proof of Theorem 29.1; the displacement 1/1000 (Proposition 31.2); $1/c$ (Section 34)	Part II's one-bit power sum $h_q(p)$, which is $f(p)$ in the proof of Proposition 35.3
y ; a/b	square root of the large-to-small ratio at $q = 1/2$, and $y = e^{t/b}$ in Theorem 32.1; $q = a/b$ there	Part II's y , $q = m/k$, $a = 1 - q$ and gradient b
L ; U	compact bound $[0, L]$ and $L(s)$; the Tsallis value $U(s)$ on the witness line and a function $U(h^2, \lambda)$	Part II's root bounds L, U
$\mathcal{D}_k$, ρ_i , $\mathcal{T}_q$	density matrices and quantum product Tsallis entropy (Section 36)	Part II's $\rho_q(p)$
AQ1–AQ9	Part III's research questions (the manuscript's Q1–Q9)	Part II's Q1–Q10

Renamed symbols. The manuscript's power-sum product $R_\alpha(p)$, its maximum $\mathcal{R}_k(\alpha)$ and the abbreviation R are $\Pi_\alpha(p)$, $\Pi_k^*(\alpha)$ and Π here, so that they cannot be read as Part II's Rényi entropy $R_{q,n}$; and its quantity $K = g(A^{-1}g)$ of Lemma 30.1 is ω , since the manuscript itself uses $K = k - 1$ from its next section on. Labels keep their names: the label of equation (30.4), which displays ω , still ends in `eq:K`. The shipped scripts and records keep the manuscript's names (for example `R_witness` in the certificate file). No normalization changed.

26.4 Where the write had to choose

- **Placement and numbering.** Part III follows Part II and precedes Part I's appendices, so no number of Parts I and II moves (the `.aux` numbers of all their labels were compared with a build of the committed text). The manuscript's Section n is Section $n + 26$ here ($n = 1, \dots, 13$), and each numbered statement and equation keeps its position: its Theorem $n.m$ is Theorem $(n + 26).m$ and its equation $(n.m)$ is $(n + 26.m)$. Its Appendix A is Section 40. Its Figures 1–2 are Figures 5–6 and its Table 1 is Table 5; its unnumbered verification table stays unnumbered, and Table 4 is new. Its questions are AQ1–AQ9.
- **Front matter.** The manuscript's title page, abstract and contents are replaced by this section (the abstract and the title-page scope statement are quoted above).
- **Bibliography.** The manuscript's entry for H. Wang (it gives the forename Haoran) is the preprint already cited by Part II and is cited as [5]; its entries for Sakai–Iwata and Reeb–Wolf are added as [7] and [8]. Its citations of the repository report (`repo`) became references to Part II's sections and theorems, and the pinned URL of that entry is in Section 26.1.
- **Figures and files.** The two figure PDFs are the delivered Matplotlib files, unchanged; like Part I's two figures they embed Type 3 fonts. File names of the delivered package are followed by the shipped names in brackets; its reproduction commands, which in this directory would run Part I's scripts, are replaced by a safe recipe (Section 37).
- **Dated notes and substitutions.** Text added at this write is marked “[Write, 29 September 2026, batch 57: ...]” in Part III and “[Added 29 September 2026, batch 57: ...]” in Part II; shorter additions are in brackets. Unmarked substitutions in the manuscript's text are only

these: “Section 23” and “Section 22” became references (Section 27); its three citations of the repository report became “(Part II above)”, “Part II (Theorem 22.1)” and “(Part II, Theorems 19.2 and 19.3)”; the two renamings above; the macro for the pinned commit and the repository path were written out; and its questions’ labels Q1–Q9 became AQ1–AQ9. No mathematical statement of the manuscript or of Parts I–II was changed, and nothing of the manuscript was omitted.

- **Status.** The placement audit of this intake checked, in high-precision or exact arithmetic and independently of the manuscript’s code, the closed form of the block curvature against a direct tangent Hessian, the identification $\Theta_2 = \Psi$, the values $3\Theta_{17}(1/2) = 0.99403\dots < 1 < 1.00680\dots = 3\Theta_{18}(1/2)$, the witness values (31.6), the identity (31.4), $U''(0)$, the finite gap (31.9), the limits $\Theta_k \rightarrow q$ and the Shannon-window limit $\lambda/(4 + \lambda)$; the quantum section (Section 36) was read but not independently checked. The write read the manuscript in full while merging it. None of this is an independent proof review. Part III is AI-assisted and unrefereed, and nothing in it is formalized: the Lean decomposition in Section 37 and question AQ9 are proposals, and ProveIt has no Lean or Rocq development of Rényi or Tsallis entropy. The four scripts were rerun on 29 September 2026 on a scratch copy with the delivered names (Python 3.13.5 via uv, NumPy 2.3.5, mpmath 1.3.0, SymPy 1.14.0, Matplotlib 3.10.8). All four finished normally, with 70 exact assertions, 4,595 numerical assertions and 7 symbolic identities; the certificate, numerical and symbolic records equal the shipped `data/03-alphabet-entropy-` files after conversion of line endings, and the regenerated profile CSV differs from the shipped one only in the last digit of 386 of its 1,245 rows (relative difference at most $7.6 \cdot 10^{-16}$, platform floating point).

27 Research target, contribution, and provenance

27.1 The explicit repository question

The ProveIt report *Nine Bits, Ten Bits*, incorporated as Part II of *Two-coin counterexamples to the proposed generalized Shepp–Olkin thresholds*, studies the entropy of an independent *bit vector*. It proves a scalar curvature criterion for binary factors and an abstract block-product criterion. Its Section 23, question Q7, asks for the block supremum on categorical probability simplices and for alphabet-dependent integer transitions. Section 22 explicitly stops short of a categorical classification (Part II above).

We solve that question for unrestricted finite categorical simplices, including unequal alphabets. The central advance is not a new rank-one matrix lemma: that part is already in the repository. It is the exact evaluation of the simplex block quantity, its unique optimizer, and the consequences that become accessible once that evaluation is available. The above-one matrix classification and the uniform Shannon transition provide further extensions.

The inspected repository snapshot is

1085b506d65e207a05b7e9c861bb1fe88432fe38.

The principal files are `article.tex` and `README.md` under

```
SetTheory/Cardinals/docs/reports/
log-concavity-and-unimodality/
bernoulli-entropy-quasiconcavity/
```

In the article, the relevant contiguous ranges inspected were source lines 1470–1770 and 2350–2530. The latter contains the block theorem and Q7. [Line numbers of the pinned `article.tex`: the first range lies in Sections 13–16, the second in Sections 21–24 of this report.] The present proofs do not require accepting unformalized repository claims as axioms: the required matrix argument is repeated.

27.2 What is proved

For an alphabet of size k , define $\Theta_k(q)$ by the unique scalar root in Theorems 29.1 and 30.3. For $0 < q < 1$, the complete answer is

$$T_q \text{ is concave on } \Delta_{k_1} \times \cdots \times \Delta_{k_n} \iff \sum_{i=1}^n \Theta_{k_i}(q) \leq 1. \quad (27.1)$$

At most one local curvature direction can violate concavity. A strict budget gives an explicit strong-concavity bound; equality gives a precise finite exceptional set and still permits strict concavity.

The half-order result is especially concrete. For three independent variables with a common alphabet, the largest universally admissible alphabet size is *seventeen*. An eighteen-symbol witness uses probabilities $81/98, 1/98, \dots, 1/98$ and a rational line segment. A sum-of-squares identity proves the positive seventeen-symbol result. This is a sharp finite-dimensional theorem, not an inference from plots.

For large alphabets, the block curvature tends to q . Consequently, concavity holds for *every* finite choice of alphabets exactly when $nq \leq 1$. This does not contradict continuity at $q = 1$ for a fixed alphabet: the approach to the Shannon limit has a shrinking window of width $\log^{-2} k$. We identify that window and its first correction. Above order one a different, explicit matrix criterion replaces the additive budget. The same classical criteria govern quantum product states when their density matrices are interpolated affinely.

27.3 Literature and limits of the novelty claim

Two-order optimization of probability power sums belongs to an existing literature. Sakai and Iwata derive sharp comparisons between distinct Rényi orders by optimizing one norm at fixed another [7]. The fixed-dimensional variance-of-surprisal problem appearing at the Shannon endpoint has a sharp classical and quantum treatment by Reeb and Wolf [8]. We do not claim those problem families, or the rank-one product-Hessian method, as new. We supply a short self-contained extremal proof adapted to the present unconstrained product of two power sums, and use it to answer the repository's specific question.

The source review was targeted, not exhaustive. For the two external papers just cited, the bibliographic records and abstracts were checked; the present derivations do not depend on an uninspected theorem in either paper. Global priority for the exact combined classification is not certified by that review.

A separate recent preprint by Haoran Wang, submitted on 23 September 2026, states a sharp threshold for the entropy of the *ordinary sum* of independent Bernoulli variables [5]. Its abstract was checked, not its proof. It is neither a premise nor a consequence of this article. The map from a vector to its sum merges outcomes, and the present product factorization is lost. No result here settles arbitrary coarsenings, partially colliding weighted sums, or all structured block families.

28 The parameter space and the meaning of concavity

For $k \geq 2$, let

$$\Delta_k = \{p \in [0, 1]^k : \sum_{j=1}^k p_j = 1\}, \quad E_k = \{v \in \mathbb{R}^k : \sum_{j=1}^k v_j = 0\}.$$

Its relative interior consists of strictly positive vectors. Derivatives on the interior are restricted to the tangent space E_k ; there is no implicit treatment of all k coordinates as independent. Put

$$S_q(p) = \sum_{j=1}^k p_j^q, \quad q > 0, \quad (28.1)$$

with $0^q = 0$. For marginals $p^{(i)} \in \Delta_{k_i}$ define

$$F_q(p^{(1)}, \dots, p^{(n)}) = \prod_{i=1}^n S_q(p^{(i)}), \quad (28.2)$$

$$T_q = \frac{F_q - 1}{1 - q}, \quad H_q = \frac{\log F_q}{1 - q} \quad (q \neq 1). \quad (28.3)$$

These are respectively the power sum, Tsallis entropy, and Rényi entropy of the independent joint law: expanding the finite product proves (28.2). At $q = 1$ both entropies have the continuous limit

$$T_1 = H_1 = - \sum_i \sum_j p_j^{(i)} \log p_j^{(i)}. \quad (28.4)$$

Natural logarithms are used throughout. Multiplication by a positive normalizing constant would not change any concavity criterion.

Concavity means that for marginal tuples P, Q and $0 \leq t \leq 1$,

$$T_q((1-t)P + tQ) \geq (1-t)T_q(P) + tT_q(Q).$$

The independent joint law associated with $(1-t)P + tQ$ is generally *not* the mixture of the two joint laws. Ordinary concavity of an entropy as a function of a joint probability vector therefore does not answer this question.

An alphabet of size one contributes the constant factor one and can be discarded. We set $\Theta_1(q) = 0$ when discussing faces. A twice continuously differentiable function on the relative interior is concave there exactly when its tangent Hessian is negative semidefinite. To see the sufficient direction, restrict to any line segment and integrate the nonpositive second derivative. Continuity then extends concavity to the closed simplex product. Conversely, a strictly positive second derivative at an interior point yields a strict midpoint violation on a sufficiently short segment.

29 The extremal power-sum product

The same symmetric power-sum problem controls both sides of $q = 1$. For $0 < \alpha < 1$, write

$$\Pi_\alpha(p) = S_{1-\alpha}(p)S_{1+\alpha}(p), \quad \Pi_k^*(\alpha) = \max_{p \in \Delta_k} \Pi_\alpha(p). \quad (29.1)$$

Cauchy–Schwarz applied to $p_j^{(1-\alpha)/2}$ and $p_j^{(1+\alpha)/2}$ gives $\Pi_\alpha \geq 1$. On the interior equality holds precisely at the uniform vector.

Theorem 29.1 (Exact optimizer and scalar reduction). *Let $k \geq 2$, $c = \log(k-1)$, and $0 < \alpha < 1$. There is a unique real $t > c$ satisfying*

$$\tanh\left(\frac{t-c}{2}\right) \tanh\left(\frac{\alpha t}{2}\right) = \alpha. \quad (29.2)$$

All maximizers in (29.1) are the k permutations of

$$p_{k,\alpha} = \left(x, \frac{1-x}{k-1}, \dots, \frac{1-x}{k-1}\right), \quad x = \frac{1}{1+e^{c-t}} > \frac{1}{2}. \quad (29.3)$$

The maximum is

$$\Pi_k^*(\alpha) = 1 + \frac{\cosh(\alpha t) - 1}{1 + \cosh(t - c)} = 1 + \frac{\sinh^2(\alpha t/2)}{\cosh^2((t - c)/2)}. \quad (29.4)$$

It is strictly increasing in the integer k . For fixed k it is real analytic in $\alpha \in (0, 1)$.

Proof. The maximum is interior. The objective is continuous on the compact simplex. Suppose a candidate has a zero coordinate, and denote its two sums by $A, B > 0$. Scale all its positive coordinates by $1 - \varepsilon$ and assign ε to one zero coordinate. The new product equals

$$(1 - \varepsilon)^2 AB + (1 - \varepsilon)^{1+\alpha} B \varepsilon^{1-\alpha} + (1 - \varepsilon)^{1-\alpha} A \varepsilon^{1+\alpha} + \varepsilon^2.$$

Relative to AB , the leading change is $B \varepsilon^{1-\alpha} + O(\varepsilon) > 0$ for small positive ε . No boundary point is a maximizer. A nonuniform interior vector has objective greater than one, so the uniform vector is not a maximizer either.

At most two coordinate values. At an interior stationary point, the Lagrange multiplier equation is

$$h(p_j) = \mu, \quad h(z) = (1 - \alpha)Bz^{-\alpha} + (1 + \alpha)Az^\alpha. \quad (29.5)$$

Multiplication by z^α turns this into a quadratic equation in z^α , so there are at most two positive coordinate values. If there are two, call them $a < b$. The function h strictly decreases and then strictly increases: its derivative changes sign exactly once. Therefore $h'(b) > 0$.

The larger value occurs once. If two coordinates equal b , transfer a small amount between those two coordinates while fixing the others. The first directional derivatives of both A and B vanish. The second derivative of their product in this direction is $2h'(b) > 0$, which is impossible at a maximum. Hence exactly one coordinate has the larger value, and all remaining coordinates are equal. This step eliminates all competing multiplicity patterns without a numerical comparison of their stationary values.

The scalar maximum is unique. Write the large-to-small ratio as e^t , so $t > 0$, and put $c = \log(k - 1)$. Direct multiplication yields

$$\Pi_\alpha(p) - 1 = \frac{\sinh^2(\alpha t/2)}{\cosh^2((t - c)/2)}.$$

This vanishes at $t = 0$ and tends to zero as $t \rightarrow \infty$. Its logarithmic derivative on $t > 0$ is

$$\alpha \coth(\alpha t/2) - \tanh((t - c)/2). \quad (29.6)$$

For $t \leq c$ this is positive. For $t > c$ the first term is strictly decreasing and the second strictly increasing. Their difference is positive at c (with the right-hand limit understood when $c = 0$) and tends to $\alpha - 1 < 0$. It has exactly one zero, equivalently (29.2). This proves all maximizing and uniqueness claims.

Embedding a maximizing k -vector as a boundary point of Δ_{k+1} and using the earlier perturbation proves strict increase in k . Finally, the derivative in t of (29.6) is strictly negative at the root, so the analytic implicit function theorem gives analytic dependence on α . The same argument permits a continuous real parameter $c \geq 0$ when asymptotic analysis requires it. $\square$

Remark 29.2 (Why two values are not by themselves enough). The Lagrange equations alone permit several coordinates at each of two values. The tangent second-variation argument is indispensable: without it, reducing immediately to one distinguished probability would leave a gap in the global optimization proof. Boundary exclusion is equally important because the tangent Hessian formulas later involve negative powers of probabilities.

30 Exact concavity below order one

Fix $0 < q < 1$, put $\alpha = 1 - q$, and abbreviate $f = S_q(p)$ and $\Pi = S_q(p)S_{2-q}(p) = \Pi_\alpha(p)$.

30.1 Evaluation of the categorical block quantity

On E_k define the positive definite bilinear form A and covector g by

$$A(v, w) = -\frac{D^2 f[v, w]}{f} = \frac{q\alpha}{f} \sum_j p_j^{q-2} v_j w_j, \quad (30.1)$$

$$g(v) = \frac{Df[v]}{f} = \frac{q}{f} \sum_j p_j^{q-1} v_j. \quad (30.2)$$

The notation $A^{-1}g$ means the unique tangent vector w satisfying $A(w, v) = g(v)$ for every tangent v .

Lemma 30.1 (Closed form for the block curvature). *With the preceding definitions, set $\omega = g(A^{-1}g)$ and $D = -D^2 \log f = A + g \otimes g$. Then*

$$(A^{-1}g)_j = \frac{1}{\alpha} \left(p_j - \frac{p_j^{2-q}}{S_{2-q}(p)} \right), \quad (30.3)$$

$$\omega = \frac{q}{\alpha} \left(1 - \frac{1}{\Pi} \right), \quad (30.4)$$

$$\tau_q(p) := g(D^{-1}g) = \frac{\omega}{1 + \omega} = \frac{q(\Pi - 1)}{\Pi - q}. \quad (30.5)$$

In particular, $\tau_q(p) \geq 0$, with equality only at the uniform vector.

Proof. The right side of (30.3) sums to zero. Substitution into (30.1) gives

$$A(A^{-1}g, v) = \frac{q}{f} \sum_j \left(p_j^{q-1} - \frac{1}{S_{2-q}(p)} \right) v_j = g(v),$$

because $\sum_j v_j = 0$. Evaluating g on that vector proves (30.4). If $w = A^{-1}g$, then $D(w, v) = (1 + \omega)g(v)$, whence $D^{-1}g = w/(1 + \omega)$. This proves (30.5). Positivity follows from $\Pi \geq 1$ and $q < 1$. $\square$

Definition 30.2 (The alphabet curvature profile). For $k \geq 2$ and $0 < q < 1$, let

$$\Theta_k(q) = \frac{q(\Pi_k^*(1 - q) - 1)}{\Pi_k^*(1 - q) - q}. \quad (30.6)$$

The map $\Pi \mapsto q(\Pi - 1)/(\Pi - q)$ has derivative $q(1 - q)/(\Pi - q)^2 > 0$. Thus Theorem 29.1 and Lemma 30.1 prove that $\Theta_k(q)$ is exactly the maximum of τ_q , attained only at the k vectors in (29.3). It is strictly increasing in k and real analytic in $q \in (0, 1)$.

30.2 The global theorem and the unstable direction

Theorem 30.3 (Complete mixed-alphabet subunit criterion). *Let $k_1, \dots, k_n \geq 2$ and $0 < q < 1$. Then T_q is jointly concave on $\prod_i \Delta_{k_i}$ if and only if*

$$\sum_{i=1}^n \Theta_{k_i}(q) \leq 1. \quad (30.7)$$

At an interior marginal tuple, put $\sigma = \sum_i \tau_q(p^{(i)})$. The tangent Hessian is negative definite if $\sigma < 1$, has a one-dimensional kernel and no positive eigenvalue if $\sigma = 1$, and has exactly one positive eigenvalue if $\sigma > 1$.

When $\sigma > 1$, an explicit positive-curvature direction is $v_i = D_i^{-1}g_i$, with D_i, g_i as in Lemma 30.1.

Proof. Let g be the concatenated covector and let D be block diagonal with blocks D_i . Differentiation of $\log F_q = \sum_i \log f_i$ gives

$$\frac{D^2 F_q}{F_q} = g \otimes g - D. \quad (30.8)$$

Choose orthonormal coordinates for each tangent space and conjugate by $D^{-1/2}$. The resulting matrix is $uu^\top - I$, where $\|u\|^2 = g(D^{-1}g) = \sigma$. It is $-I$ on $u^\perp$ and has the remaining eigenvalue $\sigma - 1$. Congruence preserves inertia, proving the local assertions. The direction $v = D^{-1}g$ satisfies

$$(g \otimes g - D)[v, v] = \sigma^2 - \sigma. \quad (30.9)$$

The positive factor $F_q/(1 - q)$ transfers these assertions to T_q .

If (30.7) holds, then $\sigma \leq 1$ everywhere. The Hessian criterion and continuity prove global concavity. If it fails, choose independently a maximizing marginal in each block. Then $\sigma > 1$ and (30.9) produces an interior midpoint violation. This also proves necessity without assuming a maximum along a restricted diagonal family. $\square$

Corollary 30.4 (Quantitative and equality cases). *If $\delta = 1 - \sum_i \Theta_{k_i}(q) > 0$, then on the interior*

$$D^2 T_q[v, v] \leq -q\delta \sum_i \|v_i\|_2^2. \quad (30.10)$$

Consequently, on the closed domain,

$$\begin{aligned} T_q((1-t)P + tQ) &\geq (1-t)T_q(P) + tT_q(Q) \\ &\quad + \frac{q\delta}{2} t(1-t) \|P - Q\|_2^2. \end{aligned} \quad (30.11)$$

If $\sum_i \Theta_{k_i}(q) = 1$, the function is strictly concave but not strongly concave. Its interior Hessian fails to be negative definite exactly at the $\prod_i k_i$ tuples of maximizing marginals, with kernel spanned by $(D_i^{-1}g_i)_i$ there.

Proof. Weighted Cauchy–Schwarz gives $g(v)^2 \leq \sigma D(v, v)$, so (30.8) is bounded above by $-\delta D(v, v)$. Since $D_i \succeq A_i$, $F_q/f_i \geq 1$, and $p_j^{q-2} \geq 1$, multiplication by $F_q/(1 - q)$ gives (30.10). Integrating along a segment proves (30.11), first in the interior and then on the boundary by continuity.

At equality in the global budget, $\sigma = 1$ requires equality in every block maximum. The optimizer theorem gives exactly the stated finite set. Every nonconstant interior segment has strictly negative second derivative except possibly at finitely many parameter values, proving strict concavity along it. More generally, a segment lies in the relative interior of the product of its minimal support faces. If this is a proper face, at least one alphabet size decreases. The strict increase of Θ_k in k , with $\Theta_1 = 0$, makes its restricted budget strictly smaller than one. The same segment argument applies on that face. A null Hessian direction at an interior maximizing tuple precludes any positive strong-concavity constant. $\square$

Corollary 30.5 (Quasiconcavity survives). *For every $0 < q < 1$ and every choice of alphabets, T_q is quasiconcave and H_q is strictly concave, whether or not (30.7) holds.*

Proof. Each $-D^2 \log f_i = D_i$ is positive definite, so $\log F_q$ is strictly concave. Its upper level sets are convex. The functions F_q and T_q are increasing transforms of $\log F_q$, which preserves upper level sets. Division by $1 - q > 0$ gives the Rényi assertion. Strict concavity on faces follows in the same way. $\square$

A failed Tsallis concavity test below one is therefore not a failure of quasiconcavity. Also, the block method in the proof is inherited from Part II (Theorem 22.1); its categorical evaluation in (30.5)–(30.6) is the step that resolves Q7 for these domains.

31 A sharp seventeen-versus-eighteen-symbol theorem

At $q = 1/2$, set $K = k - 1$ and let $y > 1$ be the square root of the large-to-small probability ratio. Then

$$\Pi(y) = \frac{(y + K)(y^3 + K)}{(y^2 + K)^2} = 1 + \frac{Ky(y - 1)^2}{(y^2 + K)^2}. \quad (31.1)$$

Differentiating shows that the optimizer is the unique root in $y > \max(1, \sqrt{K})$ of

$$y^3 - 3y^2 - 3Ky + K = 0. \quad (31.2)$$

Indeed the derivative of Π is

$$-\frac{K(y - 1)(y^3 - 3y^2 - 3Ky + K)}{(y^2 + K)^3},$$

and uniqueness also follows directly from Theorem 29.1. The curvature profile simplifies to

$$\Theta_k(1/2) = \frac{\Pi_k^*(1/2) - 1}{2\Pi_k^*(1/2) - 1}. \quad (31.3)$$

Theorem 31.1 (Three factors: exact alphabet threshold). *For three independent categorical variables, $T_{1/2}$ is strongly concave whenever all three alphabets have at most seventeen symbols. For a common alphabet of size k , it is concave if and only if $k \leq 17$. For every $k \geq 18$ and every $n \geq 3$, concavity fails on $(\Delta_k)^n$.*

Proof. By (31.3), three identical alphabets are concave exactly when $\Pi_k^*(1/2) \leq 2$. At $k = 17$, the following polynomial identity is an exact strict certificate:

$$\begin{aligned} (y^2 + 16)^2 - 16y(y - 1)^2 \\ = (y^2 - 8y - 1)^2 + 2(y - 8)^2 + 127 > 0. \end{aligned} \quad (31.4)$$

Thus $\Pi(y) < 2$ for every y , and its attained maximum is strictly below two. Hence $3\Theta_{17}(1/2) < 1$. Monotonicity in the alphabet size proves the assertion for unequal alphabets all bounded by seventeen, and Corollary 30.4 gives strong concavity.

For $k = 18$, the interior marginal

$$p_0 = \left(\frac{81}{98}, \underbrace{\frac{1}{98}, \dots, \frac{1}{98}}_{17 \text{ entries}} \right) \quad (31.5)$$

has $K = 17, y = 9$, and hence

$$\Pi(p_0) = \frac{4849}{2401}, \quad \tau_{1/2}(p_0) = \frac{2448}{7297}, \quad 3\tau_{1/2}(p_0) - 1 = \frac{47}{7297} > 0. \quad (31.6)$$

The local criterion already proves nonconcavity for three factors. Larger alphabets contain the same simplex as a face, and additional factors may be held deterministic. A concave function has concave restrictions to faces, so the failure persists. Alternatively, continuity of the strict midpoint violation below allows perturbation into the full interior for any larger finite system. $\square$

31.1 An entirely explicit finite midpoint violation

Proposition 31.2 (Rational endpoints and a certified gap). *For $|s| < 1$ let*

$$p(s) = \frac{1}{98}(81 + 17s, 1 - s, \dots, 1 - s) \in \text{int } \Delta_{18}, \quad U(s) = T_{1/2}(p(s), p(s), p(s)). \quad (31.7)$$

Then

$$U''(0) = \frac{10387\sqrt{2}}{83349} > 0. \quad (31.8)$$

At the rational displacement $h = 1/1000$, the finite midpoint deficit $\Delta = (U(h) + U(-h))/2 - U(0)$ satisfies

$$\frac{88119833628}{10^{18}} < \Delta < \frac{88119833629}{10^{18}}. \quad (31.9)$$

In particular, the endpoints and midpoint are rational marginal tuples, and their Tsallis values rigorously violate concavity.

Proof. Put

$$g(s) = \frac{\sqrt{81 + 17s} + 17\sqrt{1 - s}}{\sqrt{98}}.$$

Finite product factorization gives $U(s) = 2(g(s)^3 - 1)$. If $A(s) = \sqrt{81 + 17s} + 17\sqrt{1 - s}$, then

$$A(0) = 26, \quad A'(0) = -\frac{68}{9}, \quad A''(0) = -\frac{289}{2916} - \frac{17}{4}.$$

Substitution into $U'' = 6g(2(g')^2 + gg'')$ proves (31.8) by rational arithmetic.

For completeness, (31.9) is a finite exact-arithmetic certificate supplied by `code/verify_exact.py` [shipped here as `code/03-alphabet-entropy-verify_exact.py`]. Its enclosure algorithm is elementary. For a nonnegative rational number r , set $B = 10^{75}$ and

$$m = \left\lfloor \sqrt{\lceil rB^2 \rceil} \right\rfloor.$$

Then $m/B \leq \sqrt{r} < (m+1)/B$, as follows by squaring the endpoints. Apply this to $(81 + 17s)/98$ and $(1 - s)/98$ for $s = -h, 0, h$. Sum their intervals with weights 1, 17 to enclose $g(s)$. All endpoints are positive, so cubing preserves endpoint order. If $g_-(s), g_+(s)$ are the resulting lower and upper endpoints, the certified interval is

$$g_-(h)^3 + g_-(-h)^3 - 2g_-(0)^3 < \Delta < g_+(h)^3 + g_+(-h)^3 - 2g_+(0)^3,$$

with nonstrict endpoint enclosures also sufficient. The program checks by exact integer and rational comparison that this interval lies strictly inside (31.9). No decimal rounding or sign of a floating-point eigenvalue enters the certificate. $\square$

31.2 Other half-order dimensions

For $n \geq 2$, (31.3) gives the equivalent condition

$$(y^2 + K)^2 - (n - 2)Ky(y - 1)^2 \geq 0 \quad \text{at the maximizing root of (31.2)}. \quad (31.10)$$

Thus two factors are concave for every finite alphabet, but higher factor counts have finite thresholds. The table gives exact decisions; the root intervals underlying them are included in the certificate JSON [shipped as `data/03-alphabet-entropy-exact_certificates.json`].

The binary entry is the already known repository result. Its equality case is visible from

$$(y^2 + 1)^2 - 8y(y - 1)^2 = (y^2 - 4y + 1)^2.$$

The maximizing root $y = 2 + \sqrt{3}$ gives $\Pi_2^*(1/2) = 9/8$ and $\Theta_2(1/2) = 1/10$. This is a normalization check and recovery of an existing theorem, not a new ten-bit claim.

Alphabet k	2	3	4	5	7	8
Largest concave n	10	6	5	4	3	3
Alphabet k	10	16	17	18	32	64
Largest concave n	3	3	3	2	2	2

Table 5: Exact half-order dimension thresholds. Additional certified entries $k = 128, 256$ also have largest concave factor count two. Concavity here allows all marginal parameters, not only identical ones.

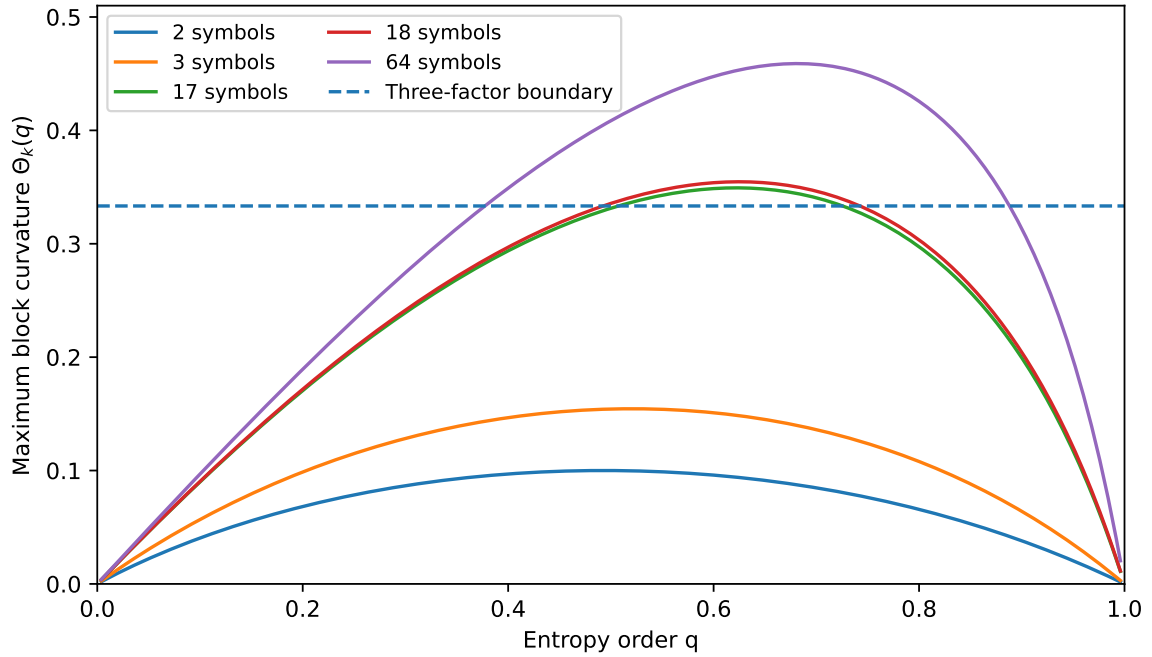


Figure 5: Numerical profiles for several alphabets. The horizontal line is the three-factor budget $1/3$. The seventeen/eighteen distinction at $q = 1/2$ is proved by Theorem 31.1, not by the plot. No global unimodality assertion is inferred from these curves.

32 Exact decisions at rational orders

Theorem 32.1 (Algebraic reduction). *Let $q = a/b \in (0, 1)$ be in lowest terms, where a, b are positive integers, and set $K = k - 1$. The maximum determining $\Theta_k(q)$ is represented by the unique real root $y > \max(1, K^{1/b})$ of*

$$ay^{2b-a} - (2b-a)y^b - (2b-a)Ky^{b-a} + aK = 0. \quad (32.1)$$

At that root,

$$\Pi_k^*(1-q) = \frac{(y^a + K)(y^{2b-a} + K)}{(y^b + K)^2}. \quad (32.2)$$

For rational q and finitely many integer alphabets, the concavity criterion (30.7), including its equality case, is decidable by exact real-algebraic computation.

Proof. Set $t = b \log y$ in (29.2). Then

$$\frac{y^b - Ky^{b-a} - 1}{y^b + Ky^{b-a} + 1} = \frac{b-a}{b}.$$

Every denominator is positive in the stated domain. Clearing them gives (32.1). The unique-root assertion follows from the unique $t > c$ proved in Theorem 29.1. The one-large-coordinate probabilities are $y^b/(y^b + K)$ and $1/(y^b + K)$, which prove (32.2).

Each profile value is therefore a specified real algebraic number. One may isolate the designated root of each integer polynomial, form the corresponding rational expression in a real algebraic number field, and determine the sign of their sum minus one. Standard exact polynomial operations and root isolation decide equality as well as strict signs. Equivalently, write a finite polynomial sign problem with isolated-root constraints and eliminate its real variables. This is a finite algorithmic assertion, not the assertion that progressively narrower strict intervals can always recognize zero. $\square$

The included exact verifier implements a smaller, transparent part of this program. At $q = 1/2$ it bisects the unique relevant cubic root using rational endpoints and evaluates (31.10) with outward rational interval arithmetic. It checks the largest admissible n and failure at $n + 1$ for fourteen alphabets. The binary equality case is handled by the exact square identity, not by a timeout. A general mixed-alphabet algebraic-number engine is *not* included or claimed implemented.

33 Large alphabets and alphabet-free concavity

33.1 A convergent expansion at fixed order

Theorem 33.1 (Large-alphabet profile). *Fix $0 < q < 1$ and let $\alpha = 1 - q$, $K = k - 1$. Define*

$$C_q = \left(\frac{2-q}{2}\right)^{2-q} \left(\frac{q}{2}\right)^q = \left(\frac{1+\alpha}{2}\right)^{1+\alpha} \left(\frac{1-\alpha}{2}\right)^{1-\alpha}. \quad (33.1)$$

As $k \rightarrow \infty$,

$$x_{k,1-q} \longrightarrow 1 - \frac{q}{2}, \quad (33.2)$$

$$\Pi_k^*(1-q) = C_q K^{1-q} + \frac{1 + (1-q)^2}{2} + O_q(K^{-(1-q)}), \quad (33.3)$$

$$\Theta_k(q) = q - \frac{q(1-q)}{C_q} K^{-(1-q)} + O_q(K^{-2(1-q)}). \quad (33.4)$$

The optimizer has a convergent power series in $\varepsilon = K^{-(1-q)}$ near zero, and Π_k^* a convergent Laurent series there. These expansions are locally uniform when q ranges over a compact subinterval of $(0, 1)$.

Proof. Write the maximizing t as $t = c + s$, where $c = \log K$, and put $u = e^s$. Clearing the hyperbolic-tangent denominators in (29.2) gives

$$(1 - \alpha)u - (1 + \alpha) - \varepsilon u^{-\alpha}((1 + \alpha)u - (1 - \alpha)) = 0. \quad (33.5)$$

At $\varepsilon = 0$ its solution is $u_0 = (1 + \alpha)/(1 - \alpha)$ and its derivative in u is $1 - \alpha > 0$. The real analytic implicit function theorem gives a convergent series $u(\varepsilon)$ near zero. This solution has $u > 1$; by uniqueness it is the desired maximizing solution for sufficiently large K . Compact parameter intervals allow a common small neighborhood by the parameter-dependent implicit function theorem and compactness. The large probability is $u/(1 + u)$, giving (33.2).

Using (29.4) before expanding yields

$$\Pi_k^* = 1 + \frac{\frac{1}{2}\varepsilon^{-1}e^{\alpha s} - 1 + \frac{1}{2}\varepsilon e^{-\alpha s}}{1 + \cosh s}. \quad (33.6)$$

The leading coefficient $L(s) = e^{\alpha s}/(2(1 + \cosh s))$ has derivative zero at $s_0 = \log((1 + \alpha)/(1 - \alpha))$ and value C_q . Since $s = s_0 + O(\varepsilon)$, its change contributes only $O(\varepsilon)$ after multiplication by ε^{-1} . The constant term in (33.6) is therefore

$$1 - \frac{1}{1 + \cosh s_0} = \frac{1 + \alpha^2}{2}.$$

This proves (33.3) and convergence of the Laurent expansion. Finally,

$$\Theta_k(q) = q - \frac{q(1 - q)}{\Pi_k^*(1 - q) - q},$$

which gives (33.4) and its asserted uniformity. $\square$

Theorem 33.2 (Sharp alphabet-independent range). *Fix a positive integer n and $0 < q < 1$. The product Tsallis entropy is concave for every finite choice of n alphabets if and only if*

$$nq \leq 1. \quad (33.7)$$

For every particular finite alphabet tuple satisfying (33.7), concavity is strong. If $nq > 1$, equal sufficiently large alphabets already fail. For each fixed q , the largest concave factor count at a common alphabet is eventually $\lfloor 1/q \rfloor$ as the alphabet grows.

Proof. The identity at the end of the preceding proof implies $0 < \Theta_k(q) < q$ for every finite k . Thus $\sum_i \Theta_{k_i}(q) < nq \leq 1$, and the strong-concavity theorem applies. Conversely (33.4) gives $\Theta_k(q) \rightarrow q$, so $n\Theta_k(q) > 1$ eventually whenever $nq > 1$. The last assertion follows by applying these two statements to $n = \lfloor 1/q \rfloor$ and $n + 1$; it also covers reciprocal integer orders. $\square$

The theorem displays a real difference from the binary nine-bit phenomenon. A factor-count statement valid uniformly over all orders for one fixed alphabet is not a statement uniform over all alphabets.

33.2 Where the first failing alphabet diverges

Corollary 33.3 (Critical alphabet near the alphabet-free endpoint). *Fix $n \geq 2$, let $q_0 = 1/n$, and let $k_{\text{fail}}(q, n)$ be the least integer common alphabet size for which $n\Theta_k(q) > 1$, for $q > q_0$ sufficiently close to q_0 . Then*

$$k_{\text{fail}}(q, n) - 1 \sim \left(\frac{q_0(1 - q_0)}{C_{q_0}(q - q_0)} \right)^{1/(1 - q_0)} \quad (q \downarrow q_0). \quad (33.8)$$

Proof. Strict increase in k makes the failure set an upper interval of integers. At each fixed k , $\Theta_k(q_0) < q_0$, so its first point escapes to infinity as $q \downarrow q_0$. By the locally uniform expansion (33.4), the sign change is governed by

$$q - q_0 = \frac{q(1 - q)}{C_q} K^{-(1 - q)} + O(K^{-2(1 - q)}).$$

For any fixed positive relative perturbation of the claimed scale, this formula has opposite strict signs on the two sides when $q - q_0$ is small. Replacing q by q_0 in the exponent causes relative error tending to zero because $(q - q_0)|\log(q - q_0)| \rightarrow 0$. Integer rounding is negligible compared with the diverging scale. These two-sided bounds prove (33.8). $\square$

33.3 Fixed-alphabet endpoints and the varentropy connection

Proposition 33.4 (The two fixed-alphabet limits). *For fixed $k \geq 2$,*

$$\lim_{q \downarrow 0} \frac{\Theta_k(q)}{q} = \frac{k - 1}{k}. \quad (33.9)$$

Set $c = \log(k - 1)$ and let $t_0 > c$ be the unique solution of

$$\tanh((t_0 - c)/2) = 2/t_0. \quad (33.10)$$

With $V_k = t_0^2/4 - 1$, as $\alpha \downarrow 0$,

$$\Pi_k^*(\alpha) = 1 + \alpha^2 V_k + O_k(\alpha^4), \quad (33.11)$$

$$\Theta_k(1 - \alpha) = \alpha V_k - \alpha^2 V_k(1 + V_k) + O_k(\alpha^3). \quad (33.12)$$

Here V_k is the largest possible variance of $\log p_j$ under the law p on k symbols.

Proof. For $0 < q < 1$, $S_q \leq k^{1 - q}$ and $S_{2 - q} \leq 1$, so $\limsup_{q \downarrow 0} \Pi_k^*(1 - q) \leq k$. For every fixed strictly positive p , the product tends to $kS_2(p)$. Taking full-support vectors arbitrarily close to a vertex shows the matching lower bound k . Substitution in (30.6) proves (33.9).

For the other endpoint, the critical equation can be written

$$\tanh((t - c)/2) = \frac{2}{t} \left(\frac{\alpha t}{2} \coth \frac{\alpha t}{2} \right).$$

The parenthesized expression extends analytically and evenly through $\alpha = 0$. At zero, the left side minus $2/t$ is strictly increasing for $t > c$, from a negative value to a positive one. Its derivative at the unique root is positive. The implicit function theorem therefore gives $t = t_0 + O_k(\alpha^2)$, analytic in α^2 near zero. Expansion of (29.4) gives

$$\frac{\alpha^2 t_0^2}{4 \cosh^2((t_0 - c)/2)} + O_k(\alpha^4) = \alpha^2(t_0^2/4 - 1) + O_k(\alpha^4),$$

using (33.10). Formula (33.12) follows by rational expansion of (30.6).

For every fixed interior p , expansion of the two power sums about order one gives $\Pi_\alpha(p) = 1 + \alpha^2 \text{Var}_p(\log p_j) + O_p(\alpha^4)$. The preceding expansion of the global maximum bounds every such variance by V_k . The limiting maximizing distribution in (29.3) attains V_k . Boundary laws follow by continuity of $p \log^m p$ for $m = 1, 2$ at zero. This proves the final interpretation. $\square$

The variance-of-surprisal extremal problem is an established subject; Reeb and Wolf give a sharp finite-dimensional bound in both classical and quantum settings [8]. The role of Proposition 33.4 here is to connect that endpoint to the exact product-entropy curvature and to prepare the nonuniform limit, not to claim the classical varentropy problem as newly solved.

34 The nonuniform Shannon window

The limits in Theorem 33.1 and Proposition 33.4 do not commute:

$$\lim_{q \uparrow 1} \lim_{k \rightarrow \infty} \Theta_k(q) = 1, \quad \lim_{k \rightarrow \infty} \lim_{q \uparrow 1} \Theta_k(q) = 0.$$

This suggests a joint limit rather than an attempt to use the fixed- q expansion uniformly up to one. The correct scale is $1 - q$ times the square of the logarithm of the alphabet size.

Theorem 34.1 (Uniform Shannon-window expansion). *Let $c = \log(k - 1) \rightarrow \infty$, put $q = 1 - \lambda/c^2$, and extend $\Theta_k(1) = 0$ and $\Pi_k^*(0) = 1$. Uniformly for λ in any fixed compact interval $[0, L]$,*

$$\Theta_k(1 - \lambda/c^2) = \frac{\lambda}{4 + \lambda} + \frac{16\lambda - 4\lambda^2 - \frac{2}{3}\lambda^3}{(\lambda + 4)^2 c^2} + O_L(c^{-4}). \quad (34.1)$$

For the optimizer $t = c + s$, with its continuous limiting selection at $\lambda = 0$,

$$s = \frac{4}{c} + \frac{\lambda^2 - 32}{3c^3} + O_L(c^{-5}), \quad (34.2)$$

$$\Pi_k^*(\lambda/c^2) - 1 = \frac{\lambda^2}{4c^2} + \frac{\lambda^2 + \lambda^4/48}{c^4} + O_L(\lambda^2 c^{-6}). \quad (34.3)$$

In fact these quantities, after their displayed removable factors are removed, have locally convergent expansions in c^{-2} .

Proof. Set $h = 1/c$ and seek $s = hz$. Dividing the critical equation in the form (29.6) by h gives

$$G(z, h, \lambda) := \frac{\tanh(hz/2)}{h} - \lambda h \coth\left(\frac{\lambda h(1 + h^2 z)}{2}\right) = 0. \quad (34.4)$$

Both apparent singularities are removable: $\tanh w/w$ and $w \coth w$ are analytic even functions near zero. Thus G is real analytic in h^2 , z , and λ near every point $(z, h, \lambda) = (4, 0, \lambda)$ with bounded λ . Its value at $h = 0$ is $z/2 - 2$, with derivative $1/2$ in z . The analytic implicit function theorem, made uniform on $0 \leq \lambda \leq L$ by compactness, gives an analytic solution $z = z(h^2, \lambda)$ near $h = 0$. It has $s > 0$ for small positive h . For $\lambda > 0$, uniqueness in Theorem 29.1 identifies it with the global optimizer. At zero it is the continuous varentropy selection.

Expanding (34.4) gives

$$z = 4 + \frac{\lambda^2 - 32}{3} h^2 + O_L(h^4),$$

proving (34.2). The exact maximum satisfies

$$r := \Pi_k^*(\lambda h^2) - 1 = \frac{\sinh^2(\lambda h(1 + h^2 z)/2)}{\cosh^2(hz/2)}. \quad (34.5)$$

Consequently

$$r = \lambda^2 h^2 \left(\frac{1}{4} + \left(1 + \frac{\lambda^2}{48} \right) h^2 + O_L(h^4) \right).$$

This proves the stronger factored error in (34.3) and is important for uniformity at $\lambda = 0$.

Write $r = \lambda^2 h^2 U(h^2, \lambda)$, where U is analytic and $U(0, \lambda) = 1/4$. Then the exact curvature formula becomes

$$\Theta_k(1 - \lambda h^2) = (1 - \lambda h^2) \frac{\lambda U(h^2, \lambda)}{1 + \lambda U(h^2, \lambda)}. \quad (34.6)$$

There is now no division by zero at $\lambda = 0$. Expanding this analytic expression through h^2 gives (34.1); its denominator is uniformly bounded away from zero for bounded nonnegative λ and small h . Analytic Taylor remainder bounds on a compact neighborhood give the stated uniform errors and convergence claims. This establishes an analytic expansion, rather than merely a formal substitution into a fixed-alphabet asymptotic series. $\square$

Corollary 34.2 (Local concavity boundary below Shannon order). *For fixed $n \geq 2$, define*

$$\lambda_n = \frac{4}{n-1}, \quad d_n = -4\lambda_n + \lambda_n^2 + \frac{\lambda_n^3}{6}.$$

For all sufficiently large common alphabets, the equation $n\Theta_k(q) = 1$ has a unique solution in a fixed small scaled neighborhood of $1 - \lambda_n/c^2$. It satisfies

$$q_{n,k}^- = 1 - \frac{\lambda_n}{c^2} - \frac{d_n}{c^4} + O_n(c^{-6}), \quad c = \log(k-1). \quad (34.7)$$

Within that neighborhood, $q > q_{n,k}^-$ gives concavity, whereas $q < q_{n,k}^-$ gives nonconcavity. For example,

$$q_{3,k}^- = 1 - \frac{2}{c^2} + \frac{8}{3c^4} + O(c^{-6}), \quad (34.8)$$

$$q_{2,k}^- = 1 - \frac{4}{c^2} - \frac{32}{3c^4} + O(c^{-6}). \quad (34.9)$$

Proof. The limit profile $f(\lambda) = \lambda/(4 + \lambda)$ satisfies $nf(\lambda_n) = 1$ and $f'(\lambda_n) = 4/(4 + \lambda_n)^2 > 0$. Apply the analytic implicit function theorem to $n\Theta_k(1 - \lambda h^2) - 1$ using (34.6). The solution has $\lambda(h) = \lambda_n - C(\lambda_n)h^2/f'(\lambda_n) + O(h^4)$, where C is the coefficient of c^{-2} in (34.1). Its second coefficient simplifies to d_n . Substituting $q = 1 - \lambda(h)h^2$ proves (34.7). The positive derivative in λ persists locally and reverses sign when expressed in q , proving the final orientation of the boundary. $\square$

Remark 34.3 (Local does not mean globally unique). Corollary 34.2 asserts uniqueness in the specified Shannon scaling neighborhood. It does not prove global unimodality of $\Theta_k(q)$ on $(0, 1)$, nor that all nonconcave orders form one interval. Those are separate questions. In particular, both endpoints $q = 0$ and $q = 1$ are concave neighborhoods for every fixed finite system, but this alone does not control the number of intervening transitions.

Corollary 34.4 (Unequal large alphabets). *Suppose n is fixed, $c \rightarrow \infty$, and $\log(k_i - 1)/c \rightarrow a_i \in (0, \infty)$. At $q = 1 - \lambda/c^2$,*

$$\sum_i \Theta_{k_i}(q) \rightarrow \sum_i \frac{\lambda a_i^2}{4 + \lambda a_i^2}. \quad (34.10)$$

A limiting sum strictly below one implies eventual concavity, and a limiting sum strictly above one implies eventual nonconcavity. Equality requires next-order information and is not decided by the limit alone.

Proof. Apply Theorem 34.1 in block i with $\lambda_i = (1 - q) \log^2(k_i - 1) \rightarrow \lambda a_i^2$ and sum the finitely many limits. Strict inequalities persist and Theorem 30.3 applies. $\square$

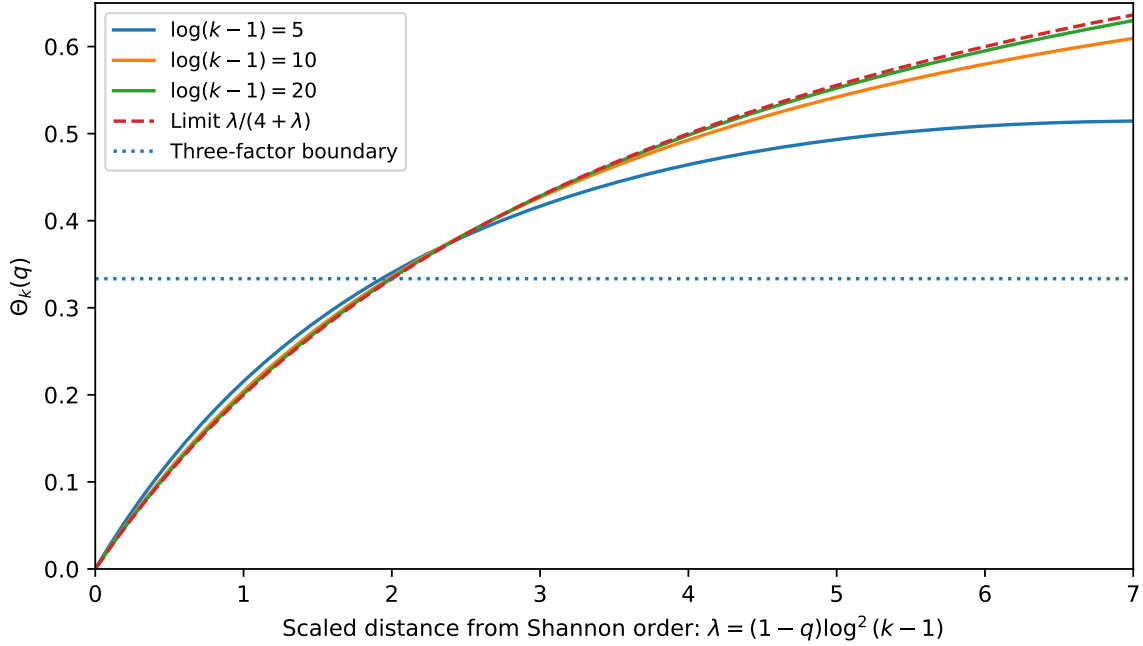


Figure 6: The Shannon-window limit and numerical finite- c profiles. For visualization, c is allowed to be continuous through the scalar optimizer equation; integer alphabets correspond to $c = \log(k - 1)$. The rigorous uniform expansion is Theorem 34.1.

35 The complete criterion above order one

Below one, concavity of T_q means concavity of F_q . Above one, $1 - q < 0$, so it means *convexity* of F_q . Treating these as the same matrix problem would give an incorrect extension of the subunit theorem.

35.1 The local compression matrix

Fix $1 < q < 2$ and let $\alpha = q - 1$. On a single tangent space define

$$A = \frac{D^2 f}{f} > 0, \quad g = \frac{Df}{f}, \quad \kappa_q(p) = g(A^{-1}g), \quad f = S_q(p).$$

The same tangent calculation as Lemma 30.1, now with $\alpha = q - 1$, gives

$$\kappa_q(p) = \frac{q}{q-1} \left(1 - \frac{1}{S_q(p)S_{2-q}(p)} \right). \quad (35.1)$$

Because the power-sum product is symmetric about order one, its maximizers are still those of Theorem 29.1. Define

$$\Gamma_k(q) = \frac{q}{q-1} \left(1 - \frac{1}{\Pi_k^*(q-1)} \right). \quad (35.2)$$

Lemma 35.1 (One scalar direction per block). *At an interior marginal tuple, convexity of F_q is equivalent to positive semidefiniteness of the $n \times n$ matrix*

$$M(\kappa)_{ii} = 1, \quad M(\kappa)_{ij} = \sqrt{\kappa_i \kappa_j} \quad (i \neq j), \quad \kappa_i = \kappa_q(p^{(i)}). \quad (35.3)$$

All remaining normalized tangent directions have positive eigenvalue one. A zero κ_i simply contributes an uncoupled identity direction.

Proof. The diagonal Hessian blocks of F_q/F_q are A_i and the off-diagonal blocks are $g_i \otimes g_j$. Normalize by the positive block forms A_i . Each normalized block gradient is a vector u_i of squared norm κ_i . The Hessian is identity on the direct sum of their orthogonal complements. On the space spanned by one unit gradient direction in each block, its entries are precisely (35.3). If $u_i = 0$, choose any unit tangent vector in that block; it is uncoupled and the same statement holds. Congruence preserves positive semidefiniteness. $\square$

Theorem 35.2 (Exact mixed-alphabet superunit criterion). *Let $1 < q < 2$ and $k_1, \dots, k_n \geq 2$. Write $\gamma_i = \Gamma_{k_i}(q)$. Then T_q is concave on $\prod_i \Delta_{k_i}$ if and only if*

$$M(\gamma) \succeq 0. \quad (35.4)$$

Equivalently, one of the following holds:

- (i) every $\gamma_i \leq 1$;
- (ii) exactly one $\gamma_j > 1$, every other $\gamma_i < 1$, and

$$(\gamma_j - 1) \sum_{i \neq j} \frac{\gamma_i}{1 - \gamma_i} \leq 1. \quad (35.5)$$

The empty sum in (ii) is zero, so the theorem also covers a single factor. For two factors the criterion is $\gamma_1 \gamma_2 \leq 1$. For $n \geq 2$ equal alphabets, it reduces to

$$\Gamma_k(q) \leq 1 \iff \Pi_k^*(q - 1) \leq q, \quad (35.6)$$

independently of the number of factors.

Proof. For any interior tuple, $0 \leq \kappa_i \leq \gamma_i$, and all $\gamma_i > 0$. Let C be diagonal with entries $\sqrt{\kappa_i/\gamma_i}$. Direct entrywise calculation gives

$$M(\kappa) = CM(\gamma)C + I - C^2. \quad (35.7)$$

Thus (35.4) implies positive semidefiniteness at every tuple. The local compression lemma, the sign $1 - q < 0$, and continuity prove global Tsallis concavity. Conversely, the block maxima are attained at interior marginals; selecting all of them realizes $M(\gamma)$. If it has a negative direction, F_q has negative curvature and T_q has positive curvature there. This proves necessity.

For the scalar classification write

$$M(\gamma) = \text{diag}(1 - \gamma_i) + vv^\top, \quad v_i = \sqrt{\gamma_i}.$$

If all $\gamma_i \leq 1$, both terms are positive semidefinite. If at least two exceed one, a vector supported on those two coordinates and orthogonal to v makes the quadratic form strictly negative. If exactly one, γ_j , exceeds one and another equals one, the corresponding 2×2 principal determinant is $1 - \gamma_j < 0$.

It remains to suppose $\gamma_j > 1$ and all the others are below one. Let $D_0 = \text{diag}(1 - \gamma_i)_{i \neq j} > 0$ and $s = \sum_{i \neq j} \gamma_i/(1 - \gamma_i)$. The remaining principal block is $D_0 + v_0 v_0^\top$. Its inverse satisfies $v_0^\top (D_0 + v_0 v_0^\top)^{-1} v_0 = s/(1 + s)$, by direct rank-one inversion. Its Schur complement in M is $1 - \gamma_j s/(1 + s)$, nonnegative precisely when $(\gamma_j - 1)s \leq 1$. This proves (ii). For two factors the determinant condition gives the stated product. For equal alphabets the eigenvalues of M are $1 - \Gamma_k$ with multiplicity $n - 1$ and $1 + (n - 1)\Gamma_k$ once. This proves (35.6). $\square$

This theorem has a useful asymmetric feature: one block may have a superunit quantity exceeding one without destroying product Tsallis concavity, provided the other blocks satisfy the compensation inequality (35.5). Replacing the matrix criterion by a requirement that every block separately have a concave logarithm would therefore be sufficient but not necessary for unequal alphabets.

35.2 Order two, orders above two, and Rényi entropy

Proposition 35.3 (The remaining positive orders). *Discard all size-one alphabets and let n count the remaining factors.*

- (i) *If $n = 1$, T_q is concave for every $q > 0$.*
- (ii) *At $q = 1$, $T_1 = H_1$ is concave for all finite alphabet tuples.*
- (iii) *At $q = 2$ and $n \geq 2$, T_2 is concave if and only if every alphabet is binary.*
- (iv) *At $q > 2$ and $n \geq 2$, T_q is not concave. In fact it fails quasiconcavity already on two binary faces.*

For product Rényi entropy, all finite alphabet tuples are concave at $0 < q \leq 1$. At $1 < q < 2$, the exact criterion is $\Gamma_{k_i}(q) \leq 1$ for every block. At $q = 2$ it is that every nontrivial alphabet be binary. For $q > 2$, one nontrivial alphabet already precludes concavity.

Proof. For one factor, S_q has negative semidefinite Hessian below one and positive semidefinite Hessian above one. The denominator $1 - q$ changes sign accordingly. The $q = 1$ claim follows by differentiating $-\sum p_j \log p_j$, or by taking the continuous limit of its standard concavity inequality.

At $q = 2$, the tangent calculation for (35.1) gives

$$\kappa_2(p) = 2 \left(1 - \frac{1}{k S_2(p)} \right), \quad \sup_{\text{int } \Delta_k} \kappa_2 = 2(1 - 1/k).$$

The supremum is approached at a vertex. For binary factors it is one, so (35.7) with all $\gamma_i = 1$ proves sufficiency. If one alphabet has size greater than two and there is any other nontrivial block, their supremal quantities have product greater than one. Interior parameters sufficiently close to vertices make the 2×2 compression determinant negative. This proves necessity.

For $q > 2$, restrict two blocks to binary faces and write $f(p) = p^q + (1 - p)^q$. Along $(p + s, p - s)$ the product has second derivative $2(ff'' - (f')^2)$ at $s = 0$. As $p \downarrow 0$,

$$ff'' - (f')^2 \longrightarrow q(q - 1) - q^2 = -q < 0.$$

For sufficiently small positive p , T_q has positive second derivative. Its values at s and $-s$ are equal by symmetry and strictly larger than its midpoint value for small nonzero s , proving failure of quasiconcavity. Hold other blocks fixed at vertices.

Finally $H_q = (1/(1 - q)) \sum_i \log f_i$, so its Hessian is block diagonal. Above one its concavity is equivalent to convexity of each $\log f_i$. For $1 < q < 2$, this means $A_i - g_i \otimes g_i \succeq 0$, or $\kappa_i \leq 1$ everywhere, giving the criterion involving Γ . The $q = 2$ supremum gives the binary criterion. The same binary limit $ff'' - (f')^2 \rightarrow -q$ proves failure for $q > 2$, even in one block. $\square$

The binary specializations agree with the repository's previously established classifications (Part II, Theorems 19.2 and 19.3). The new content here is the explicit higher-alphabet and mixed-alphabet evaluation, not the binary endpoint statements.

35.3 The upper Shannon boundary

Corollary 35.4 (A local boundary above order one). *For $c = \log(k - 1) \rightarrow \infty$ and bounded nonnegative λ ,*

$$\Gamma_k(1 + \lambda/c^2) = \frac{\lambda}{4} + \frac{\lambda + \lambda^2/4 - \lambda^3/24}{c^2} + O_L(c^{-4}), \quad (35.8)$$

with the continuous extension at $\lambda = 0$. For every fixed $n \geq 2$ equal alphabets, the unique local concavity boundary above one is

$$q_k^+ = 1 + \frac{4}{c^2} - \frac{64}{3c^4} + O(c^{-6}). \quad (35.9)$$

The boundary is independent of n : below it locally T_q is concave, and above it locally it is not.

Proof. Use the same $r = \Pi_k^*(\lambda/c^2) - 1$ as in (34.3), now in

$$\Gamma_k(1 + \lambda/c^2) = \frac{1 + \lambda/c^2}{\lambda/c^2} \frac{r}{1 + r}.$$

Its analytic removable extension gives (35.8). The equal-alphabet criterion is $\Gamma_k \leq 1$. The limiting root is $\lambda = 4$, with derivative $1/4$. The implicit function theorem gives $\lambda(c) = 4 - 64/(3c^2) + O(c^{-4})$, proving (35.9) and the local orientation. $\square$

Together Corollaries 34.2 and 35.4 quantify how a finite-alphabet neighborhood of Shannon concavity narrows on both sides as the alphabet grows. They do not assert a global interval theorem at all orders.

36 Quantum product-state parameters have the same thresholds

Let $\mathcal{D}_k$ denote the compact convex set of $k \times k$ positive semidefinite Hermitian matrices of trace one. For density matrices $\rho_i \in \mathcal{D}_{k_i}$, define

$$\mathcal{T}_q(\rho_1, \dots, \rho_n) = \frac{\prod_i \text{Tr}(\rho_i^q) - 1}{1 - q}, \quad q \neq 1. \quad (36.1)$$

Eigenvalues of a tensor product are products of eigenvalues, so this is the Tsallis entropy of $\rho_1 \otimes \dots \otimes \rho_n$. The parameter domain is $\prod_i \mathcal{D}_{k_i}$; interpolation is affine in the individual matrices. It is not a claim about a new entanglement threshold or arbitrary mixtures in the full joint state space.

Theorem 36.1 (Classical–quantum equivalence for this parameterization). *For every $q > 0$ and finite alphabet dimensions $k_1, \dots, k_n$, the quantum product entropy (36.1), with its Shannon extension at one, is concave on $\prod_i \mathcal{D}_{k_i}$ if and only if the classical product entropy is concave on $\prod_i \Delta_{k_i}$. Thus Theorems 30.3 and 35.2 and Proposition 35.3 give its exact classification. The same equivalence holds for product Rényi entropy.*

Below one, the maximal block curvature occurs exactly at states whose spectra have the one-large-eigenvalue form (29.3). Noncommuting tangent directions cannot create an additional instability.

Proof. Necessity follows by restricting every matrix to be diagonal in a fixed basis. For sufficiency, work first at positive definite states. At any such state choose a unitary eigenbasis and write its eigenvalues as $p_1, \dots, p_k > 0$. A Hermitian trace-zero tangent matrix H decomposes into its real diagonal part and its off-diagonal part. For $f(\rho) = \text{Tr} \rho^q$,

$$Df[H] = q \sum_j p_j^{q-1} H_{jj}, \quad (36.2)$$

$$\begin{aligned} D^2 f[H, H] &= q(q-1) \sum_j p_j^{q-2} H_{jj}^2 \\ &\quad + 2q \sum_{j < \ell} \frac{p_j^{q-1} - p_\ell^{q-1}}{p_j - p_\ell} |H_{j\ell}|^2. \end{aligned} \quad (36.3)$$

For distinct eigenvalues these identities follow by differentiating the eigenvalue equation: $p'_j = H_{jj}$ and $p''_j = 2 \sum_{\ell \neq j} |H_{j\ell}|^2 / (p_j - p_\ell)$, and then summing $qp_j^{q-1} p''_j + q(q-1)p_j^{q-2} (p'_j)^2$. For repeated eigenvalues, take limits; the divided difference extends to $(q-1)p_j^{q-2}$. This also follows by approximating the positive definite state by ones with simple spectrum, since the power function is smooth on a compact positive spectral interval.

The first derivative in every off-diagonal direction vanishes. Thus all cross-block product-Hessian terms involve only the diagonal parts. The diagonal contribution to the entropy Hessian is exactly the classical tangent Hessian. The divided difference in (36.3) has the sign of $q - 1$. After multiplying by the product of the other positive factors and dividing by $1 - q$, its contribution to the Tsallis Hessian is strictly negative in every nonzero off-diagonal direction. It can only improve concavity. For Rényi entropy the additional gradient-square term is also purely diagonal, and the same sign argument applies. Hence the classical criterion is sufficient at every positive definite tuple.

At $q = 1$, apply the identical divided-difference argument to $-\text{Tr } \rho \log \rho$; its diagonal second derivatives and off-diagonal divided differences are negative. Continuity on the compact density matrix sets extends all concavity inequalities from positive definite states to the boundary.

For the last claim, below one the positive form $-D^2 \log f$ splits into the classical diagonal form and a positive off-diagonal form, whereas its gradient is diagonal. Its inverse applied to that gradient has no off-diagonal component. The block curvature is therefore exactly the classical $\tau_q(p)$, and Theorem 29.1 identifies its maximum. $\square$

The eighteen-symbol witness from Proposition 31.2 immediately gives a commuting density-matrix witness in dimension eighteen. Conversely, three quantum systems of dimensions at most seventeen satisfy the same half-order concavity theorem, even for noncommuting interpolation directions. The classical extremizers lift to unitary orbits; unlike the classical permutation set, this quantum extremal set is not finite.

37 Verification, implementation, and a formalization route

37.1 What the included checks establish

The proofs of the optimizer, global criteria, analytic expansions, and quantum equivalence are mathematical arguments in this article. The programs are independent implementation checks and exact finite certificates; they are not substitutes for those arguments.

The supplied package contains three verification layers.

Layer	Completed	Scope
Exact rational/integer	70	Sum-of-squares identities, rational curvature witness, finite Jensen enclosure, and fourteen certified half-order alphabet/dimension entries.
Symbolic	7	Half-order derivative, rational-witness second derivative, and Shannon-window coefficients.
Numerical	4,595	Direct tangent Hessians, curvature contractions, inertia tests, 2,880 random extremal comparisons, and twelve quantum finite-difference checks.

All three layers completed successfully in preparing this manuscript. The exact layer uses only Python's standard library. The symbolic layer uses SymPy; the numerical layer uses NumPy and mpmath with a recorded random seed. The maximum observed absolute error in the independently computed block contraction was approximately $3.6 \cdot 10^{-15}$; the maximum Hessian-identity entry error was approximately $2.9 \cdot 10^{-14}$. These are numerical diagnostics, not uniform error bounds for all inputs.

The exact script outputs both an aggregate status and the rational isolating intervals used in its half-order decisions. The midpoint certificate is based on outward square-root intervals and rational arithmetic only. The algebraic equality at the binary ten-factor threshold is treated by

a polynomial square identity. The scripts do not misinterpret a small decimal or a long-running interval calculation as an exact zero.

The numerical Shannon-window table records scaled remainders for several values of λ and c . It checks the derived expressions but does not supply the uniform analytic remainder theorem; that theorem follows from (34.4)–(34.6). The figures are illustrations made from deterministic scalar-root computations.

37.2 A reproducible package

From the archive root, the principal commands are [of the delivered package, whose layout this report does not keep]

```
python code/verify_exact.py
python code/verify_symbolic.py
python code/verify_numerical.py
python code/make_figures.py
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
```

[Do not run these commands in this report's directory: there code/verify_exact.py, code/verify_symbolic.py and code/make_figures.py are Part I's scripts, code/verify_numerical.py does not exist, and the last command builds the whole report. The shipped Makefile code/03-alphabet-entropy-Makefile names the same delivered paths and must not be used here either. A safe rerun restores the delivered names in a scratch copy:]

```
W=/path/to/scratch
mkdir -p "$W/03/code" "$W/03/verification" "$W/03/data" "$W/03/figures"
for f in code/03-alphabet-entropy-*.py; do
  cp "$f" "$W/03/code/${f#code/03-alphabet-entropy-}"
done
cd "$W/03"
python code/verify_exact.py
python code/verify_symbolic.py
python code/verify_numerical.py
python code/make_figures.py
```

The package contains the compiled PDF, editable source, figures and figure data, exact and numerical logs, a claim ledger, bibliography and repository provenance notes, and a build file. No repository mutation or upload is required to reproduce these checks. No Lean installation or external network access is needed for the exact certificate script.

[Write, 29 September 2026, batch 57: where the delivered package's files are in this report.

- Compiled PDF and source: the report's own `article.pdf` and `article.tex`.
- Figures, as delivered: `figures/03-alphabet-entropy-alphabet_profiles.pdf`, `figures/03-alphabet-entropy-shannon_window.pdf`; figure data: `data/03-alphabet-entropy-alphabet_profiles.csv`.
- The delivered `verification/` records and `requirements.txt`: `data/03-alphabet-entropy-*` (ten files with the figure data); its layout check: `03-alphabet-entropy-LAYOUT_QA.md`.
- Claim ledger and source notes: `03-alphabet-entropy-CLAIM_LEDGER.md`, `03-alphabet-entropy-SOURCES.md`; build file: `code/03-alphabet-entropy-Makefile`.
- Not shipped: the delivered PDF, README and checksum file.

The scripts write `verification/exact_certificates.json`, `verification/numerical_checks.json`, `verification/symbolic_checks.txt`, `data/alphabet_profiles.csv` and two PDFs in `figures/`, all below their parent directory, which in the scratch copy above is `$W/03`; compare them with the shipped `data/03-alphabet-entropy-` files. On Windows they write CRLF line endings. The rerun made for this write is recorded in Section 26.4.]

37.3 Suggested Lean decomposition

The most efficient initial target is the half-order seventeen/eighteen result, rather than the entire analytic classification at once.

Finite algebra and probability. Define the product mass function on finite types and prove the power-sum factorization. Prove the seventeen-symbol polynomial identity with a ring-normalization tactic. Check the rational values in (31.6) and the normalized marginal line in (31.7). This layer does not require transcendental asymptotics.

Tangent linear algebra. Represent the simplex tangent as the kernel of coordinate summation. Prove (30.3) by direct substitution, then the rank-one contraction formula and the Cauchy–Schwarz concavity criterion. The core equivalence can be proved without a spectral theorem; the exact one-positive-eigenvalue statement can be added later. For the above-one theorem, the finite compression matrix and identity (35.7) isolate the algebraic core.

Global optimization. Formalize boundary improvement, Lagrange multipliers on the simplex interior, and the two-equal-large-coordinate second variation. The last argument is the critical logical step that justifies the one-large-coordinate reduction. The unique-root theorem then needs only elementary monotonicity of hyperbolic functions.

Certified computation. A verified integer square-root checker would turn the finite Jensen enclosure into a kernel-checked witness. A verified rational interval evaluator and a uniqueness theorem for the cubic would certify the half-order table. General rational-order algebraic equality requires additional root-isolation and sign machinery; it is not delivered by the current small script.

Analytic and quantum extensions. Only after these core results should one formalize the parameter-dependent analytic implicit function theorem, uniform Taylor bounds, and spectral divided differences. The source file contains ordinary proofs but no claimed Lean code or kernel certificate. A successful Python run is not a proof-assistant bridge from code execution to a real-analytic theorem.

38 Further research questions

The categorical-simplex instance of repository Q7 is resolved above. The following continuations are not claimed solved. [Write, 29 September 2026, batch 57: “repository Q7” is Part II’s question Q7 (Section 23). The manuscript numbers its own questions Q1–Q9; they are printed as AQ1–AQ9 so that they cannot be confused with Part II’s Q1–Q10.]

AQ1. Global shape and the number of concavity transitions. Is $\Theta_k(q)$ strictly unimodal on $(0, 1)$ for every k ? More generally, how many crossings of one can $\sum_i \Theta_{k_i}(q)$ have? The exact optimizer allows implicit differentiation, and the present endpoint and Shannon-window expansions fix several signs, but neither fact rules out additional interior critical points. A proof would convert the present pointwise criterion into a global interval classification.

AQ2. Equality orders and exact exceptional systems. Classify rational orders and integer alphabet tuples with $\sum_i \Theta_{k_i}(q) = 1$. At half order, the cubic and the polynomial in (31.10) make each equal-alphabet instance elementary real algebra, but a uniform classification of rational orders is a different problem. Useful first targets are all equal-alphabet equality

cases at $q = 1/2, 1/3, 2/3$ and small factor counts, with compact exact certificates rather than decimal lists.

- AQ3. Optimal stability of the extremal distribution.** Find sharp constants and exponents relating $\Pi_k^*(\alpha) - \Pi_\alpha(p)$ to the distance from p to the maximizing permutation orbit. How do they degenerate as $k \rightarrow \infty$ or $\alpha \rightarrow 0, 1$? Such bounds would yield robust near-extremizer theorems and stronger certified approximations than a scalar root value alone. The strong-concavity constant in (30.10) is useful but not asserted optimal.
- AQ4. Affine constraints and conserved means.** Determine the exact criterion when each marginal or the combined system is restricted by prescribed linear statistics. The tangent inverse must then be projected onto a smaller constraint kernel. The unrestricted one-large-coordinate optimizer need not survive, because the Lagrange multiplier equation acquires coordinate-dependent terms. A promising first problem is one additional mean constraint with two distinct statistic values, followed by general prescribed spectra or costs.
- AQ5. Coarsenings of independent joint laws.** Which partitions of the product alphabet preserve subunit power-sum concavity in the marginal parameters? Singleton cells give the present problem; sums and other statistics merge cells and lose product factorization. The ordinary-sum claim in [5] concerns a separate special structure, not a general answer. A useful classification should specify precisely which merging patterns remove or retain the rank-one obstruction.
- AQ6. The small-order, large-alphabet transition.** The endpoint law (33.9) is for fixed k , whereas (33.4) is for fixed q . Identify uniform joint asymptotics as $q \downarrow 0$ and $k \rightarrow \infty$, including possible scales involving $q \log k$. This would complement the Shannon window and quantify the first failing alphabet when both the order and the factor count vary.
- AQ7. Infinite alphabets and energy-constrained quantum states.** The finite-alphabet theorem suggests the bound $nq \leq 1$ as a starting point, but infinite sums may diverge and entropy need not be continuous in a weak topology. Formulate the right integrability or energy conditions, then determine which finite-dimensional arguments pass to limits. No unrestricted infinite-alphabet continuity assertion is implied by the finite result.
- AQ8. Geometry above one and structured blocks.** For unequal alphabets, understand the order dependence of the Schur condition (35.5) and classify its equality loci. Extend the block evaluation from full simplices to exponential families, grouped categories, or constrained density matrices. The abstract product criterion alone does not compute those new suprema; this is the part of the broader block question that remains.
- AQ9. A small verified decision engine.** Implement the rational-order root isolation and equality branch of Theorem 32.1 with checkable certificates, then formalize the optimizer and finite-dimensional curvature theorem in Lean. A useful milestone would be a kernel-checked seventeen-symbol inequality, eighteen-symbol rational midpoint witness, and exact decision for every half-order entry in Table 5.

39 Conclusion and claim boundary

The higher-alphabet question becomes tractable after a specific variational reduction. The tangent inverse of a categorical power sum is explicit; its curvature depends only on $S_q S_{2-q}$; the global maximum of that product has exactly one large coordinate; and the remaining optimization has one monotone scalar critical equation. These facts produce an exact classification rather than a collection of sufficient estimates.

The sharp seventeen/eighteen-symbol transition is a finite consequence with rational failure data and a polynomial positivity certificate. The large-alphabet theorem and the uniform Shannon window then explain how this finite-dimensional geometry changes when the alphabet itself diverges. Above order one the correct object is a compression matrix, not the below-one additive budget. The spectral calculation shows that noncommuting quantum directions add no further obstruction for this product-state parameterization.

The categorical-simplex portion of the repository’s Q7 is therefore answered, with further quantitative and quantum extensions. The article does not claim global profile unimodality, a classification of arbitrary coarsenings or constraints, a proof of the ordinary-sum Shepp–Olkin problem, a fully implemented general algebraic decision engine, or a new Lean formalization. Those boundaries are mathematical distinctions, not qualifications on the exact criteria proved here.

40 Auditable coefficients and implementation details

40.1 The Shannon-window coefficients before division

The implicit equation (34.4) has the expansion

$$G = \frac{z}{2} - 2 + h^2 \left(-\frac{z^3}{24} + 2z - \frac{\lambda^2}{6} \right) + O(h^4).$$

Writing $z = 4 + z_2 h^2 + O(h^4)$ gives $z_2 = (\lambda^2 - 32)/3$. Substituting into (34.5), stationarity eliminates any dependence of the following coefficient on the next unknown optimizer term:

$$\begin{aligned} r = \frac{\lambda^2}{4} h^2 + \left(\lambda^2 + \frac{\lambda^4}{48} \right) h^4 \\ + \left(\frac{\lambda^6}{1440} + \frac{\lambda^4}{4} - \frac{4\lambda^2}{3} \right) h^6 + O_L(\lambda^2 h^8). \end{aligned} \quad (40.1)$$

These coefficients are checked symbolically in the supplied script. Their analytic validity and uniform remainder follow from the implicit function argument, not from symbolic series operations alone.

For the lower boundary, division of the profile correction by its leading derivative gives

$$-\frac{C(\lambda)}{f'(\lambda)} = -4\lambda + \lambda^2 + \frac{\lambda^3}{6}, \quad f(\lambda) = \frac{\lambda}{4 + \lambda}.$$

For the upper boundary, the correction at $\lambda = 4$ in (35.8) is $16/3$, and the derivative of the leading term is $1/4$, giving the coefficient $-64/3$.

40.2 Exact half-order root isolation

For $K \geq 1$ let $P_K(y) = y^3 - 3y^2 - 3Ky + K$. It is negative at $y = 1$ and positive at $y = 2K + 10$. Its derivative is $3((y - 1)^2 - K - 1)$, so it first decreases and then increases on $y > 1$, and has exactly one root there. This root is greater than $\sqrt{K}$: on $1 \leq y \leq \sqrt{K}$ the polynomial is decreasing from a negative value when that interval is nonempty. Thus ordinary rational bisection between 1 and $2K + 10$ isolates exactly the optimizer root.

The verifier performs 300 such bisections, then checks both the sign bracket and the domain inequalities. To decide a strict threshold it evaluates the quartic (31.10) using interval Horner arithmetic with rational endpoints. A positive lower endpoint proves concavity; a negative upper endpoint proves failure. A separate square identity covers the binary equality. The exact intervals are saved in `verification/exact_certificates.json` [shipped as `data/03-alphabet-entropy-exact_certificates.json`], so their precision and root choice can be independently inspected.

40.3 Why the large-alphabet constant term is stable

In (33.6), write the expression as $\varepsilon^{-1}L(s) + B(s) + \varepsilon C(s)$. The optimizer satisfies $s = s_0 + s_1\varepsilon + \cdots$ and $L'(s_0) = 0$. Hence

$$\varepsilon^{-1}L(s) = \varepsilon^{-1}L(s_0) + \frac{1}{2}L''(s_0)s_1^2\varepsilon + O(\varepsilon^2).$$

There is no constant contribution from moving the optimizer. The constant term is exactly $B(s_0) = (1 + \alpha^2)/2$. Ignoring this stationarity check is a common way to obtain an incorrect second term in a parameter-dependent extremal expansion.

A Random area selection and manifest exclusion

The area list was fixed before a single call to Python’s `secrets.randbelow(96)`. The returned zero-based index was 57, hence the selected one-based area was 58: *Discrete probability*. The original draw is in `notes/selection.json`; its full generating script is in `code/select_area.py`. There was no redraw. Re-executing that script makes a *new* random draw; it does not reproduce the original OS randomness. The preserved JSON is the historical selection record. The area labels overlap, so uniformity is over this declared list of labels, not over an intrinsic partition of all mathematics.

The supplied `manifest(1).tex`, dated 20 September 2026, catalogues 71 research packages. All entries were read as exclusion targets, without attempting to revalidate their claimed proofs. None concerns the numerical Rényi/Tsallis thresholds, the two-coin fixed-mean problem, or the particular Hillion–Johnson conjecture selected here. The exclusion audit is recorded in `notes/manifest_exclusion.md`.

The complete 96-area list

- | | | |
|--------------------------------|------------------------------|---------------------------------------|
| 1. Additive combinatorics | 33. Ring theory | 65. Complex analysis |
| 2. Analytic number theory | 34. Commutative algebra | 66. Geometric function theory |
| 3. Algebraic number theory | 35. Noncommutative algebra | 67. Harmonic analysis |
| 4. Diophantine approximation | 36. Representation theory | 68. Functional analysis |
| 5. Finite fields | 37. Lie algebras | 69. Banach space geometry |
| 6. Arithmetic dynamics | 38. Universal algebra | 70. Operator theory |
| 7. Combinatorial number theory | 39. Lattice theory | 71. Matrix inequalities |
| 8. Integer partitions | 40. Order theory | 72. Tensor algebra |
| 9. Permutation patterns | 41. Category theory | 73. Optimization theory |
| 10. Algebraic combinatorics | 42. Type theory | 74. Discrete optimization |
| 11. Enumerative combinatorics | 43. Proof theory | 75. Numerical analysis |
| 12. Extremal set theory | 44. Computability theory | 76. Approximation theory |
| 13. Ramsey theory | 45. Model theory | 77. Combinatorial game theory |
| 14. Spectral graph theory | 46. Set-theoretic topology | 78. Mathematical logic of games |
| 15. Structural graph theory | 47. General topology | 79. Information theory |
| 16. Graph polynomials | 48. Algebraic topology | 80. Boolean functions |
| 17. Graph labeling | 49. Knot theory | 81. Hypergraph theory |
| 18. Design theory | 50. Low-dimensional topology | 82. Incidence geometry |
| 19. Coding theory | 51. Differential geometry | 83. Finite geometry |
| 20. Matroid theory | 52. Metric geometry | 84. Topological combinatorics |
| 21. Combinatorial geometry | 53. Geometric group theory | 85. Cluster algebras |
| 22. Discrete geometry | 54. Dynamical systems | 86. Tropical geometry |
| 23. Convex geometry | 55. Symbolic dynamics | 87. Algebraic statistics |
| 24. Lattice polytopes | 56. Ergodic theory | 88. Combinatorial commutative algebra |
| 25. Tiling theory | 57. Probability inequalities | 89. Symmetric functions |
| 26. Combinatorics on words | 58. Discrete probability | 90. q-series |
| 27. Automata theory | 59. Random walks | 91. Difference equations |
| 28. Formal language theory | 60. Stochastic processes | 92. Functional equations |
| 29. Semigroup theory | 61. Special functions | 93. Fractional calculus |
| 30. Finite group theory | 62. Orthogonal polynomials | 94. Integral inequalities |
| 31. Infinite group theory | 63. Continued fractions | 95. Potential theory |
| 32. Combinatorial group theory | 64. Asymptotic analysis | 96. Fractal geometry |

B A compact independent certificate

The following Python fragment is a complete exact verification of the order-two Jensen obstruction. It uses neither the derivative formulas nor any third-party package.

```
from fractions import Fraction as Q

def collision(a, b):
    f = ((1-a)*(1-b), a+b-2*a*b, a*b)
    return sum(x*x for x in f)

center = collision(Q(1,10), Q(1,10))
endpoint = collision(Q(3,20), Q(1,20))
assert center == Q(55088,80000)
assert endpoint == Q(54907,80000)
assert center - endpoint == Q(181,80000) > 0
```

Both endpoint entropies are equal by exchanging the coins. For Tsallis order two, subtracting the power sum from one reverses the displayed inequality. For Rényi order two, applying the decreasing function $-\log$ reverses it. Thus the fragment certifies a genuine Jensen failure, not just a numerical approximation to one.

References

- [1] E. Hillion and O. Johnson, *A proof of the Shepp–Olkin entropy concavity conjecture*, Bernoulli **23**(4B) (2017), 3638–3649. doi:10.3150/16-BEJ860. Original version: arXiv:1503.01570v1 (5 March 2015), especially Section 4 and Conjecture 4.2 on PDF page 7.
- [2] O. Johnson, *Entropy and thinning of discrete random variables*, The IMA Volumes in Mathematics and its Applications **161** (2017), 33–53. arXiv:1510.05390v3 (7 June 2016), Section 8, item 5.
- [3] E. Hillion and O. Johnson, *A proof of the Shepp–Olkin entropy monotonicity conjecture*, Electronic Journal of Probability **24** (2019), paper 126, 1–14. arXiv:1810.09791. The Rényi and Tsallis monotonicity discussion is in Section 4.
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- [5] H. Wang, *The Sharp Rényi and Tsallis Threshold in the Shepp–Olkin Concavity Problem*, preprint, submitted 23 September 2026, arXiv:2609.27433v1. Cited by Part II’s manuscript, which consulted its abstract and HTML and records its stated ordinary-sum theorem for context without relying on it; not read or verified for this report. [Added 29 September 2026, batch 57: Part III’s manuscript cites the same preprint, gives the author’s forename, Haoran, and likewise inspected only its abstract.]
- [6] C. Tsallis, *Possible generalization of Boltzmann–Gibbs statistics*, Journal of Statistical Physics **52** (1988), 479–487. doi:10.1007/BF01016429.
- [7] Y. Sakai and K. Iwata, *Sharp Bounds Between Two Rényi Entropies of Distinct Positive Orders*, preprint, 2016, arXiv:1605.00019v2. Cited by Part III’s manuscript, which inspected the bibliographic record and abstract, for the antecedent two-order extremal problem, not as an uninspected proof dependency.

- [8] D. Reeb and M. M. Wolf, *Tight bound on relative entropy by entropy difference*, IEEE Transactions on Information Theory **61** (2015), 1458–1473. [arXiv:1304.0036v3](#). Cited by Part III’s manuscript, which inspected the bibliographic record and abstract; includes a sharp finite-dimensional variance-of-surprisal bound.